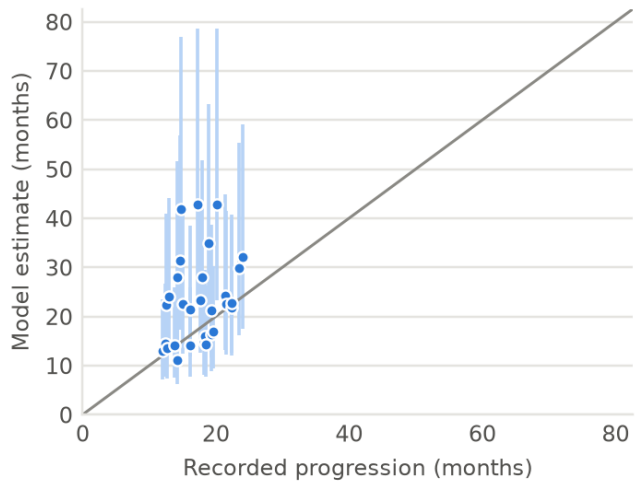
Group B · Progression in year 2 recorded progression-free survival: 12–24 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
30	**81**	**17.8 mo**	**60%**
2.8% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 12.0–24.0 mo	estimate in this window · median |error| 4 mo



Most frequent path end-points in this group	Patients
hereditary breast ovarian cancer syndrome	5
Signaling by plasma membrane FGFR1 fusions	3
Striated Muscle Contraction	3
breast carcinoma	2
Olfactory Signaling Pathway	2
Separation of Sister Chromatids	2
PKMTs methylate histone lysines	2
TFAP2 (AP-2) family regulates transcription of growth factors and their receptors	2

Logic formulas firing most often in this group	Mean weight	Patients
NStage(p, N0) ⇒ Progress(p)	-0.182	8
Stage(p, II) ⇒ Progress(p)	-0.097	7
TStage(p, T4) ⇒ Progress(p)	+0.121	5
Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	+0.079	3
Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ⇒ Progress(p)	+0.088	2
Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p)	-0.025	2

Most common driver alterations

TP53 mutated (43%), MYC amplified (27%), PIK3CA mutated (23%), ERBB2 amplified (17%), KMT2C mutated (17%), CCND1 amplified (7%), RUNX1 mutated (7%), FOXA1 mutated (7%), GATA3 mutated (7%), BRCA1 mutated (7%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-A8-A08X	43	III	Her2	12.0 mo	13 mo	✓	90	B_progression_12-24m/TCGA-A8-A08X.pdf
TCGA-D8-A1XC	85	III	LumA	12.4 mo	14 mo	✓	91	B_progression_12-24m/TCGA-D8-A1XC.pdf
TCGA-A2-A0CM	40	II	Basal	12.6 mo	22 mo	✓	53	B_progression_12-24m/TCGA-A2-A0CM.pdf
TCGA-C8-A12Q	78	III	Her2	12.7 mo	13 mo	✓	97	B_progression_12-24m/TCGA-C8-A12Q.pdf
TCGA-BH-A1EY	79	II	LumA	13.0 mo	24 mo	✗	25	B_progression_12-24m/TCGA-BH-A1EY.pdf
TCGA-A7-A425	70	III	LumA	13.8 mo	14 mo	✓	92	B_progression_12-24m/TCGA-A7-A425.pdf
TCGA-D8-A1JA	60	missing	Her2	14.2 mo	11 mo	✗	27	B_progression_12-24m/TCGA-D8-A1JA.pdf
TCGA-AR-A2LH	55	III	Normal	14.3 mo	28 mo	✗	90	B_progression_12-24m/TCGA-AR-A2LH.pdf
TCGA-E9-A1N6	52	II	missing	14.6 mo	31 mo	✗	72	B_progression_12-24m/TCGA-E9-A1N6.pdf
TCGA-B6-A0WW	58	X	LumB	14.7 mo	42 mo	✗	78	B_progression_12-24m/TCGA-B6-A0WW.pdf
TCGA-B6-A409	44	III	Basal	15.0 mo	23 mo	✓	85	B_progression_12-24m/TCGA-B6-A409.pdf
TCGA-B6-A0IK	63	III	Her2	16.1 mo	14 mo	✓	94	B_progression_12-24m/TCGA-B6-A0IK.pdf
TCGA-B6-A0X1	48	II	Basal	16.1 mo	21 mo	✓	98	B_progression_12-24m/TCGA-B6-A0X1.pdf
TCGA-E9-A2JS	72	II	LumB	17.2 mo	43 mo	✗	37	B_progression_12-24m/TCGA-E9-A2JS.pdf
TCGA-GM-A2D9	69	I	LumA	17.6 mo	23 mo	✓	4	B_progression_12-24m/TCGA-GM-A2D9.pdf
TCGA-AR-A0TY	54	II	LumB	17.9 mo	28 mo	✗	26	B_progression_12-24m/TCGA-AR-A0TY.pdf
TCGA-A7-A13E	62	II	Basal	18.3 mo	15 mo	✓	50	B_progression_12-24m/TCGA-A7-A13E.pdf
TCGA-A2-A0YJ	39	III	Basal	18.4 mo	16 mo	✓	90	B_progression_12-24m/TCGA-A2-A0YJ.pdf
TCGA-B6-A0WX	40	III	Basal	18.5 mo	14 mo	✓	94	B_progression_12-24m/TCGA-B6-A0WX.pdf
TCGA-A2-A3XU	35	II	Basal	18.9 mo	35 mo	✗	53	B_progression_12-24m/TCGA-A2-A3XU.pdf
TCGA-A7-A13G	79	II	LumA	19.3 mo	21 mo	✓	31	B_progression_12-24m/TCGA-A7-A13G.pdf
TCGA-AR-A1AI	47	II	Basal	19.3 mo	16 mo	✓	29	B_progression_12-24m/TCGA-AR-A1AI.pdf
TCGA-E2-A2P5	78	III	missing	19.6 mo	17 mo	✓	93	B_progression_12-24m/TCGA-E2-A2P5.pdf
TCGA-BH-A18J	56	IV	LumA	20.1 mo	43 mo	✗	99	B_progression_12-24m/TCGA-BH-A18J.pdf
TCGA-A2-A0SW	82	IV	LumB	21.5 mo	24 mo	✗	99	B_progression_12-24m/TCGA-A2-A0SW.pdf
TCGA-AC-A2B8	84	II	LumA	21.5 mo	23 mo	✓	28	B_progression_12-24m/TCGA-AC-A2B8.pdf
TCGA-LQ-A4E4	73	III	missing	22.4 mo	22 mo	✓	90	B_progression_12-24m/TCGA-LQ-A4E4.pdf
TCGA-A2-A3XS	62	III	missing	22.4 mo	23 mo	✓	92	B_progression_12-24m/TCGA-A2-A3XS.pdf
TCGA-B6-A0X4	62	II	LumA	23.5 mo	30 mo	✗	43	B_progression_12-24m/TCGA-B6-A0X4.pdf
TCGA-EW-A1P7	59	II	Normal	24.0 mo	32 mo	✗	57	B_progression_12-24m/TCGA-EW-A1P7.pdf

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**43**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**Her2**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T4D**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A8-A08X-01** · Aneuploidy score: **23** · TMB (nonsyn., /Mb): **1.1** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

90th

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 32%

Model-estimated progression time

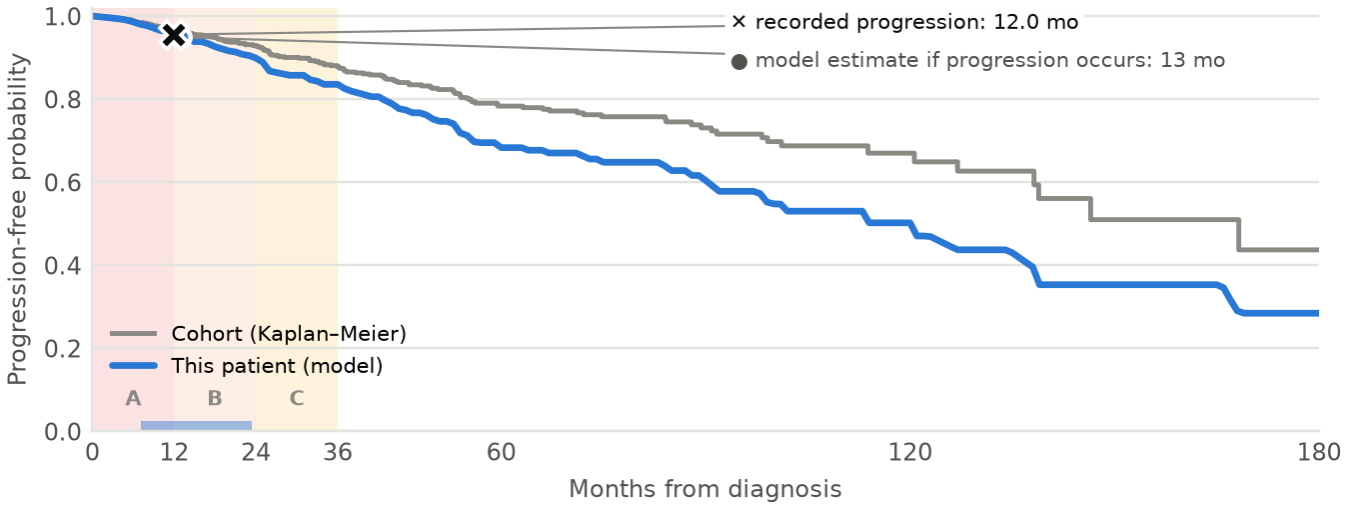
≈ 13 months

Window 12–24 months · IQR 7–23 mo if progression occurs

Recorded progression

12.0 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 1 mo



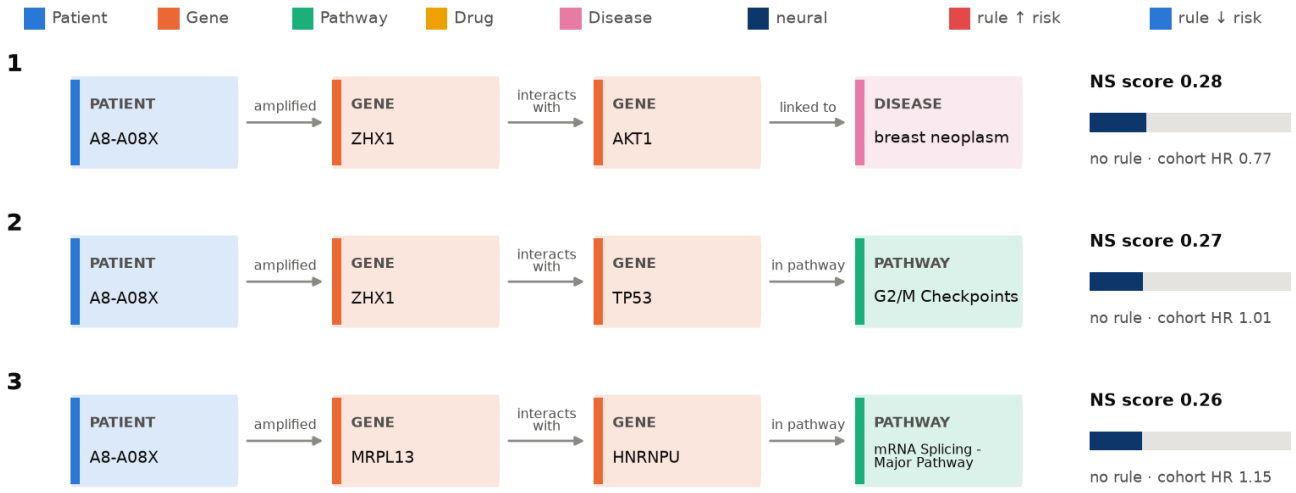
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 43-year-old patient has a Her2 tumour, AJCC stage III (T4, N3, M0). The fused neuro-symbolic model places them in the **90th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4) ⇒ Progress(p)*) raises the progression hazard (weight +0.096 per SD). The highest-scoring knowledge-graph path runs from **ZHX1** to **breast neoplasm**. Progression was recorded at 12.0 months; the model's estimate given progression was 13 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

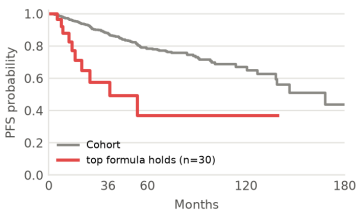


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A8-A08X** amplified → **ZHX1** interacts with → **AKT1** linked to → **breast neoplasm**	47	No learned formula fires on this path; its score is neural only. cohort: HR 0.77, n = 794 carriers	**0.28**
2	**TCGA-A8-A08X** amplified → **ZHX1** interacts with → **TP53** in pathway → **G2/M Checkpoints**	44	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 289 carriers	**0.27**
3	**TCGA-A8-A08X** amplified → **MRPL13** interacts with → **HNRNPU** in pathway → **mRNA Splicing - Major Pathway**	44	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 237 carriers	**0.26**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

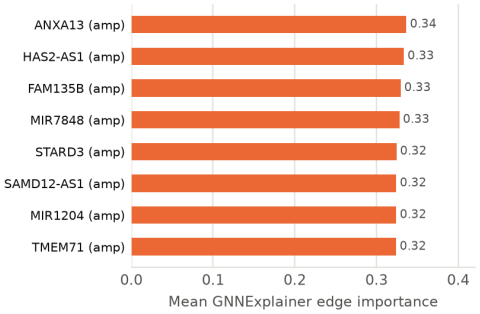
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.096 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	28 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.C176F	57	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
ANXA13	High-level amplification	GISTIC ±2	0	0.34
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.33
FAM135B	High-level amplification	GISTIC ±2	0	0.33
MIR7848	High-level amplification	GISTIC ±2	0	0.33
STARD3	High-level amplification	GISTIC ±2	1	0.32
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.32
MIR1204	High-level amplification	GISTIC ±2	0	0.32
TMEM71	High-level amplification	GISTIC ±2	0	0.32
ZHX1-C8ORF76	High-level amplification	GISTIC ±2	0	0.32



110 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 45 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CX	52	II	Her2	0.75	no progression, followed 56.8 mo
TCGA-AO-A0JE	53	III	Her2	0.71	no progression, followed 76.8 mo
TCGA-E2-A1B0	50	III	Her2	0.71	no progression, followed 53.6 mo
TCGA-BH-A1F2	53	III	LumB	0.70	no progression, followed 31.5 mo
TCGA-AN-A04C	51	II	Her2	0.69	no progression, followed 1.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-D8-A1XC · Group B

Report date 31 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**85**	Tumour type	**Breast Invasive Mixed Mucinous Carcinoma**
Sex	**Female**	OncoTree code	**IMMC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8480/3**

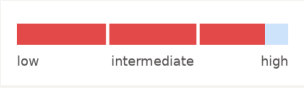
Sample: **TCGA-D8-A1XC-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.87** · MSIsensor score: **0.29** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 6% · 24 mo 11% · 60 mo 32%

Model-estimated progression time

≈ 14 months

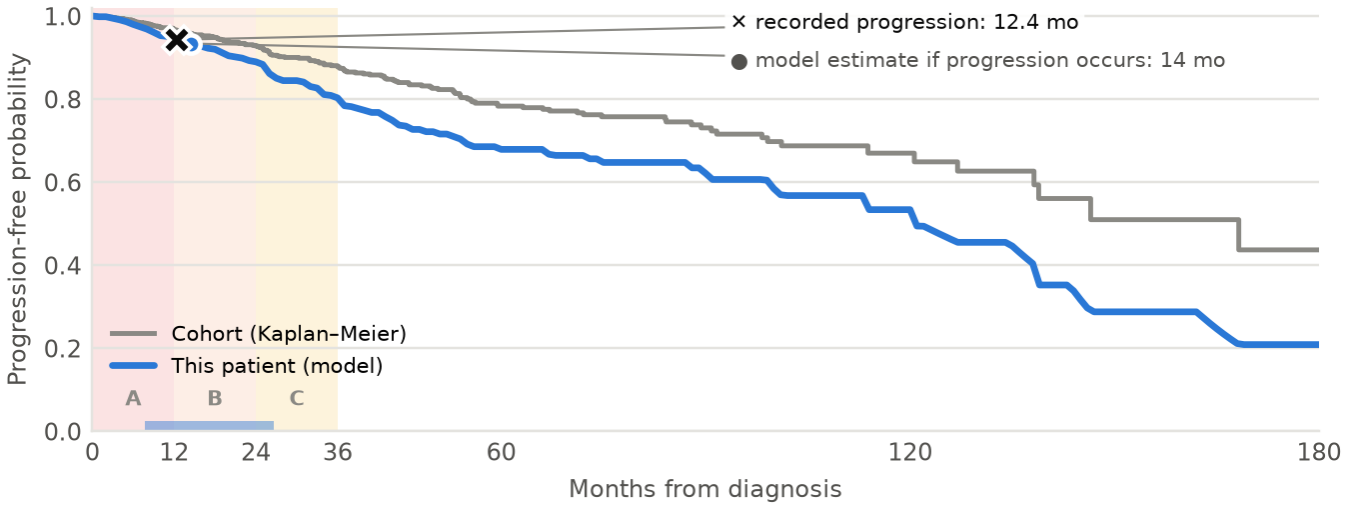
Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

12.4 months

Group B · Progression in year 2

✓ estimate in recorded window · |error| 2 mo



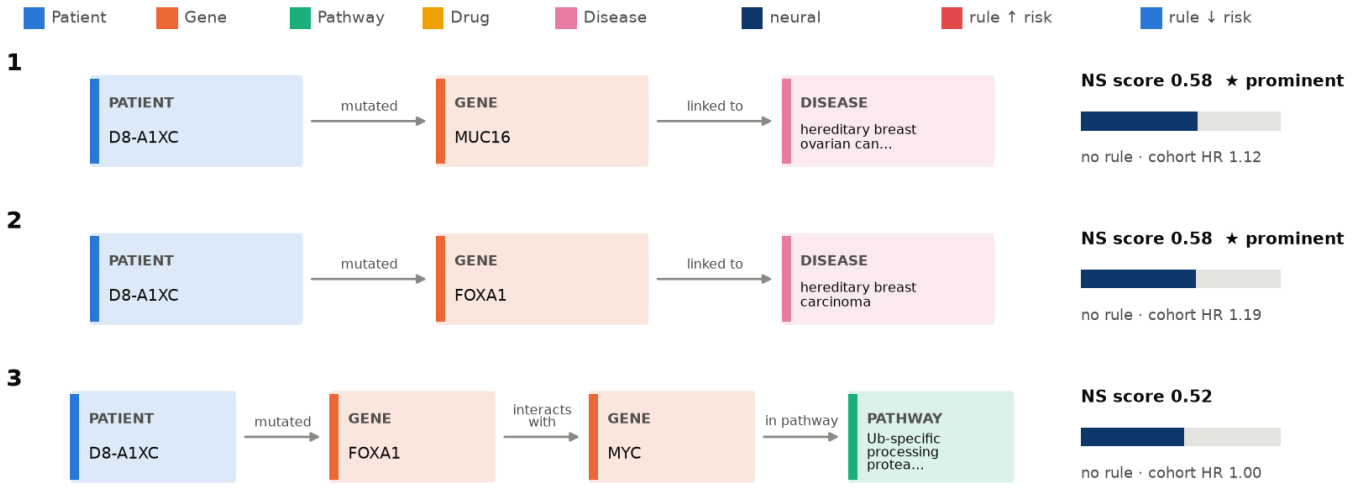
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 85-year-old patient has a LumA tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.119 per SD). The highest-scoring knowledge-graph path runs from **MUC16** to **hereditary breast ovarian cancer syndrome**. Progression was recorded at 12.4 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

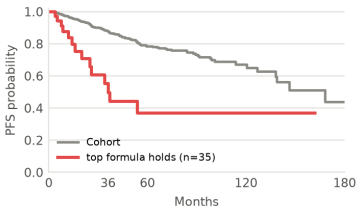


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-D8-A1XC** mutated → **MUC16** linked to → **hereditary breast ovarian cancer syndrome**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 800 carriers	**0.58**
2 ★	**TCGA-D8-A1XC** mutated → **FOXA1** linked to → **hereditary breast carcinoma**	96	No learned formula fires on this path; its score is neural only. cohort: HR 1.19, n = 797 carriers	**0.58**
3	**TCGA-D8-A1XC** mutated → **FOXA1** interacts with → **MYC** in pathway → **Ub-specific processing proteases**	86	No learned formula fires on this path; its score is neural only. cohort: HR 1.00, n = 457 carriers	**0.52**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

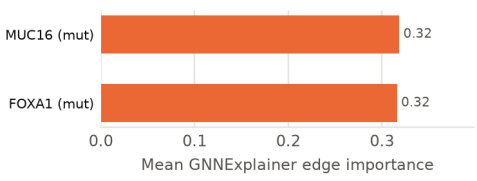
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.119 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	35 pts
+0.008 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.E410Afs*2	9	0.32
MUC16	Mutation	p.E6944D	8	0.32



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MUC16 → Oregovomab (DB04964). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1FR	73	III	LumA	0.94	progressed at 54.0 mo (group D)
TCGA-A2-A3KD	47	III	LumA	0.66	no progression, followed 39.6 mo
TCGA-D8-A73W	79	III	LumA	0.64	progressed at 0.7 mo (group A)
TCGA-C8-A133	65	III	LumA	0.63	no progression, followed 0.0 mo
TCGA-B6-A0RO	71	III	LumA	0.62	no progression, followed 162.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A0CM-01** · Aneuploidy score: **24** · TMB (nonsyn., /Mb): **1.73** · MSIsensor score: **0.39** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

53rd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

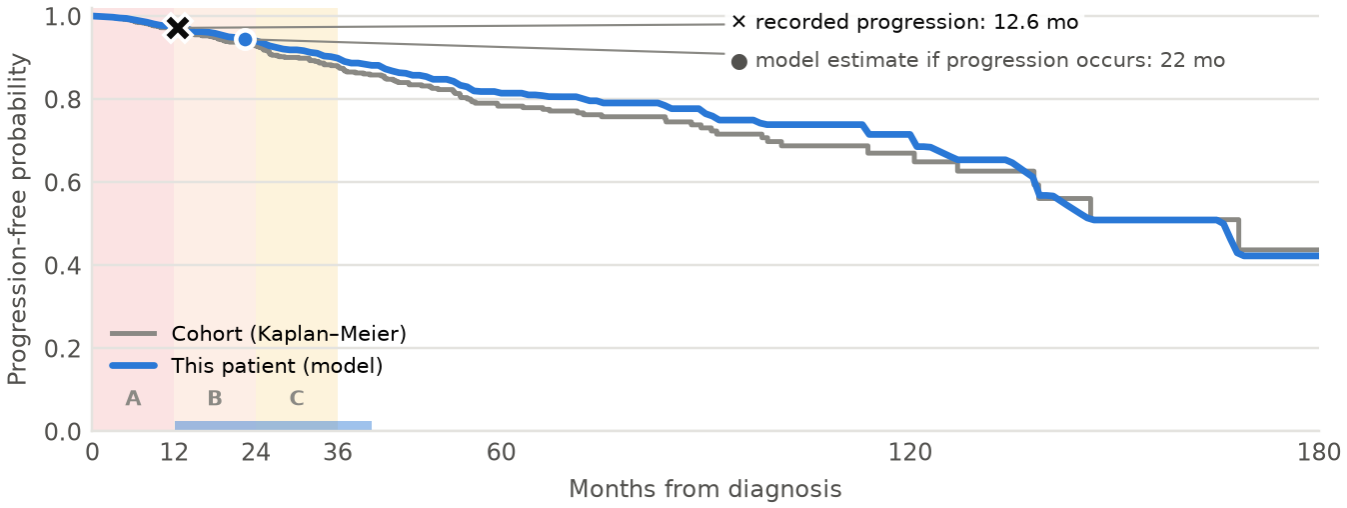
≈ 22 months

Window 12–24 months · IQR 12–41 mo
if progression occurs

Recorded progression

12.6 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 10 mo



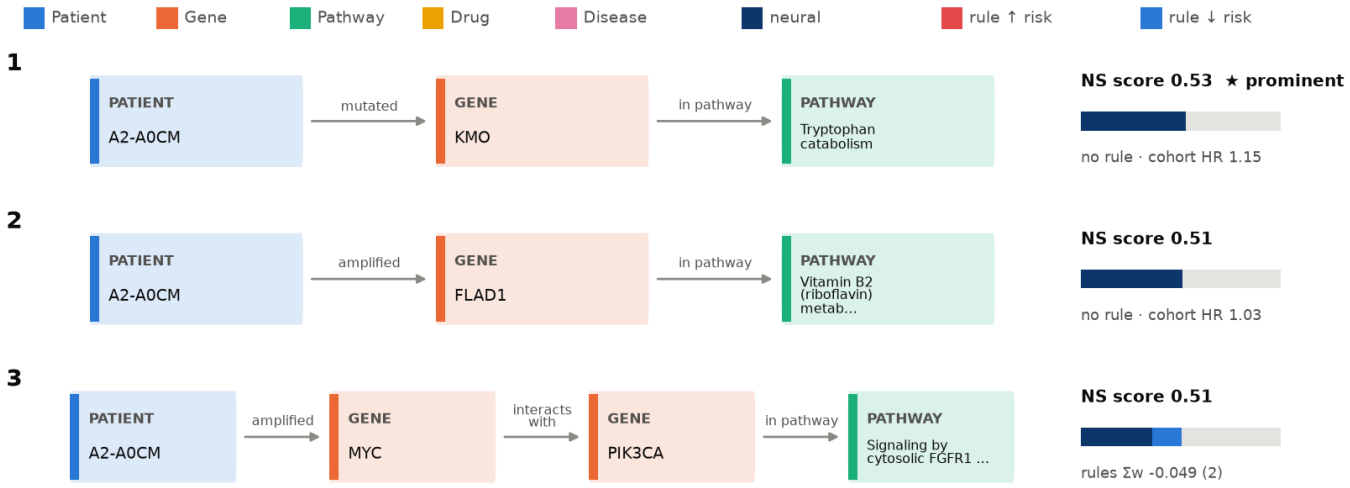
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **53rd risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **KMO** to **Tryptophan catabolism**. Progression was recorded at 12.6 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

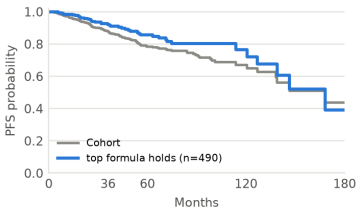


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A0CM** mutated → **KMO** in pathway → **Tryptophan catabolism**	88	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 93 carriers	**0.53**
2	**TCGA-A2-A0CM** amplified → **FLAD1** in pathway → **Vitamin B2 (riboflavin) metabolism**	85	No learned formula fires on this path; its score is neural only. cohort: HR 1.03, n = 171 carriers	**0.51**
3	**TCGA-A2-A0CM** amplified → **MYC** interacts with → **PIK3CA** in pathway → **Signaling by cytosolic FGFR1 fusion mutants**	60	↓ **slower progression** · Σw -0.049 over 2 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.86, n = 309 carriers	**0.51**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

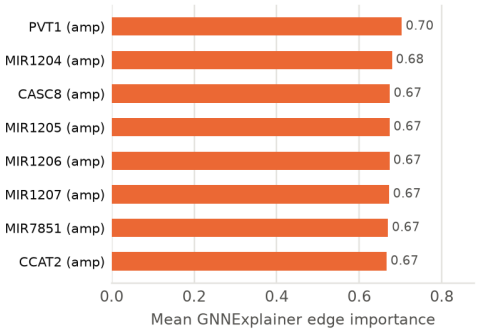
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.007 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	306 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TP53 · driver	Mutation	p.E204Sfs*43	57	0.31
KMT2C · driver	Homozygous deletion	GISTIC ±2	4	0.31
PVT1	High-level amplification	GISTIC ±2	0	0.70
MIR1204	High-level amplification	GISTIC ±2	0	0.68
CASC8	High-level amplification	GISTIC ±2	0	0.67
MIR1205	High-level amplification	GISTIC ±2	0	0.67
MIR1206	High-level amplification	GISTIC ±2	0	0.67
MIR1207	High-level amplification	GISTIC ±2	0	0.67
MIR7851	High-level amplification	GISTIC ±2	0	0.67
CCAT2	High-level amplification	GISTIC ±2	0	0.67
POU5F1B	High-level amplification	GISTIC ±2	0	0.66



22 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1N8	48	II	Basal	0.82	no progression, followed 34.2 mo
TCGA-AR-A1AQ	49	II	Basal	0.80	no progression, followed 99.3 mo
TCGA-E2-A1B6	44	II	Basal	0.79	no progression, followed 28.5 mo
TCGA-A2-A3XX	49	II	Basal	0.79	no progression, followed 47.3 mo
TCGA-AR-A1AI	47	II	Basal	0.78	progressed at 19.3 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**78**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Asian**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**—**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-C8-A12Q-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **3.27** · MSIsensor score: **0.35** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

97th

High risk tier · fused neuro-symbolic model



Probability of progression

28% by 36 mo

12 mo 8% · 24 mo 16% · 60 mo 44%

Model-estimated progression time

≈ 13 months

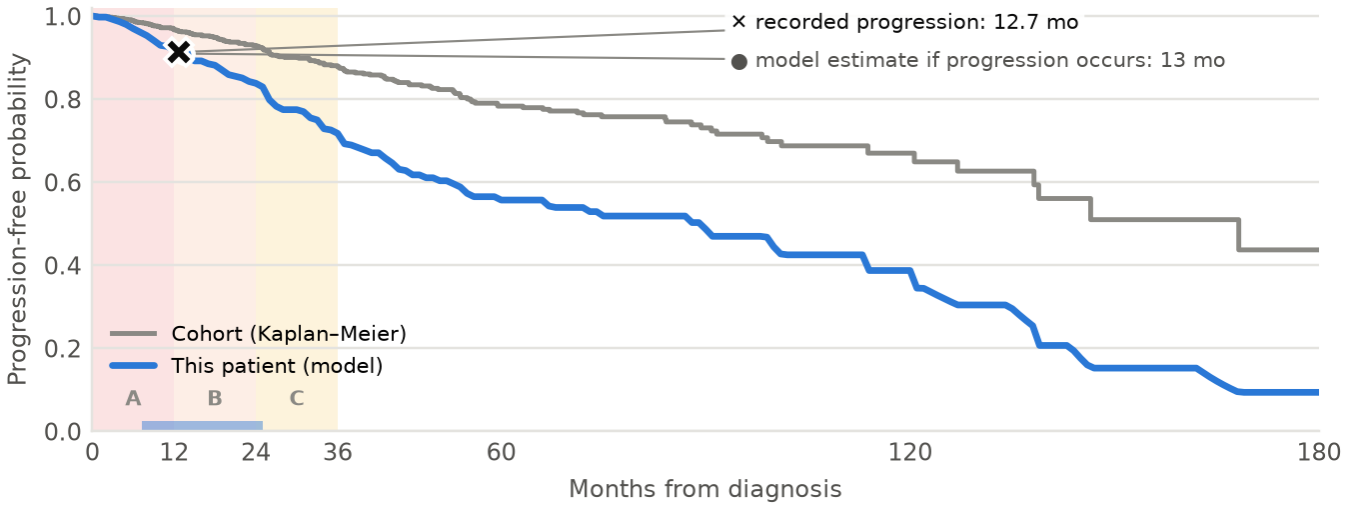
Window 12–24 months · IQR 7–25 mo if progression occurs

Recorded progression

12.7 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 1 mo



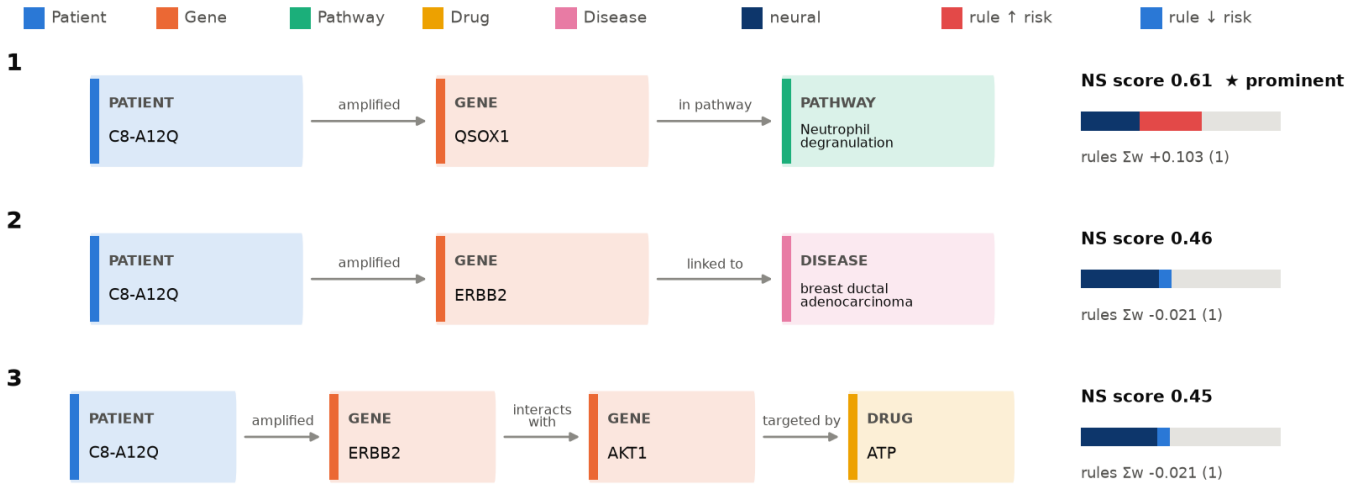
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 78-year-old patient has a Her2 tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **97th risk percentile** (high tier). The strongest learned formula that holds for them (*Subtype(p, Her2)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Platelet degranulation ')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.103 per SD). The highest-scoring knowledge-graph path runs from **QSOX1** to **Neutrophil degranulation**; the learned formulas it instantiates point to faster progression (Σw +0.103). Progression was recorded at 12.7 months; the model's estimate given progression was 13 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

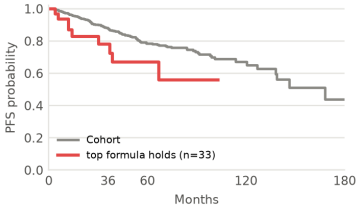


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-C8-A12Q** amplified → **QSOX1** in pathway → **Neutrophil degranulation**	49	↑ **faster progression** · Σw +0.103 over 1 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ⇒ Progress(p) cohort: HR 1.35, n = 306 carriers	**0.61**
2	**TCGA-C8-A12Q** amplified → **ERBB2** linked to → **breast ductal adenocarcinoma**	65	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 1.31, n = 529 carriers	**0.46**
3	**TCGA-C8-A12Q** amplified → **ERBB2** interacts with → **AKT1** targeted by → **ATP**	64	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 0.83, n = 326 carriers	**0.45**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

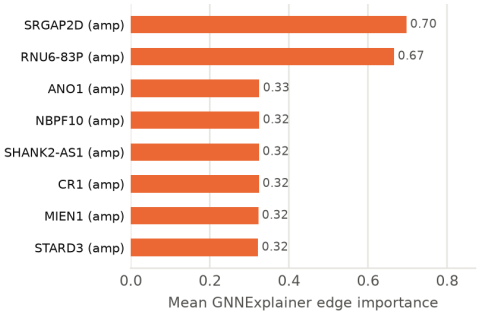
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.103 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ⇒ Progress(p) [KG]	mined pair	32 pts
+0.098 ↑ risk	Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	mined pair	31 pts
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
+0.025 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Interaction between L1 and Ankyrins') ⇒ Progress(p) [KG]	mined pair	28 pts
-0.021 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) [KG]	RL-proposed	73 pts
+0.012 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	mined pair	45 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
KMT2C · driver	Mutation	p.I323=	4	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
TP53 · driver	Mutation	p.Y205N	57	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.70
RNU6-83P	High-level amplification	GISTIC ±2	0	0.67
ANO1	High-level amplification	GISTIC ±2	2	0.33
NBPF10	High-level amplification	GISTIC ±2	0	0.32
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
CR1	High-level amplification	GISTIC ±2	3	0.32
MIEN1	High-level amplification	GISTIC ±2	0	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32



67 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021) … and 55 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0HY	60	I	LumB	0.71	no progression, followed 50.8 mo
TCGA-C8-A138	54	III	Her2	0.67	no progression, followed 12.5 mo
TCGA-A8-A07I	69	III	Her2	0.66	no progression, followed 14.0 mo
TCGA-B6-A0I9	62	IV	Her2	0.66	progressed at 11.9 mo (group A)
TCGA-AR-A254	50	III	Her2	0.66	no progression, followed 85.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

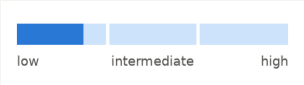
Sample: **TCGA-BH-A1EY-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.23** · MSIsensor score: **0.05** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

25th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

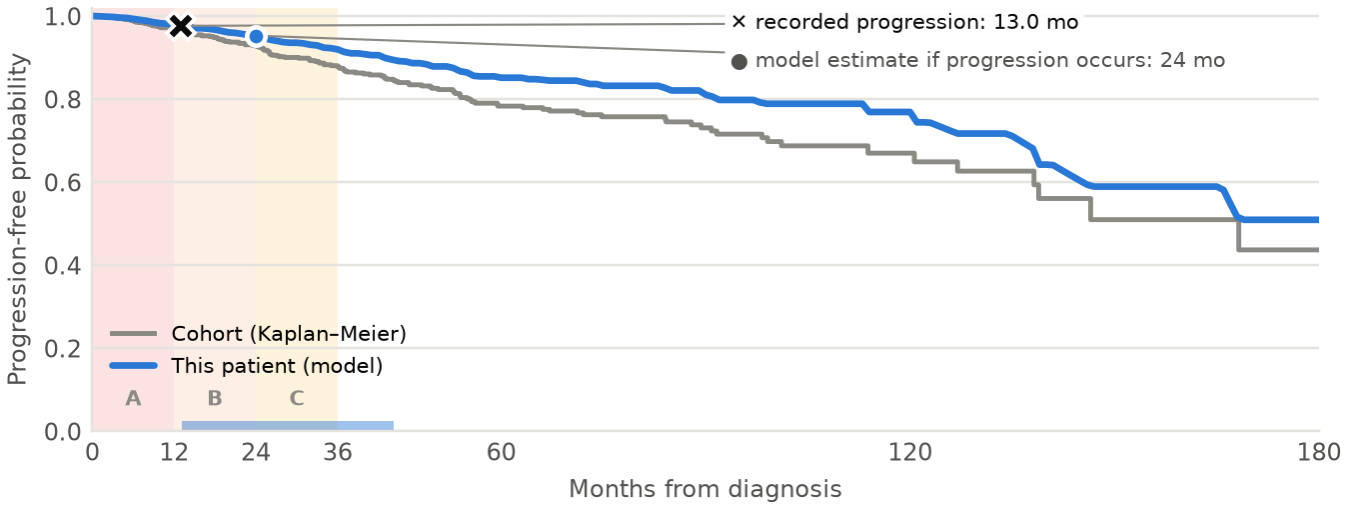
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

13.0 months

■ **Group B** · Progression in year 2
✗ **recorded window differs** · |error| 11 mo



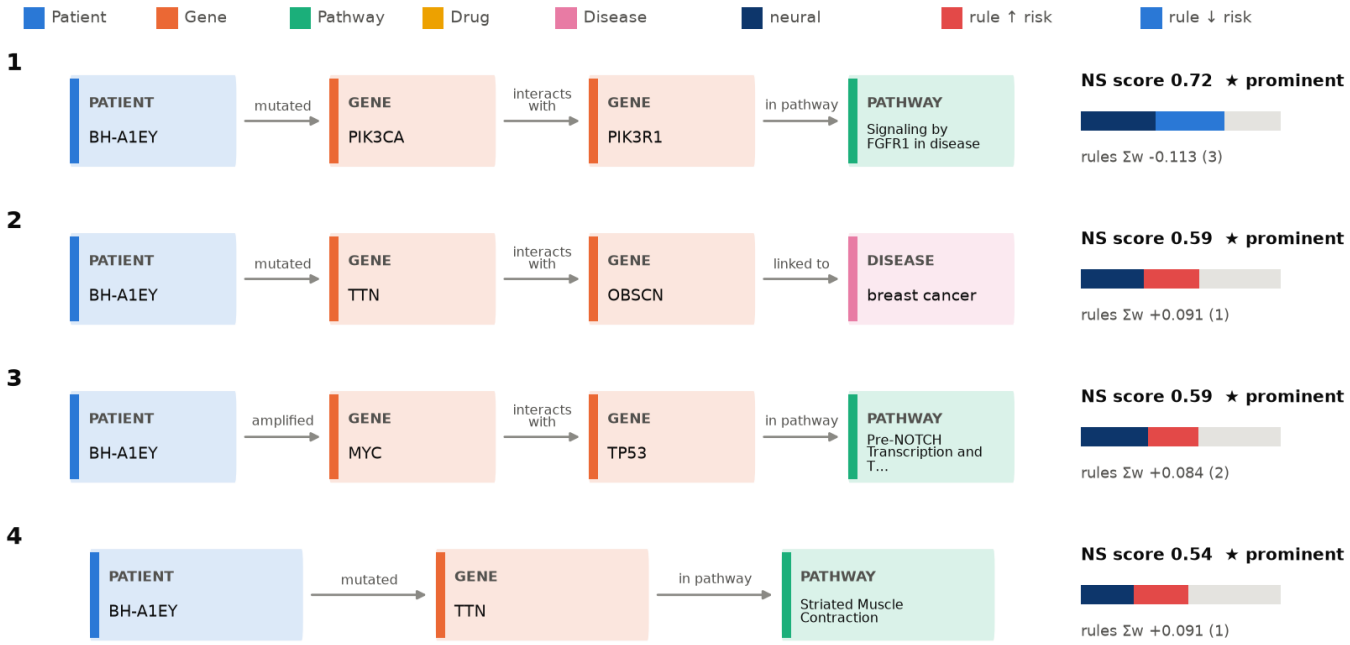
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **25th risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Signaling by FGFR1 in disease**; the learned formulas it instantiates point to slower progression (Σw -0.113). Progression was recorded at 13.0 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

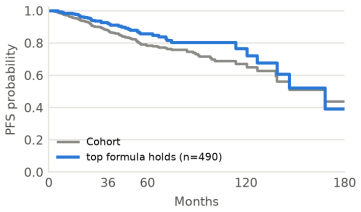


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1EY** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Signaling by FGFR1 in disease**	63	↓ **slower progression** · Σw -0.113 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.75, n = 459 carriers	**0.72**
2 ★	**TCGA-BH-A1EY** mutated → **TTN** interacts with → **OBSCN** linked to → **breast cancer**	53	↑ **faster progression** · Σw +0.091 over 1 formula(s) Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) cohort: HR 0.97, n = 792 carriers	**0.59**
3 ★	**TCGA-BH-A1EY** amplified → **MYC** interacts with → **TP53** in pathway → **Pre-NOTCH Transcription and Translation**	56	↑ **faster progression** · Σw +0.084 over 2 formula(s) Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) cohort: HR 0.98, n = 490 carriers	**0.59**
4 ★	**TCGA-BH-A1EY** mutated → **TTN** in pathway → **Striated Muscle Contraction**	44	↑ **faster progression** · Σw +0.091 over 1 formula(s) Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) cohort: HR 0.85, n = 332 carriers	**0.54**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

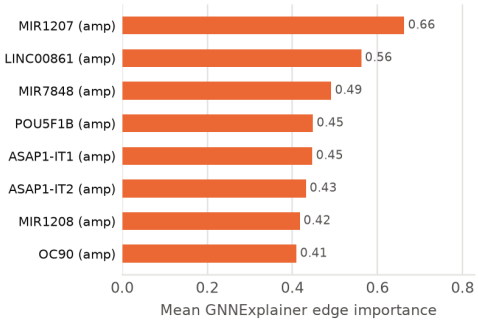
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
+0.091 ↑ risk	Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) [KG]	RL-proposed	84 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.024 ↓ risk	Stage(p, II) ∧ Mutated(p, TP53) ∧ Altered(p, g) ∧ InPathway(g, 'Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants') ⇒ Progress(p) [KG]	RL-proposed	71 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.Y126C	57	0.32
KMT2C · driver	Mutation	p.S3394N	4	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
MAP2K4 · driver	Mutation	p.X363_splice	12	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
MIR1207	High-level amplification	GISTIC ±2	0	0.66
LINC00861	High-level amplification	GISTIC ±2	0	0.56
MIR7848	High-level amplification	GISTIC ±2	0	0.49
POU5F1B	High-level amplification	GISTIC ±2	0	0.45
ASAP1-IT1	High-level amplification	GISTIC ±2	0	0.45
ASAP1-IT2	High-level amplification	GISTIC ±2	0	0.43
MIR1208	High-level amplification	GISTIC ±2	0	0.42



49 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813) … and 18 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1RH	63	II	Her2	0.79	no progression, followed 46.6 mo
TCGA-BH-A0H9	69	II	LumA	0.78	no progression, followed 41.0 mo
TCGA-BH-A0GY	67	II	LumA	0.78	no progression, followed 30.3 mo
TCGA-A8-A07Z	85	II	LumA	0.76	no progression, followed 45.1 mo
TCGA-S3-AA10	65	II	Basal	0.76	no progression, followed 19.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**70**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

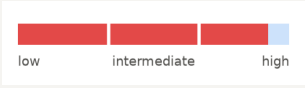
Sample: **TCGA-A7-A425-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **0.93** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

92nd

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 6% · 24 mo 11% · 60 mo 33%

Model-estimated progression time

≈ 14 months

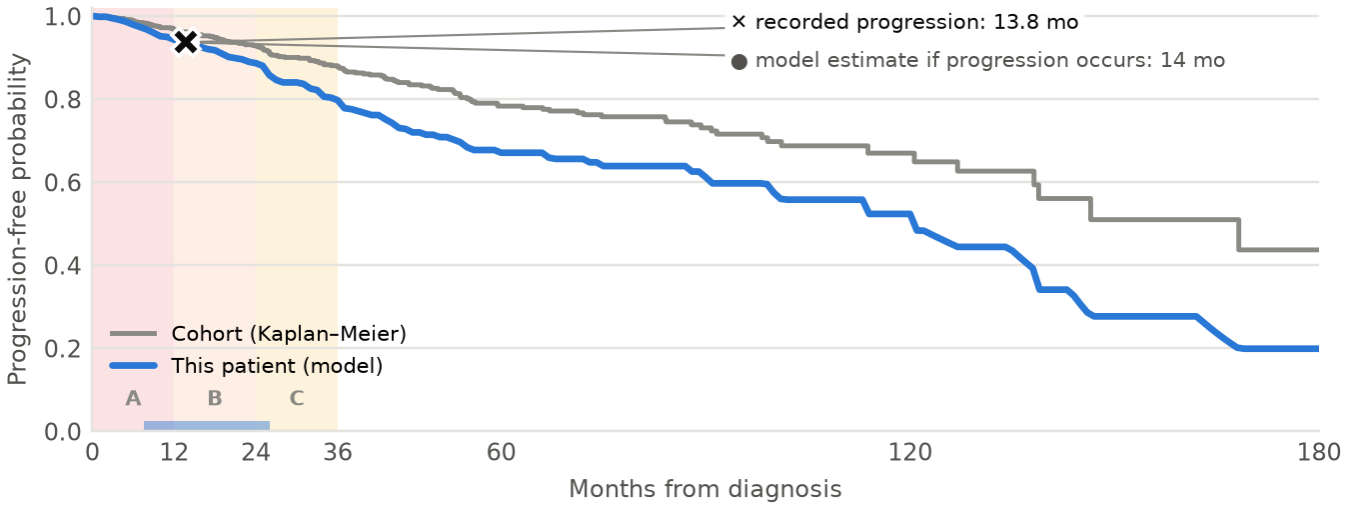
Window 12–24 months · IQR 8–26 mo if progression occurs

Recorded progression

13.8 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 0 mo



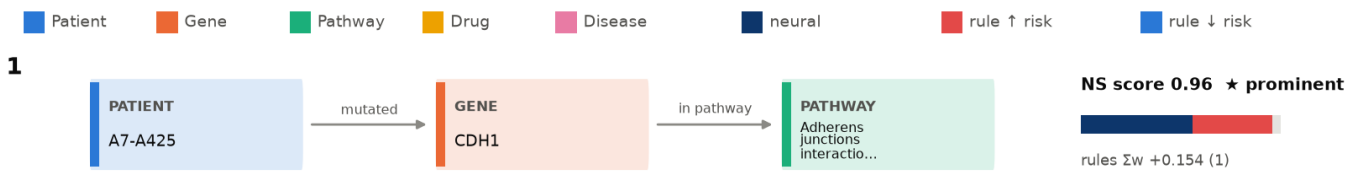
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 70-year-old patient has a LumA tumour, AJCC stage III (T3, N3, MX). The fused neuro-symbolic model places them in the **92nd risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p)*) raises the progression hazard (weight +0.154 per SD). The highest-scoring knowledge-graph path runs from **CDH1** to **Adherens junctions interactions**; the learned formulas it instantiates point to faster progression (Σw +0.154). Progression was recorded at 13.8 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



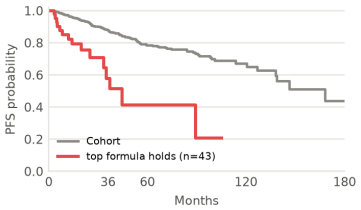
1 of 17 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's

gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A7-A425** mutated → **CDH1** in pathway → **Adherens junctions interactions**	93	↑ **faster progression** · Σw +0.154 over 1 formula(s) NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 1.05, n = 206 carriers	**0.96**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

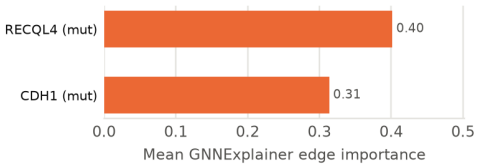
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.154 ↑ risk	NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	43 pts
+0.026 ↑ risk	TStage(p, T3) ⇒ Progress(p)	single atom	102 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.K397Mfs*17	17	0.31
RECQL4	Mutation	p.P160S	0	0.40



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-Z7-A8R5	61	III	LumA	0.79	progressed at 6.0 mo (group A)
TCGA-GM-A4E0	67	III	LumA	0.78	no progression, followed 72.0 mo
TCGA-E9-A6HE	45	III	LumA	0.77	no progression, followed 27.8 mo
TCGA-D8-A3Z6	56	III	LumA	0.70	no progression, followed 18.5 mo
TCGA-E9-A5FK	60	III	LumA	0.70	no progression, followed 26.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**—**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

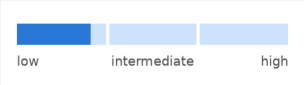
Sample: **TCGA-D8-A1JA-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **26.6** · MSIsensor score: **0.38** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

27th

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

≈ 11 months

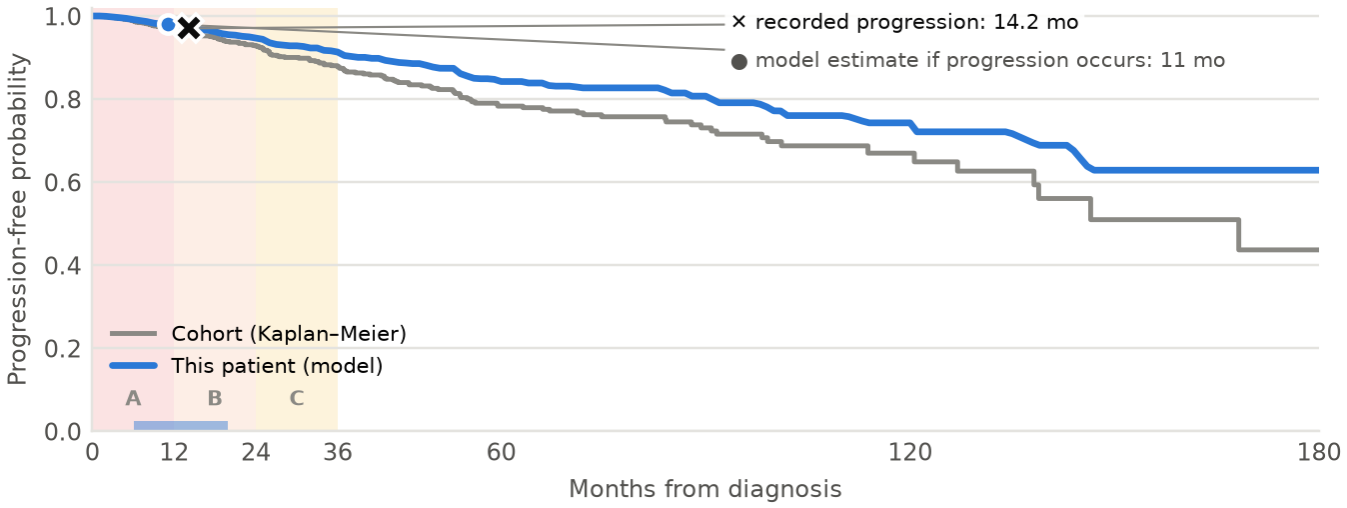
Window ≤ 12 months · IQR 6–20 mo if progression occurs

Recorded progression

14.2 months

Group B · Progression in year 2

✗ recorded window differs · |error| 3 mo



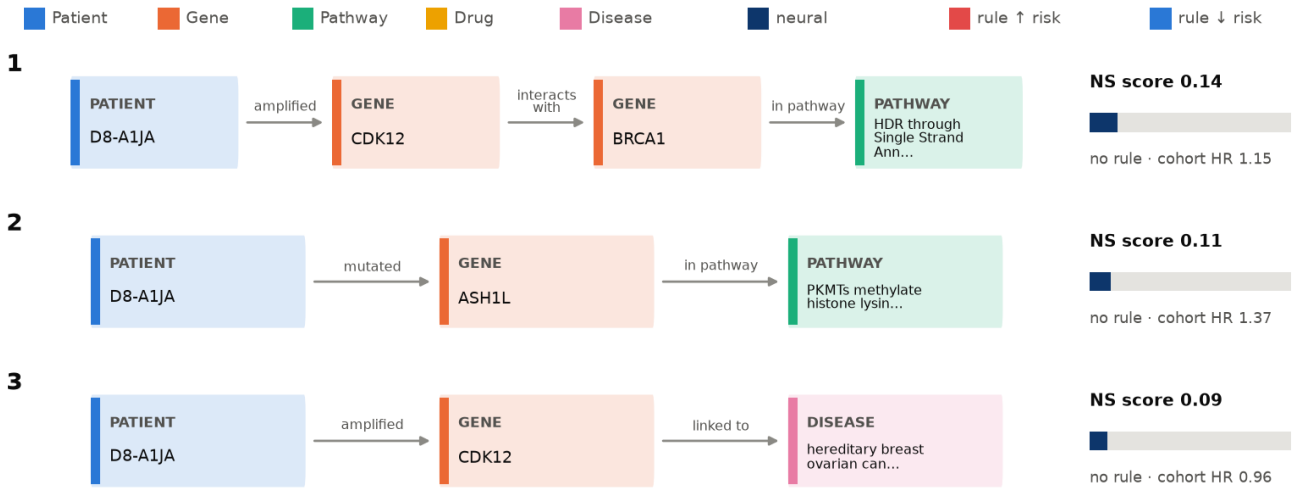
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a Her2 tumour, AJCC stage missing (T4, missing, M0). The fused neuro-symbolic model places them in the **27th risk percentile** (low tier). The strongest learned formula that holds for them (*TStage(p, T4) ⇒ Progress(p)*) raises the progression hazard (weight +0.140 per SD). The highest-scoring knowledge-graph path runs from **CDK12** to **HDR through Single Strand Annealing (SSA)**. Progression was recorded at 14.2 months; the model's estimate given progression was 11 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

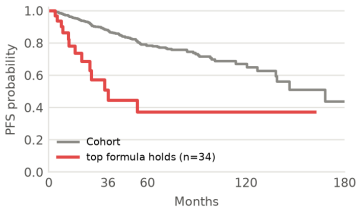


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-D8-A1JA** amplified → **CDK12** interacts with → **BRCA1** in pathway → **HDR through Single Strand Annealing (SSA)**	23	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 158 carriers	**0.14**
2	**TCGA-D8-A1JA** mutated → **ASH1L** in pathway → **PKMTs methylate histone lysines**	18	No learned formula fires on this path; its score is neural only. cohort: HR 1.37, n = 323 carriers	**0.11**
3	**TCGA-D8-A1JA** amplified → **CDK12** linked to → **hereditary breast ovarian cancer syndrome**	15	No learned formula fires on this path; its score is neural only. cohort: HR 0.96, n = 798 carriers	**0.09**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

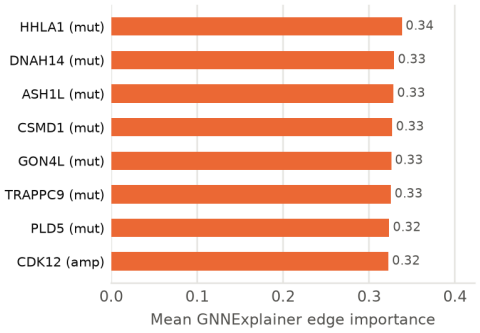
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.140 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	32 pts
+0.072 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ⇒ Progress(p) [KG]	mined pair	32 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
ERBB2 · driver	Mutation	p.V91=	55	0.31
FGFR1 · driver	Mutation	p.L675=	43	0.31
RB1 · driver	Mutation	p.V714=	25	0.31
HHLA1	Mutation	p.W420C	0	0.34
DNAH14	Mutation	p.*742*	0	0.33
ASH1L	Mutation	p.E2148Q, p.E2569Q, p.K2155N	1	0.33
CSMD1	Mutation	p.R2141K, p.S1027L	0	0.33
GON4L	Mutation	p.E2209K	0	0.33
TRAPPC9	Mutation	p.E600K	2	0.33
PLD5	Mutation	p.Q6=	0	0.32
CDK12	High-level amplification	GISTIC ±2	2	0.32



56 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 90 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AC-A23H	90	II	Her2	0.55	progressed at 5.7 mo (group A)
TCGA-BH-A1EN	78	II	Her2	0.54	no progression, followed 69.9 mo
TCGA-A8-A09G	79	III	Her2	0.53	no progression, followed 0.0 mo
TCGA-C8-A1HK	53	II	Her2	0.52	no progression, followed 12.0 mo
TCGA-A2-A04X	34	II	Her2	0.51	no progression, followed 55.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**55**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE III**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

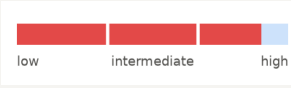
Sample: **TCGA-AR-A2LH-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **2** · MSIsensor score: **0.01** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

90th

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 32%

Model-estimated progression time

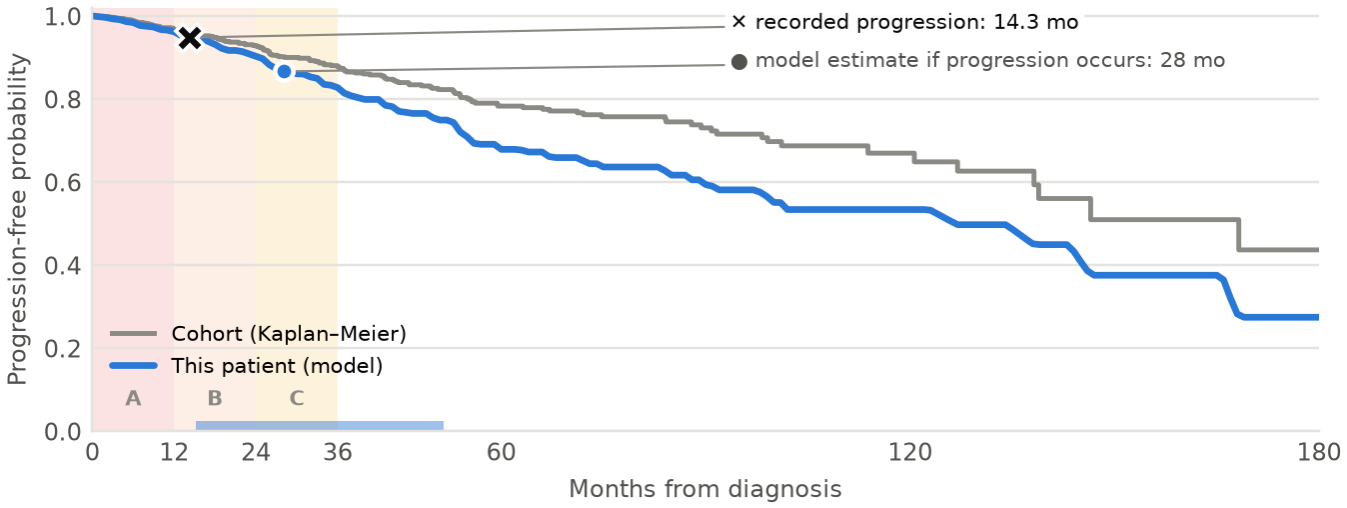
≈ 28 months

Window 24–36 months · IQR 15–52 mo
if progression occurs

Recorded progression

14.3 months

Group B · Progression in year 2
✗ recorded window differs · |error| 14 mo



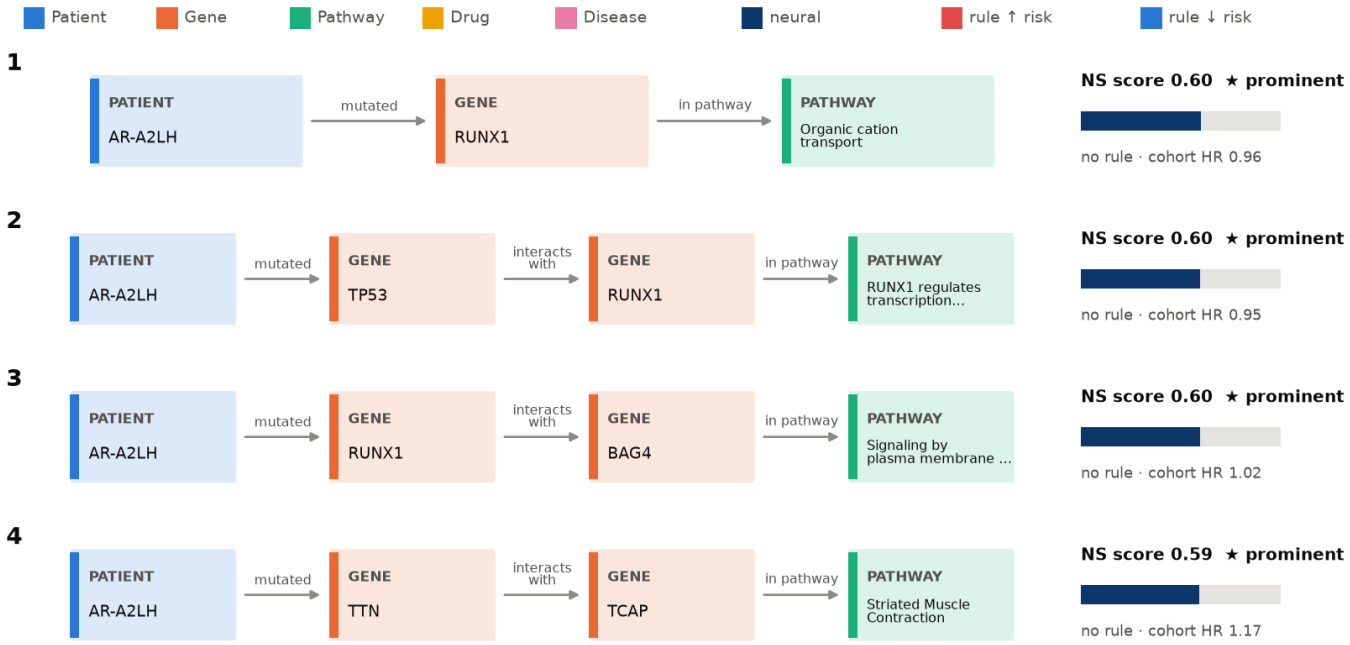
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 55-year-old patient has a Normal tumour, AJCC stage III (T3, N3, M0). The fused neuro-symbolic model places them in the **90th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N3) ⇒ Progress(p)*) raises the progression hazard (weight +0.077 per SD). The highest-scoring knowledge-graph path runs from **RUNX1** to **Organic cation transport**. Progression was recorded at 14.3 months; the model's estimate given progression was 28 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

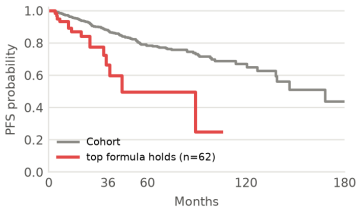


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AR-A2LH** mutated → **RUNX1** in pathway → **Organic cation transport**	100	No learned formula fires on this path; its score is neural only. cohort: HR 0.96, n = 56 carriers	**0.60**
2 ★	**TCGA-AR-A2LH** mutated → **TP53** interacts with → **RUNX1** in pathway → **RUNX1 regulates transcription of genes involved in interleukin signaling**	100	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 86 carriers	**0.60**
3 ★	**TCGA-AR-A2LH** mutated → **RUNX1** interacts with → **BAG4** in pathway → **Signaling by plasma membrane FGFR1 fusions**	100	No learned formula fires on this path; its score is neural only. cohort: HR 1.02, n = 139 carriers	**0.60**
4 ★	**TCGA-AR-A2LH** mutated → **TTN** interacts with → **TCAP** in pathway → **Striated Muscle Contraction**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.17, n = 329 carriers	**0.59**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

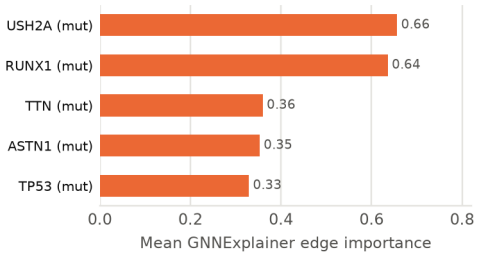
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.077 ↑ risk	NStage(p, N3) ⇒ Progress(p)	single atom	61 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
RUNX1 · driver	Mutation	p.Q264*	18	0.64
TP53 · driver	Mutation	p.R273H	57	0.33
USH2A	Mutation	p.E1739Q	0	0.66
TTN	Mutation	p.G12310E	2	0.36
ASTN1	Mutation	p.I825=	0	0.35



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A0JB	50	III	Normal	0.64	no progression, followed 50.7 mo
TCGA-BH-A0HP	65	III	LumA	0.63	no progression, followed 13.6 mo
TCGA-A8-A09X	62	III	Her2	0.62	no progression, followed 14.0 mo
TCGA-LD-A74U	79	III	LumA	0.62	no progression, followed 13.2 mo
TCGA-E9-A6HE	45	III	LumA	0.60	no progression, followed 27.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**52**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E9-A1N6-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **—** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

72nd

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 24%

Model-estimated progression time

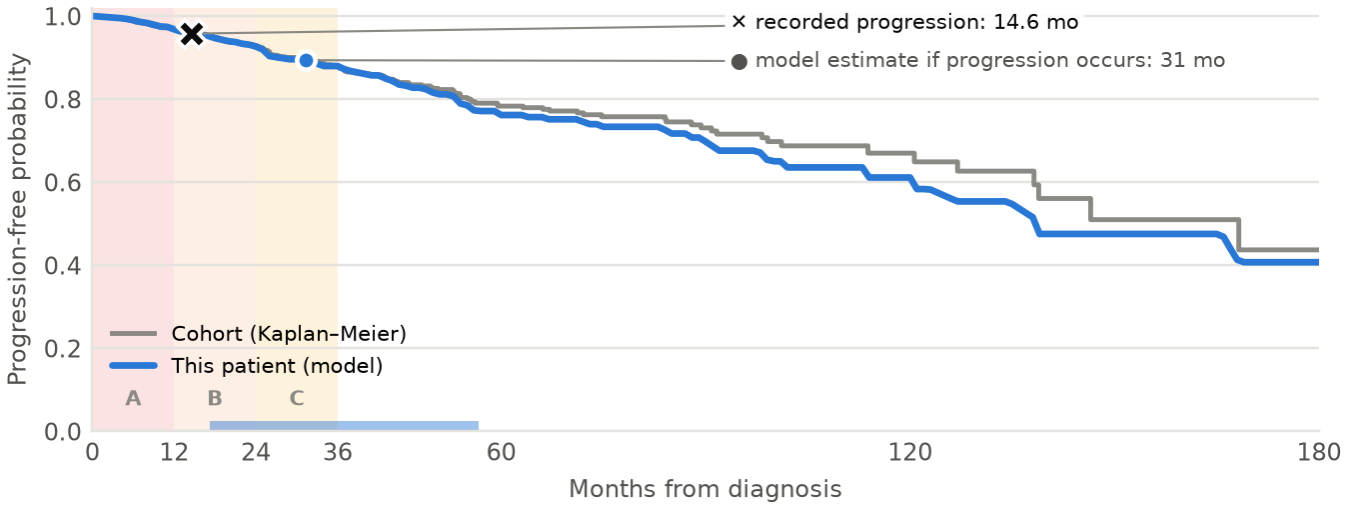
≈ 31 months

Window 24–36 months · IQR 17–57 mo
if progression occurs

Recorded progression

14.6 months

Group B · Progression in year 2
✗ recorded window differs · |error| 17 mo



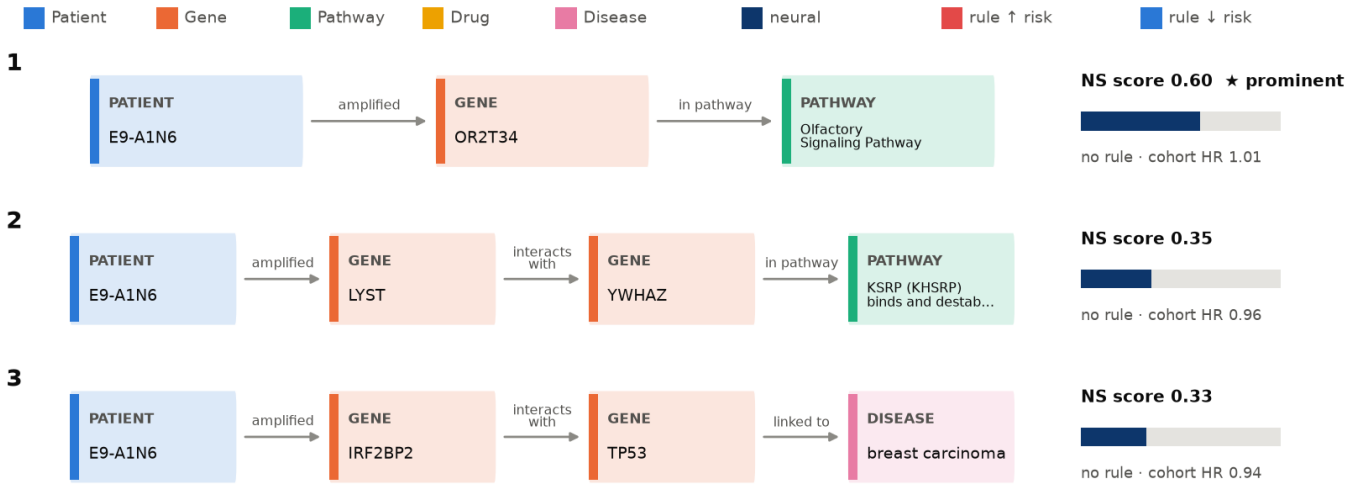
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 52-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **72nd risk percentile** (high tier). The strongest learned formula that holds for them (*Stage(p, II)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.025 per SD). The highest-scoring knowledge-graph path runs from **OR2T34** to **Olfactory Signaling Pathway**. Progression was recorded at 14.6 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

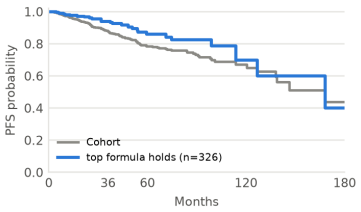


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E9-A1N6** amplified → **OR2T34** in pathway → **Olfactory Signaling Pathway**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 158 carriers	**0.60**
2	**TCGA-E9-A1N6** amplified → **LYST** interacts with → **YWHAZ** in pathway → **KSRP (KHSRP) binds and destabilizes mRNA**	59	No learned formula fires on this path; its score is neural only. cohort: HR 0.96, n = 170 carriers	**0.35**
3	**TCGA-E9-A1N6** amplified → **IRF2BP2** interacts with → **TP53** linked to → **breast carcinoma**	54	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 784 carriers	**0.33**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

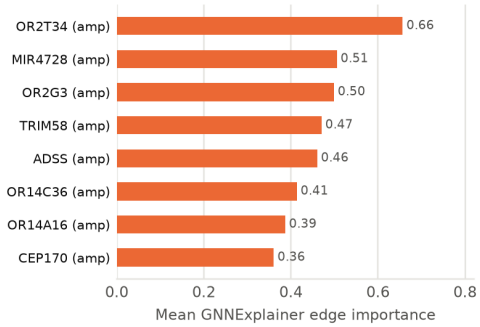
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
NF1 · driver	High-level amplification	GISTIC ±2	6	0.31
OR2T34	High-level amplification	GISTIC ±2	1	0.66
MIR4728	High-level amplification	GISTIC ±2	0	0.51
OR2G3	High-level amplification	GISTIC ±2	1	0.50
TRIM58	High-level amplification	GISTIC ±2	0	0.47
ADSS	High-level amplification	GISTIC ±2	0	0.46
OR14C36	High-level amplification	GISTIC ±2	1	0.41
OR14A16	High-level amplification	GISTIC ±2	1	0.39
CEP170	High-level amplification	GISTIC ±2	0	0.36
ZPBP2	High-level amplification	GISTIC ±2	0	0.35



70 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 60 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A0U1	36	II	missing	0.82	no progression, followed 133.2 mo
TCGA-D8-A1JB	54	II	LumA	0.78	no progression, followed 55.5 mo
TCGA-BH-A0B7	42	II	Her2	0.77	no progression, followed 84.1 mo
TCGA-AO-A03N	59	II	LumB	0.74	progressed at 54.1 mo (group D)
TCGA-AN-A0FT	63	II	LumA	0.73	no progression, followed 7.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**58**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0WW-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.13** · MSIsensor score: **0.67** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

78th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 3% · 24 mo 8% · 60 mo 26%

Model-estimated progression time

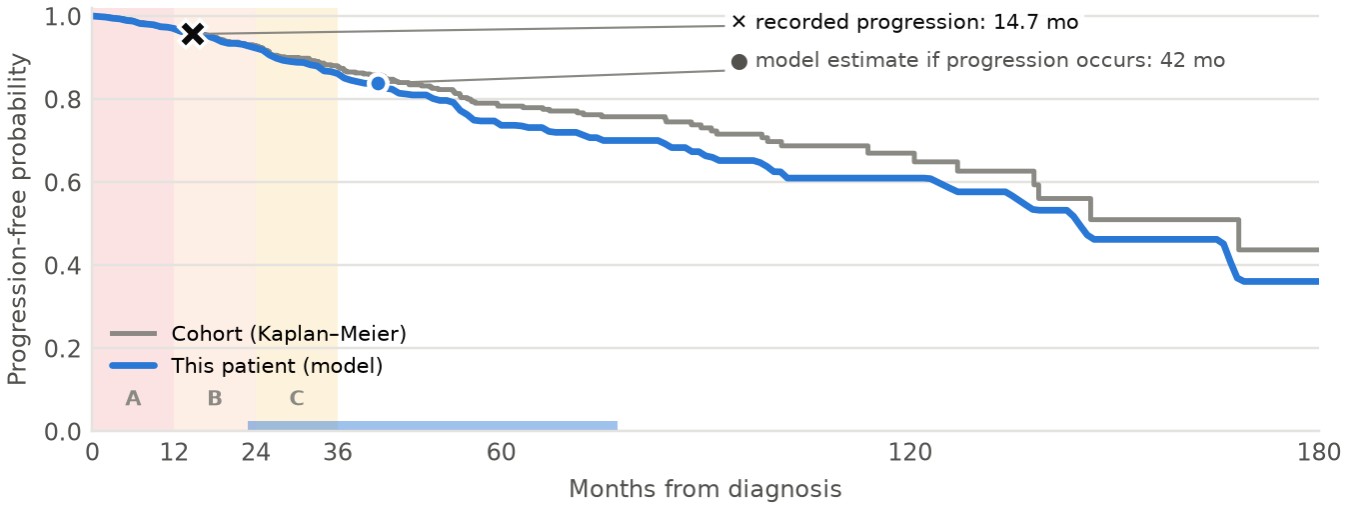
≈ 42 months

Window > 36 months · IQR 23–77 mo if progression occurs

Recorded progression

14.7 months

Group B · Progression in year 2
✗ recorded window differs · |error| 27 mo



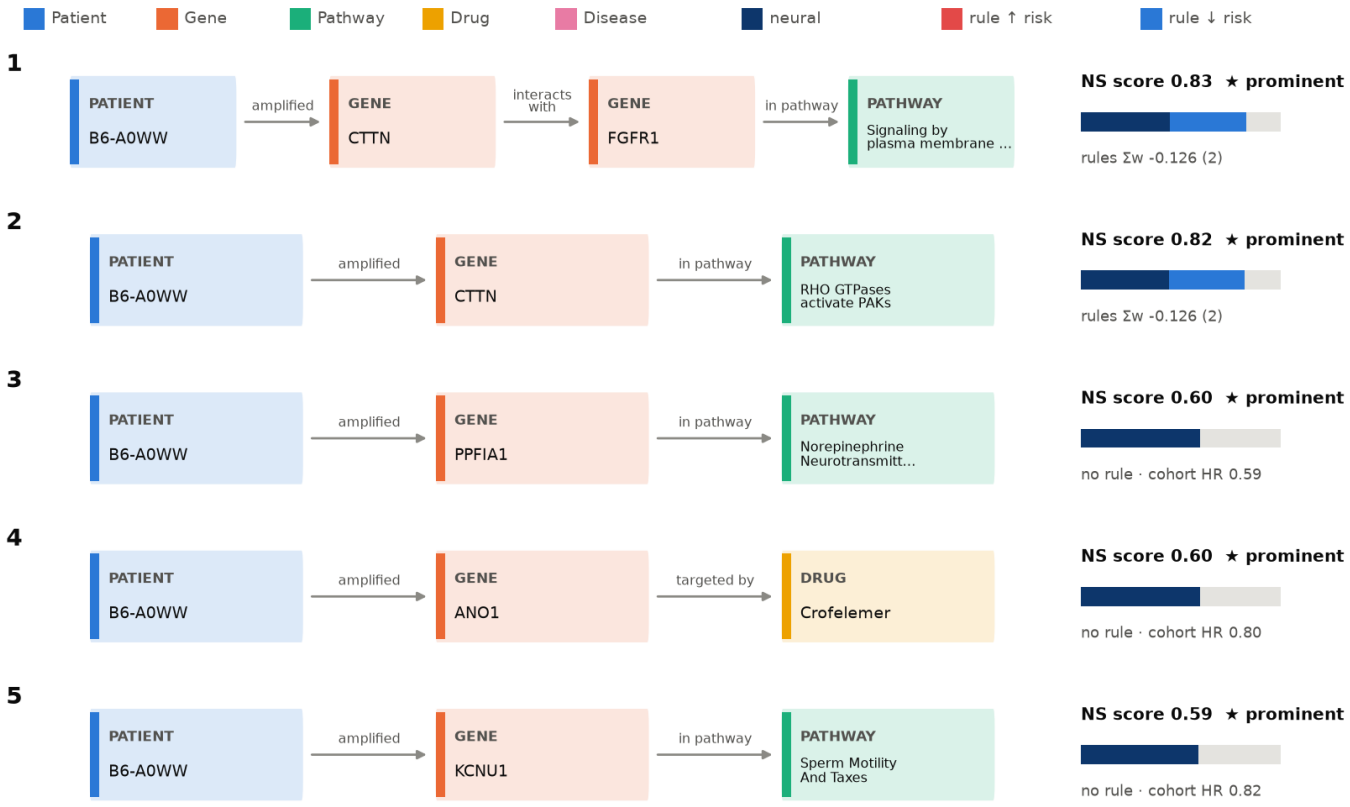
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 58-year-old patient has a LumB tumour, AJCC stage X (T2, N1, MX). The fused neuro-symbolic model places them in the **78th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'Clathrin-mediated endocytosis') ⇒ Progress(p)*) lowers the progression hazard (weight -0.101 per SD). The highest-scoring knowledge-graph path runs from **CTTN** to **Signaling by plasma membrane FGFR1 fusions**; the learned formulas it instantiates point to slower progression (Σw -0.126). Progression was recorded at 14.7 months; the model's estimate given progression was 42 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

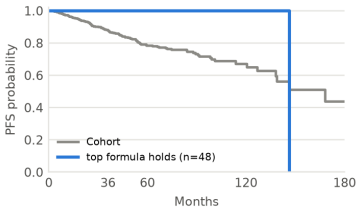


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A0WW** amplified → **CTTN** interacts with → **FGFR1** in pathway → **Signaling by plasma membrane FGFR1 fusions**	74	↓ **slower progression** · Σw -0.126 over 2 formula(s) NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'Clathrin-mediated endocytosis') ⇒ Progress(p) cohort: HR 1.02, n = 139 carriers	**0.83**
2 ★	**TCGA-B6-A0WW** amplified → **CTTN** in pathway → **RHO GTPases activate PAKs**	74	↓ **slower progression** · Σw -0.126 over 2 formula(s) NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'Clathrin-mediated endocytosis') ⇒ Progress(p) cohort: HR 0.60, n = 123 carriers	**0.82**
3 ★	**TCGA-B6-A0WW** amplified → **PPFIA1** in pathway → **Norepinephrine Neurotransmitter Release Cycle**	100	No learned formula fires on this path; its score is neural only. cohort: HR 0.59, n = 129 carriers	**0.60**
4 ★	**TCGA-B6-A0WW** amplified → **ANO1** targeted by → **Crofelemer**	99	No learned formula fires on this path; its score is neural only. cohort: HR 0.80, n = 134 carriers	**0.60**
5 ★	**TCGA-B6-A0WW** amplified → **KCNU1** in pathway → **Sperm Motility And Taxes**	98	No learned formula fires on this path; its score is neural only. cohort: HR 0.82, n = 108 carriers	**0.59**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

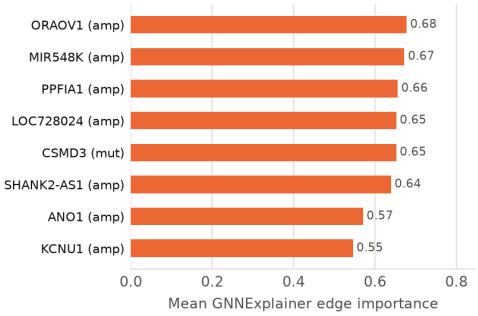
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.101 ↓ risk	NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'Clathrin-mediated endocytosis') ⇒ Progress(p) [KG]	mined pair	48 pts
-0.067 ↓ risk	NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'RMTs methylate histone arginines') ⇒ Progress(p) [KG]	mined pair	51 pts
-0.053 ↓ risk	NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX1 Expression and Activity') ⇒ Progress(p) [KG]	mined pair	89 pts
-0.025 ↓ risk	NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'RHO GTPases activate PAKs') ⇒ Progress(p) [KG]	mined pair	43 pts
-0.017 ↓ risk	NStage(p, N1) ∧ Altered(p, g) ∧ TargetedBy(g, 'Arsenic trioxide') ⇒ Progress(p) [KG]	mined pair	51 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
PTEN · driver	Mutation	p.K266*, p.N48Tfs*6	22	0.31
ORAOV1	High-level amplification	GISTIC ±2	0	0.68
MIR548K	High-level amplification	GISTIC ±2	0	0.67
PPFIA1	High-level amplification	GISTIC ±2	6	0.66
LOC728024	High-level amplification	GISTIC ±2	0	0.65
CSMD3	Mutation	p.S3666L	0	0.65
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.64
ANO1	High-level amplification	GISTIC ±2	2	0.57
KCNU1	High-level amplification	GISTIC ±2	1	0.55
ZNF703	High-level amplification	GISTIC ±2	1	0.50



26 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 9 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1JI	54	II	LumB	0.72	no progression, followed 19.0 mo
TCGA-AC-A2BM	41	II	LumB	0.70	no progression, followed 99.4 mo
TCGA-E2-A15K	58	II	LumB	0.69	no progression, followed 9.0 mo
TCGA-AN-A03Y	66	II	LumB	0.66	no progression, followed 0.3 mo
TCGA-A1-A0SQ	45	II	LumA	0.66	no progression, followed 18.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**44**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A409-01** · Aneuploidy score: **30** · TMB (nonsyn., /Mb): **0.97** · MSIsensor score: **0.07** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

85th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 29%

Model-estimated progression time

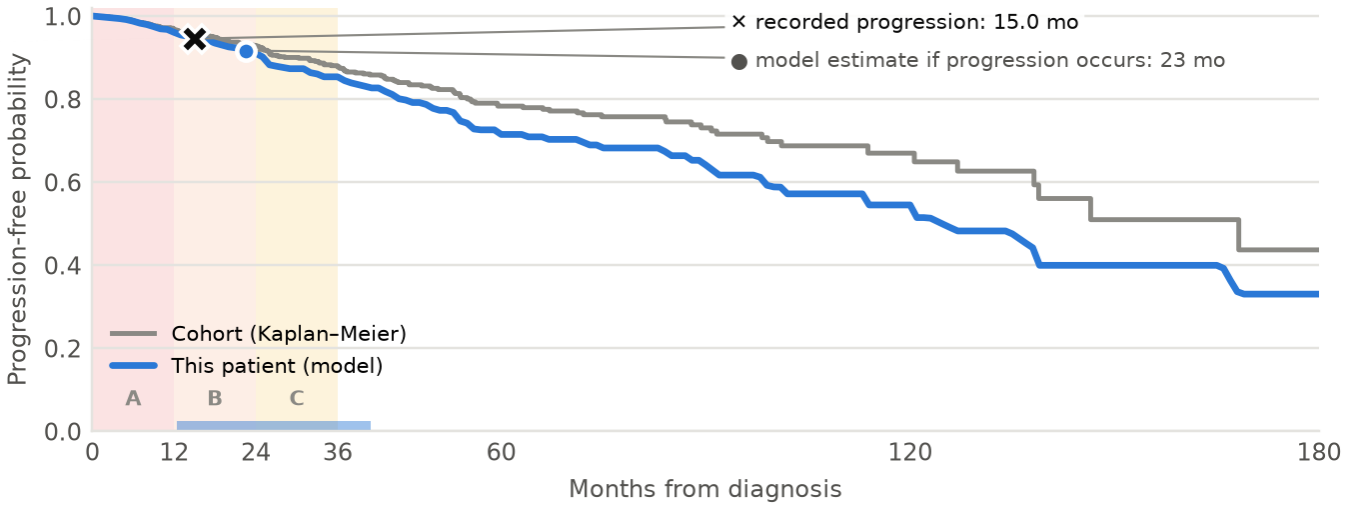
≈ 23 months

Window 12–24 months · IQR 12–41 mo
if progression occurs

Recorded progression

15.0 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 8 mo



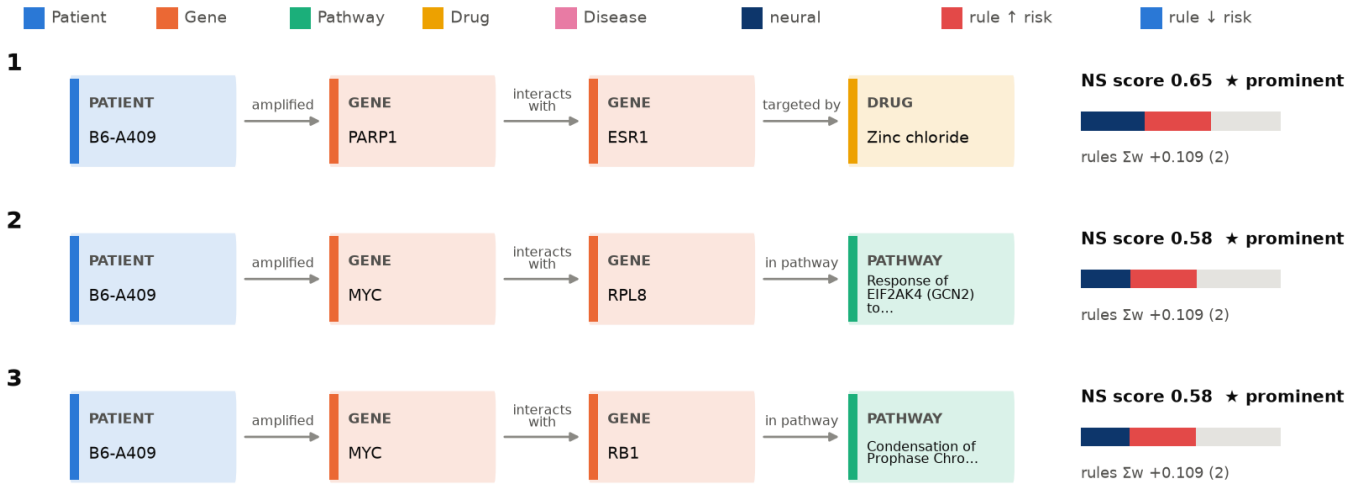
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 44-year-old patient has a Basal tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **85th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'Neurofascin interactions')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Metalloprotease DUBs')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Estrogen-dependent gene expression')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.102 per SD). The highest-scoring knowledge-graph path runs from **PARP1** to **Zinc chloride**; the learned formulas it instantiates point to faster progression (Σw +0.109). Progression was recorded at 15.0 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

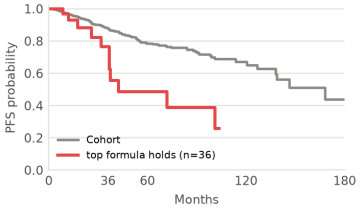


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A409** amplified → **PARP1** interacts with → **ESR1** targeted by → **Zinc chloride**	53	↑ **faster progression** · Σw +0.109 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Metalloprotease DUBs') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ⇒ Progress(p) cohort: HR 1.00, n = 414 carriers	**0.65**
2 ★	**TCGA-B6-A409** amplified → **MYC** interacts with → **RPL8** in pathway → **Response of EIF2AK4 (GCN2) to amino acid deficiency**	42	↑ **faster progression** · Σw +0.109 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Metalloprotease DUBs') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ⇒ Progress(p) cohort: HR 0.91, n = 94 carriers	**0.58**
3 ★	**TCGA-B6-A409** amplified → **MYC** interacts with → **RB1** in pathway → **Condensation of Prophase Chromosomes**	41	↑ **faster progression** · Σw +0.109 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Metalloprotease DUBs') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ⇒ Progress(p) cohort: HR 1.20, n = 52 carriers	**0.58**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

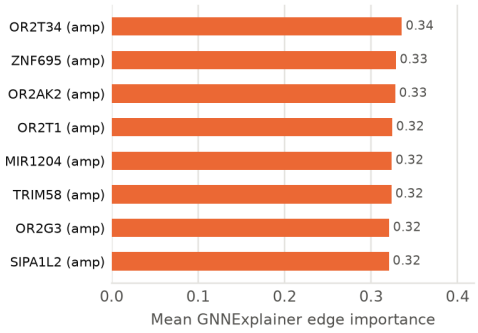
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.102 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Metalloprotease DUBs') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ⇒ Progress(p) [KG]	RL-proposed	36 pts
+0.040 ↑ risk	Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	mined pair	33 pts
+0.007 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ∧ Altered(p, g) ∧ TargetedBy(g, 'Zinc') ⇒ Progress(p) [KG]	RL-proposed	65 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
TP53 · driver	Mutation	p.T253Hfs*11	57	0.31
OR2T34	High-level amplification	GISTIC ±2	1	0.34
ZNF695	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
OR2T1	High-level amplification	GISTIC ±2	1	0.32
MIR1204	High-level amplification	GISTIC ±2	0	0.32
TRIM58	High-level amplification	GISTIC ±2	0	0.32
OR2G3	High-level amplification	GISTIC ±2	1	0.32
SIPA1L2	High-level amplification	GISTIC ±2	0	0.32
HEATR1	High-level amplification	GISTIC ±2	2	0.32
OR11L1	High-level amplification	GISTIC ±2	1	0.32



77 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 35 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A6VW	48	II	Basal	0.81	no progression, followed 9.4 mo
TCGA-E9-A22G	47	II	Basal	0.79	no progression, followed 40.7 mo
TCGA-A2-A0D0	60	II	Basal	0.78	no progression, followed 67.3 mo
TCGA-A2-A0SX	48	I	Basal	0.76	progressed at 40.7 mo (group D)
TCGA-B6-A0WX	40	III	Basal	0.76	progressed at 18.5 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**63**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

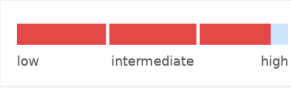
Sample: **TCGA-B6-A0IK-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **3.8** · MSIsensor score: **0.25** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

94th

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 6% · 24 mo 12% · 60 mo 35%

Model-estimated progression time

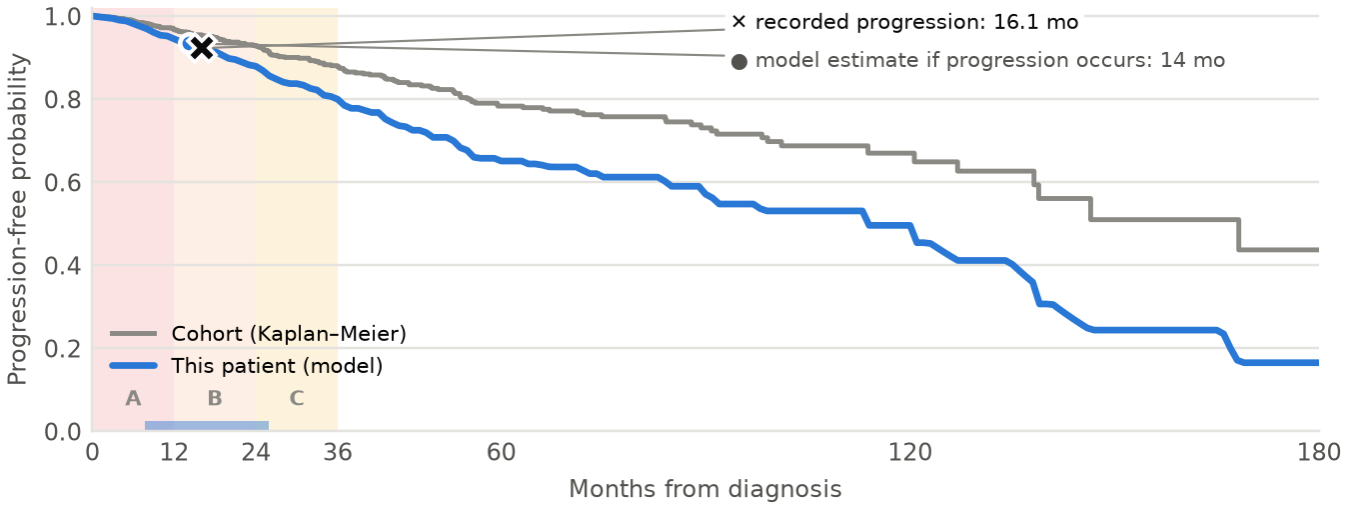
≈ 14 months

Window 12–24 months · IQR 8–26 mo if progression occurs

Recorded progression

16.1 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 2 mo



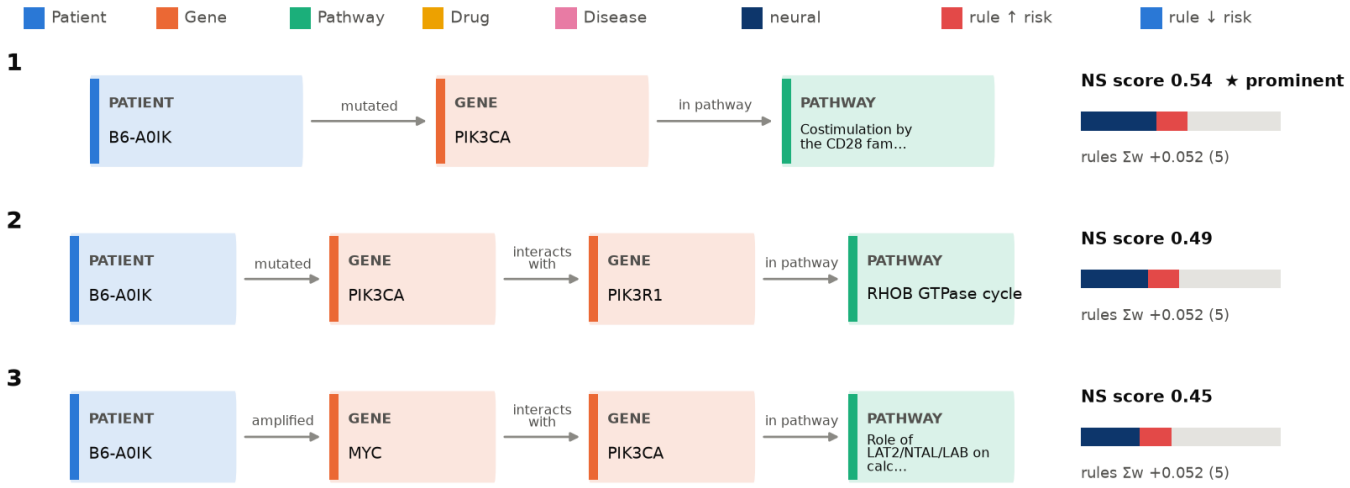
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 63-year-old patient has a Her2 tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **94th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4) ⇒ Progress(p)*) raises the progression hazard (weight +0.113 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Costimulation by the CD28 family**; the learned formulas it instantiates point to faster progression (Σw +0.052). Progression was recorded at 16.1 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

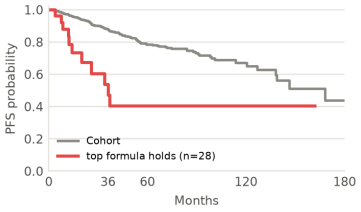


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A0IK** mutated → **PIK3CA** in pathway → **Costimulation by the CD28 family**	63	↑ **faster progression** · Σw +0.052 over 5 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K Cascade') ⇒ Progress(p) cohort: HR 0.86, n = 309 carriers	**0.54**
2	**TCGA-B6-A0IK** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **RHOB GTPase cycle**	56	↑ **faster progression** · Σw +0.052 over 5 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K Cascade') ⇒ Progress(p) cohort: HR 0.78, n = 133 carriers	**0.49**
3	**TCGA-B6-A0IK** amplified → **MYC** interacts with → **PIK3CA** in pathway → **Role of LAT2/NTAL/LAB on calcium mobilization**	49	↑ **faster progression** · Σw +0.052 over 5 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K Cascade') ⇒ Progress(p) cohort: HR 0.86, n = 309 carriers	**0.45**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

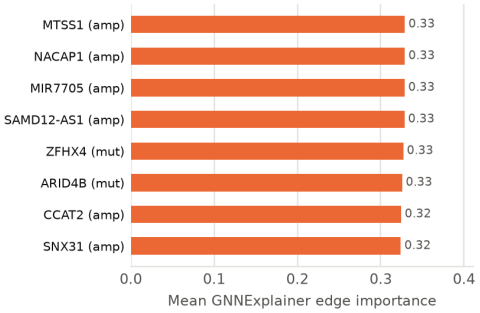
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.113 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	26 pts
+0.026 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K Cascade') ⇒ Progress(p) [KG]	mined pair	31 pts
+0.023 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ⇒ Progress(p) [KG]	mined pair	55 pts
-0.008 ↓ risk	Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	134 pts
+0.005 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	28 pts
+0.005 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR4') ⇒ Progress(p) [KG]	mined pair	28 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.32
NF1 · driver	Mutation	p.P2696Tfs*14	6	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
BAP1 · driver	Homozygous deletion	GISTIC ±2	9	0.31
TP53 · driver	Mutation	p.L194R	57	0.31
MTSS1	High-level amplification	GISTIC ±2	0	0.33
NACAP1	High-level amplification	GISTIC ±2	0	0.33
MIR7705	High-level amplification	GISTIC ±2	0	0.33
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.33
ZFHX4	Mutation	p.L2845V	0	0.33
ARID4B	Mutation	p.S1018*	3	0.33
CCAT2	High-level amplification	GISTIC ±2	0	0.32



81 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945) … and 16 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1B0	50	III	Her2	0.73	no progression, followed 53.6 mo
TCGA-A8-A09Q	83	III	LumB	0.69	no progression, followed 25.0 mo
TCGA-BH-A1F2	53	III	LumB	0.69	no progression, followed 31.5 mo
TCGA-A8-A08R	52	II	Basal	0.69	no progression, followed 1.0 mo
TCGA-C8-A12P	55	II	Her2	0.69	no progression, followed 11.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0X1 · Group B

Report date 31 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**48**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0X1-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **2.27** · MSIsensor score: **1.54** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

34% by 36 mo

12 mo 10% · 24 mo 22% · 60 mo 54%

Model-estimated progression time

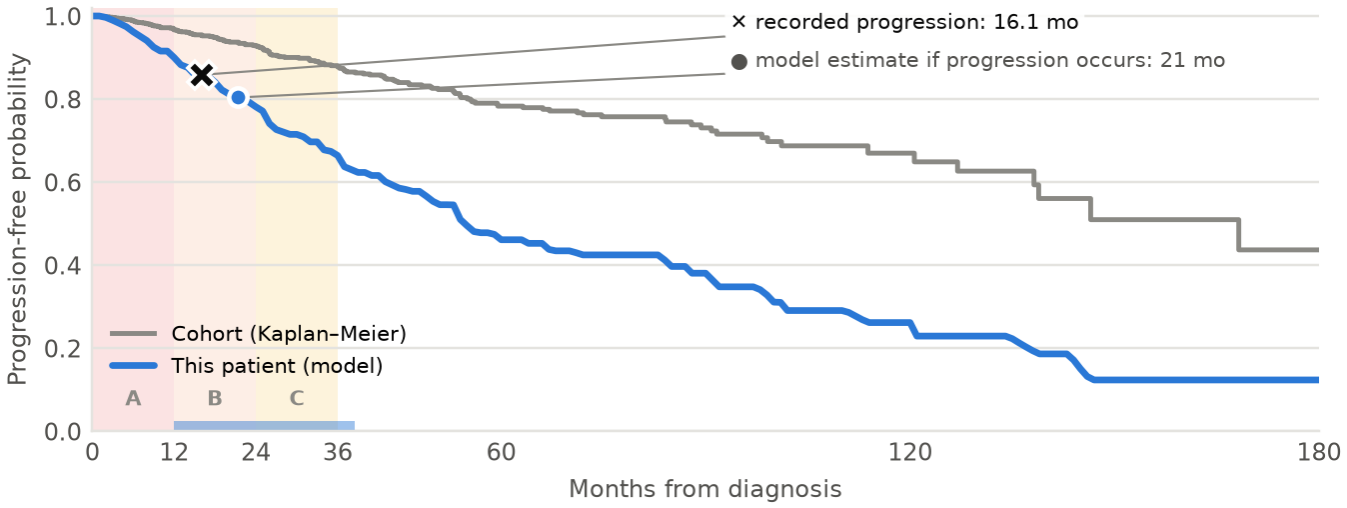
≈ 21 months

Window 12–24 months · IQR 12–38 mo
if progression occurs

Recorded progression

16.1 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 5 mo



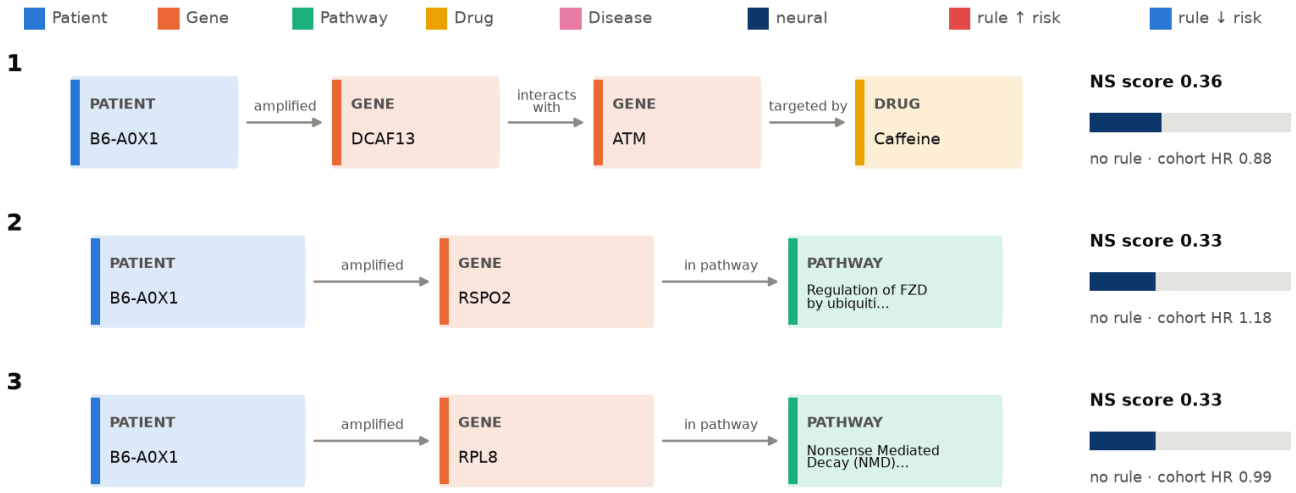
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 48-year-old patient has a Basal tumour, AJCC stage II (T2, N1, M1). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The strongest learned formula that holds for them (*Stage(p, II)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.006 per SD). The highest-scoring knowledge-graph path runs from **DCAF13** to **Caffeine**. Progression was recorded at 16.1 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

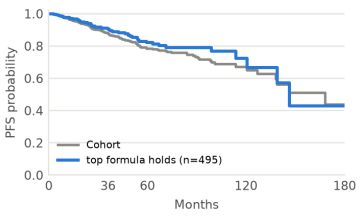


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0X1** amplified → **DCAF13** interacts with → **ATM** targeted by → **Caffeine**	59	No learned formula fires on this path; its score is neural only. cohort: HR 0.88, n = 323 carriers	**0.36**
2	**TCGA-B6-A0X1** amplified → **RSPO2** in pathway → **Regulation of FZD by ubiquitination**	55	No learned formula fires on this path; its score is neural only. cohort: HR 1.18, n = 107 carriers	**0.33**
3	**TCGA-B6-A0X1** amplified → **RPL8** in pathway → **Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC)**	54	No learned formula fires on this path; its score is neural only. cohort: HR 0.99, n = 139 carriers	**0.33**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

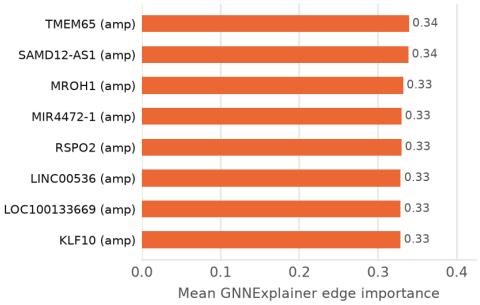
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BRCA1 · driver	Mutation	p.E23Sfs*8	29	0.31
TP53 · driver	Mutation	p.X331_splice	57	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TMEM65	High-level amplification	GISTIC ±2	0	0.34
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.34
MROH1	High-level amplification	GISTIC ±2	0	0.33
MIR4472-1	High-level amplification	GISTIC ±2	0	0.33
RSPO2	High-level amplification	GISTIC ±2	1	0.33
LINC00536	High-level amplification	GISTIC ±2	0	0.33
LOC100133669	High-level amplification	GISTIC ±2	0	0.33
KLF10	High-level amplification	GISTIC ±2	0	0.33
MROH5	High-level amplification	GISTIC ±2	0	0.33



280 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813) … and 92 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A0TS	46	II	Basal	0.89	no progression, followed 84.1 mo
TCGA-C8-A12K	80	II	Basal	0.88	no progression, followed 0.0 mo
TCGA-A2-A3Y0	57	II	Basal	0.87	no progression, followed 50.8 mo
TCGA-A2-A0T0	59	II	Basal	0.86	no progression, followed 17.5 mo
TCGA-AN-A0FL	62	II	Basal	0.85	no progression, followed 7.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**72**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

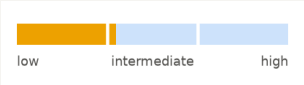
Sample: **TCGA-E9-A2JS-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **1.93** · MSIsensor score: **0.81** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

37th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

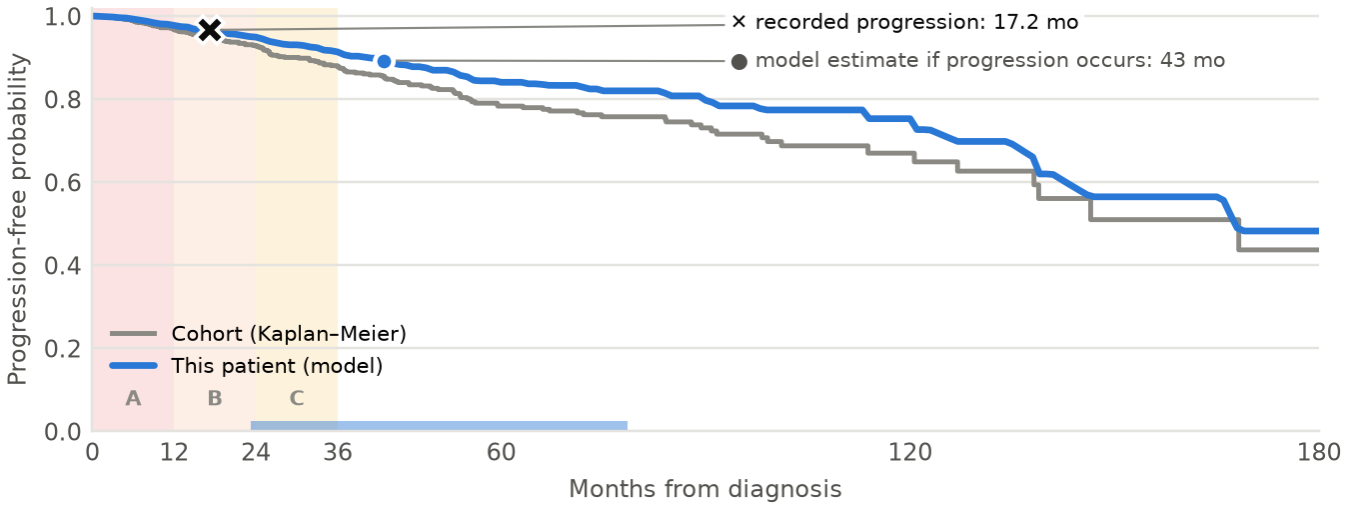
≈ 43 months

Window > 36 months · IQR 23–79 mo if progression occurs

Recorded progression

17.2 months

Group B · Progression in year 2
✗ recorded window differs · |error| 26 mo



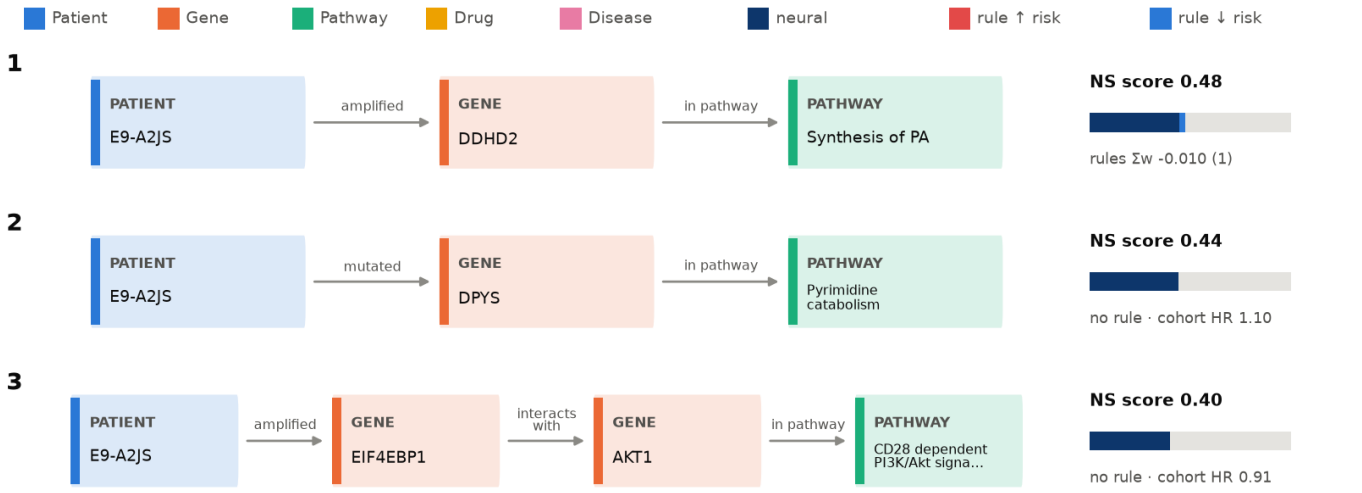
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 72-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **37th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **DDHD2** to **Synthesis of PA**; the learned formulas it instantiates point to slower progression (Σw -0.010). Progression was recorded at 17.2 months; the model's estimate given progression was 43 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

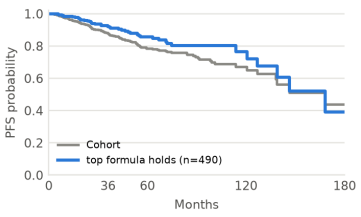


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-E9-A2JS** amplified → **DDHD2** in pathway → **Synthesis of PA**	74	↓ **slower progression** · Σw -0.010 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) cohort: HR 0.80, n = 189 carriers	**0.48**
2	**TCGA-E9-A2JS** mutated → **DPYS** in pathway → **Pyrimidine catabolism**	74	No learned formula fires on this path; its score is neural only. cohort: HR 1.10, n = 99 carriers	**0.44**
3	**TCGA-E9-A2JS** amplified → **EIF4EBP1** interacts with → **AKT1** in pathway → **CD28 dependent PI3K/Akt signaling**	67	No learned formula fires on this path; its score is neural only. cohort: HR 0.91, n = 394 carriers	**0.40**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

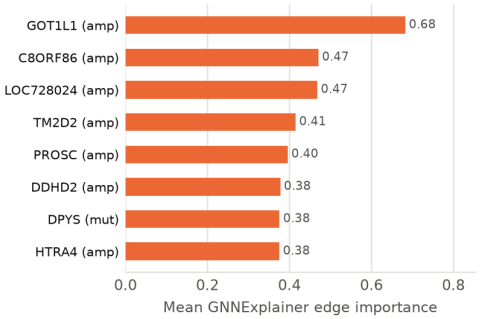
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.017 ↓ risk	Histology(p, lobular) ⇒ Progress(p)	single atom	154 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.68
C8ORF86	High-level amplification	GISTIC ±2	0	0.47
LOC728024	High-level amplification	GISTIC ±2	0	0.47
TM2D2	High-level amplification	GISTIC ±2	0	0.41
PROSC	High-level amplification	GISTIC ±2	0	0.40
DDHD2	High-level amplification	GISTIC ±2	1	0.38
DPYS	Mutation	Splice_Region	1	0.38
HTRA4	High-level amplification	GISTIC ±2	0	0.38
LETM2	High-level amplification	GISTIC ±2	0	0.35
STAR	High-level amplification	GISTIC ±2	1	0.33
PLEKHA2	High-level amplification	GISTIC ±2	1	0.33



40 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 45 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A204	80	II	LumB	0.84	no progression, followed 83.3 mo
TCGA-E2-A1IE	61	II	LumA	0.78	progressed at 59.5 mo (group D)
TCGA-A2-A0ES	52	II	LumA	0.78	no progression, followed 72.0 mo
TCGA-E9-A2JT	63	II	LumA	0.73	no progression, followed 9.5 mo
TCGA-HN-A2OB	45	II	LumA	0.73	progressed at 50.7 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-GM-A2D9 · Group B

Report date 31 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**69**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

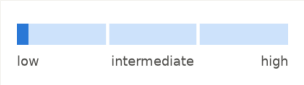
Sample: **TCGA-GM-A2D9-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **14.57** · MSIsensor score: **0.07** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

4th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 12%

Model-estimated progression time

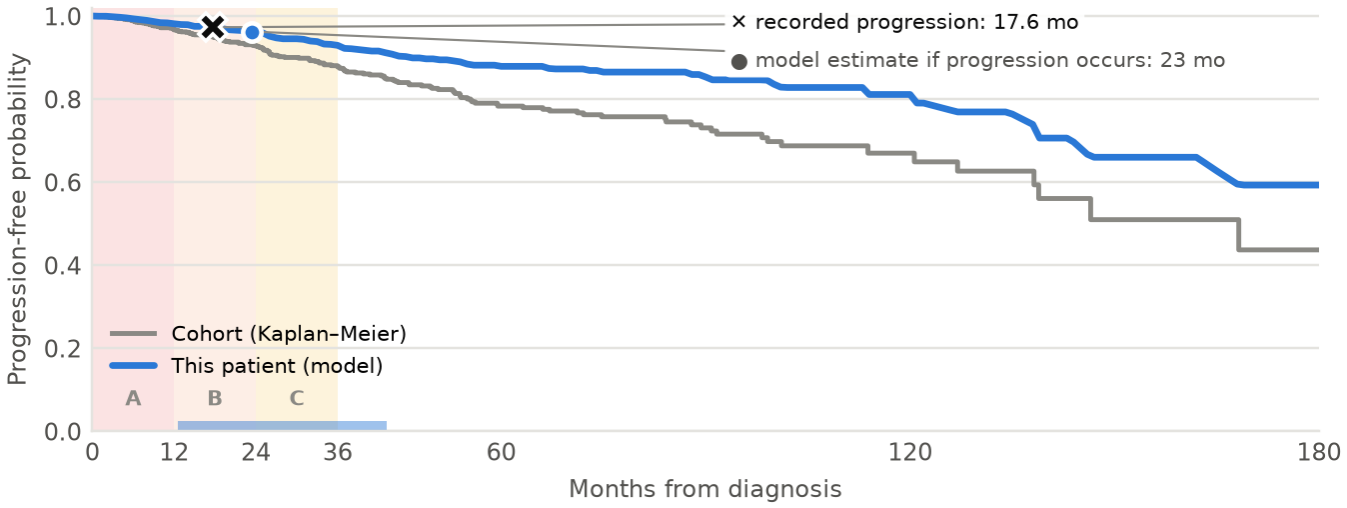
≈ 23 months

Window 12–24 months · IQR 13–43 mo
if progression occurs

Recorded progression

17.6 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 6 mo



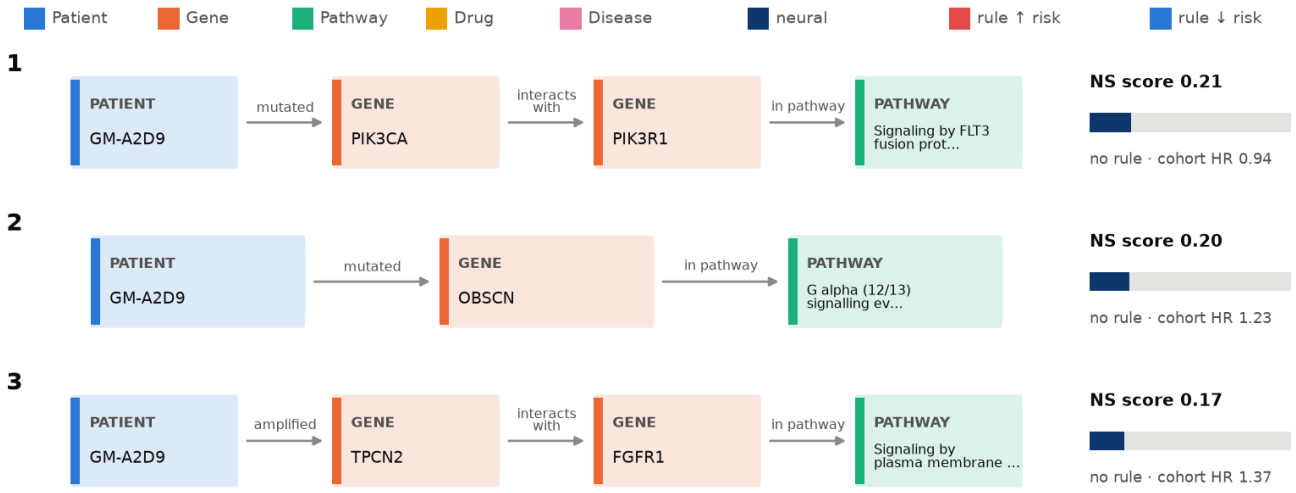
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 69-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **4th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Signaling by FLT3 fusion proteins**. Progression was recorded at 17.6 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

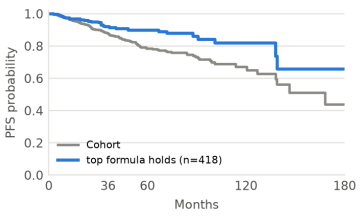


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-GM-A2D9** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Signaling by FLT3 fusion proteins**	35	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 321 carriers	**0.21**
2	**TCGA-GM-A2D9** mutated → **OBSCN** in pathway → **G alpha (12/13) signalling events**	33	No learned formula fires on this path; its score is neural only. cohort: HR 1.23, n = 121 carriers	**0.20**
3	**TCGA-GM-A2D9** amplified → **TPCN2** interacts with → **FGFR1** in pathway → **Signaling by plasma membrane FGFR1 fusions**	29	No learned formula fires on this path; its score is neural only. cohort: HR 1.37, n = 138 carriers	**0.17**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

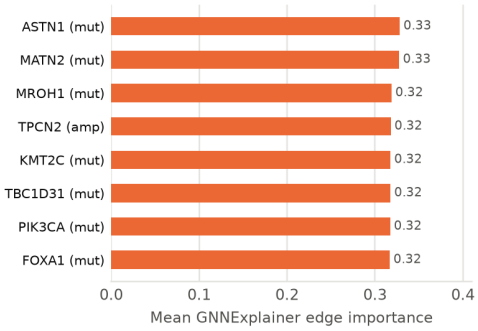
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.028 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	133 pts
-0.021 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	139 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Stage(p, I) ⇒ Progress(p)	mined pair	138 pts
-0.003 ↓ risk	Subtype(p, LumA) ∧ Mutated(p, MAP3K1) ⇒ Progress(p)	mined pair	56 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.Q982K	4	0.32
PIK3CA · driver	Mutation	p.E453K, p.H1047R	70	0.32
FOXA1 · driver	Mutation	p.K237E, p.K237N, p.K237R	9	0.32
RUNX1 · driver	Mutation	p.E316K	18	0.32
CBFB · driver	Mutation	p.G123C	29	0.32
BRCA1 · driver	Mutation	p.S915C	29	0.32
MAP3K1 · driver	Mutation	p.Q345*	13	0.31
ASTN1	Mutation	p.D1287N	0	0.33
MATN2	Mutation	p.E804Q	0	0.33
MROH1	Mutation	p.X239_splice	0	0.32
TPCN2	High-level amplification	GISTIC ±2	1	0.32
TBC1D31	Mutation	p.R374K	0	0.32



35 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0EO	54	I	LumA	0.79	no progression, followed 80.3 mo
TCGA-D8-A1JP	73	I	LumA	0.75	no progression, followed 21.0 mo
TCGA-B6-A0RN	60	I	LumA	0.74	no progression, followed 263.3 mo
TCGA-E9-A1R5	63	I	LumA	0.73	no progression, followed 3.0 mo
TCGA-BH-A0H3	46	I	LumA	0.70	no progression, followed 63.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**54**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

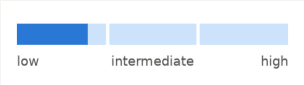
Sample: **TCGA-AR-A0TY-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **2.7** · MSIsensor score: **0.75** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

26th

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

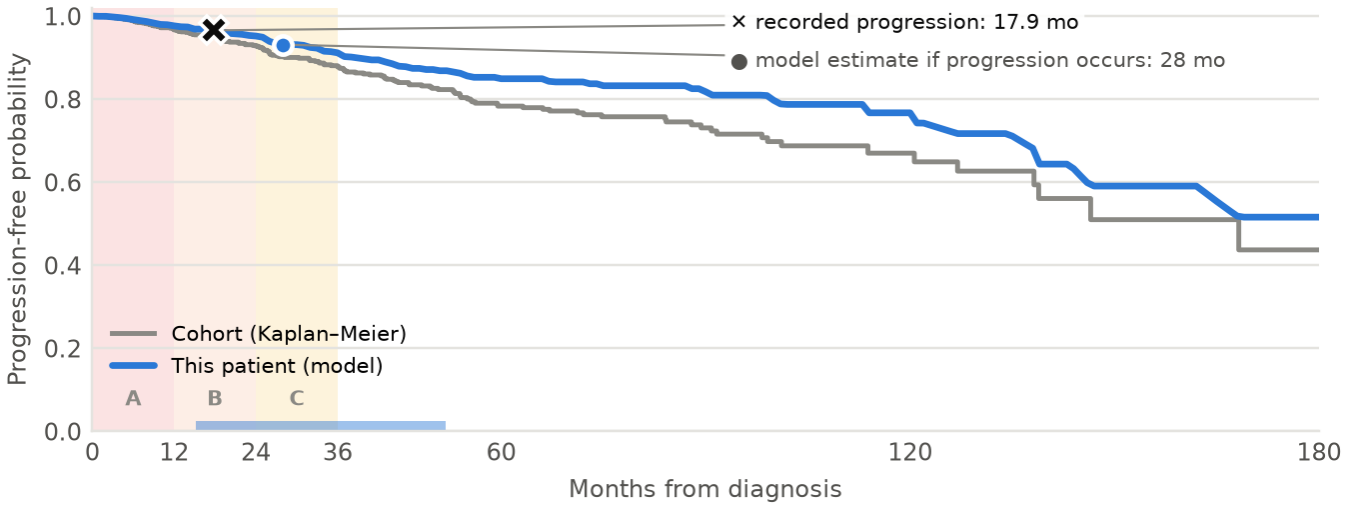
≈ 28 months

Window 24–36 months · IQR 15–52 mo
if progression occurs

Recorded progression

17.9 months

■ **Group B** · Progression in year 2
✗ **recorded window differs** · |error| 10 mo



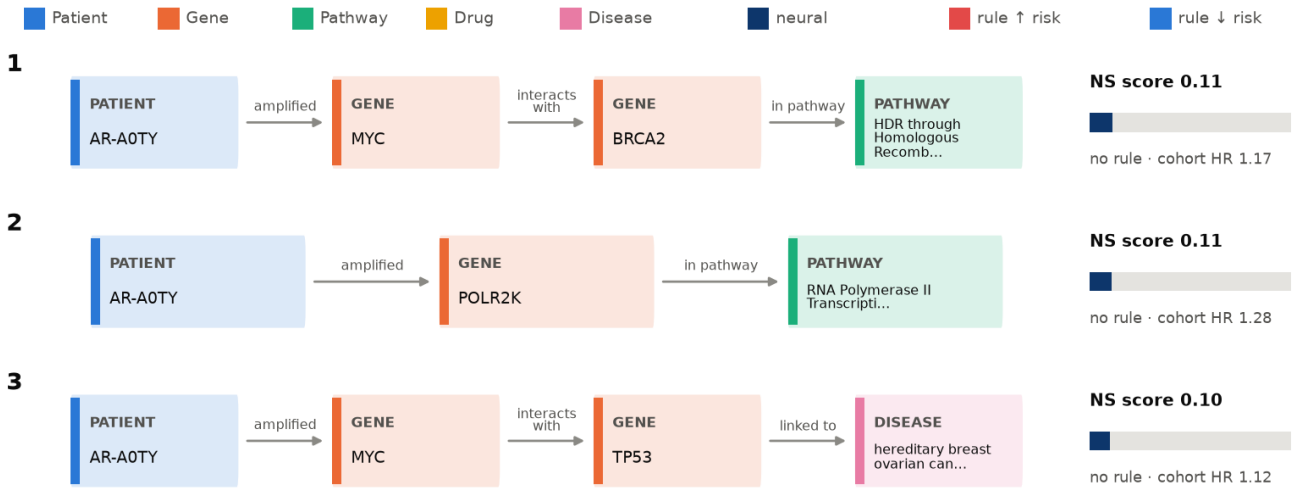
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 54-year-old patient has a LumB tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **26th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **MYC** to **HDR through Homologous Recombination (HRR)**. Progression was recorded at 17.9 months; the model's estimate given progression was 28 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

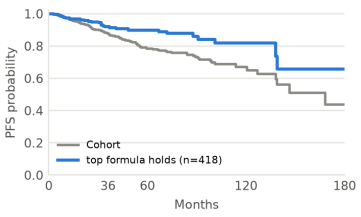


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A0TY** amplified → **MYC** interacts with → **BRCA2** in pathway → **HDR through Homologous Recombination (HRR)**	19	No learned formula fires on this path; its score is neural only. cohort: HR 1.17, n = 180 carriers	**0.11**
2	**TCGA-AR-A0TY** amplified → **POLR2K** in pathway → **RNA Polymerase II Transcription Pre-Initiation And Promoter Opening**	18	No learned formula fires on this path; its score is neural only. cohort: HR 1.28, n = 129 carriers	**0.11**
3	**TCGA-AR-A0TY** amplified → **MYC** interacts with → **TP53** linked to → **hereditary breast ovarian cancer syndrome**	17	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 800 carriers	**0.10**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

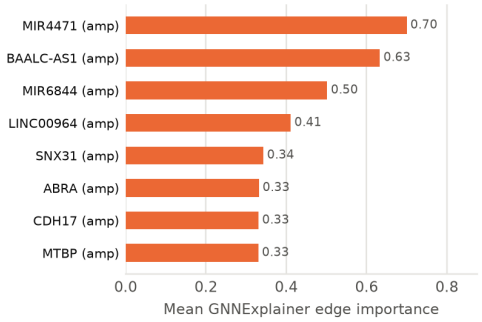
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
-0.004 ↓ risk	Altered(p, g) ∧ InPathway(g, 'G alpha (s) signalling events') ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 regulates genes involved in megakaryocyte differentiation and platelet function') ∧ Altered(p, g) ∧ InPathway(g, 'FOXO-mediated transcription of oxidative stress, metabolic and neuronal genes') ⇒ Progress(p) [KG]	RL-proposed	37 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.H434Pfs*42	13	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
MAP2K4 · driver	Homozygous deletion	GISTIC ±2	12	0.31
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.31
MIR4471	High-level amplification	GISTIC ±2	0	0.70
BAALC-AS1	High-level amplification	GISTIC ±2	0	0.63
MIR6844	High-level amplification	GISTIC ±2	0	0.50
LINC00964	High-level amplification	GISTIC ±2	0	0.41
SNX31	High-level amplification	GISTIC ±2	0	0.34
ABRA	High-level amplification	GISTIC ±2	0	0.33
CDH17	High-level amplification	GISTIC ±2	1	0.33
MTBP	High-level amplification	GISTIC ±2	0	0.33



136 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); STK3 → Fostamatinib (DB12010); NDUFB9 → NADH (DB00157); SQLE → Naftifine (DB00735); SQLE → Terbinafine (DB00857); SQLE → Butenafine (DB01091); SQLE → Ellagic acid (DB08846); YWHAZ → Phenethyl Isothiocyanate (DB12695); ATP6V1C1 → Tiludronic acid (DB01133) … and 6 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A14T	52	II	LumA	0.78	no progression, followed 76.0 mo
TCGA-AR-A0U2	47	III	LumB	0.77	progressed at 52.9 mo (group D)
TCGA-BH-A0AY	62	II	LumA	0.76	no progression, followed 25.5 mo
TCGA-C8-A12N	58	II	LumA	0.76	no progression, followed 11.8 mo
TCGA-A2-A0CU	73	II	LumA	0.76	no progression, followed 5.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A7-A13E-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **3.1** · MSIsensor score: **4.87** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

50th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

≈ 15 months

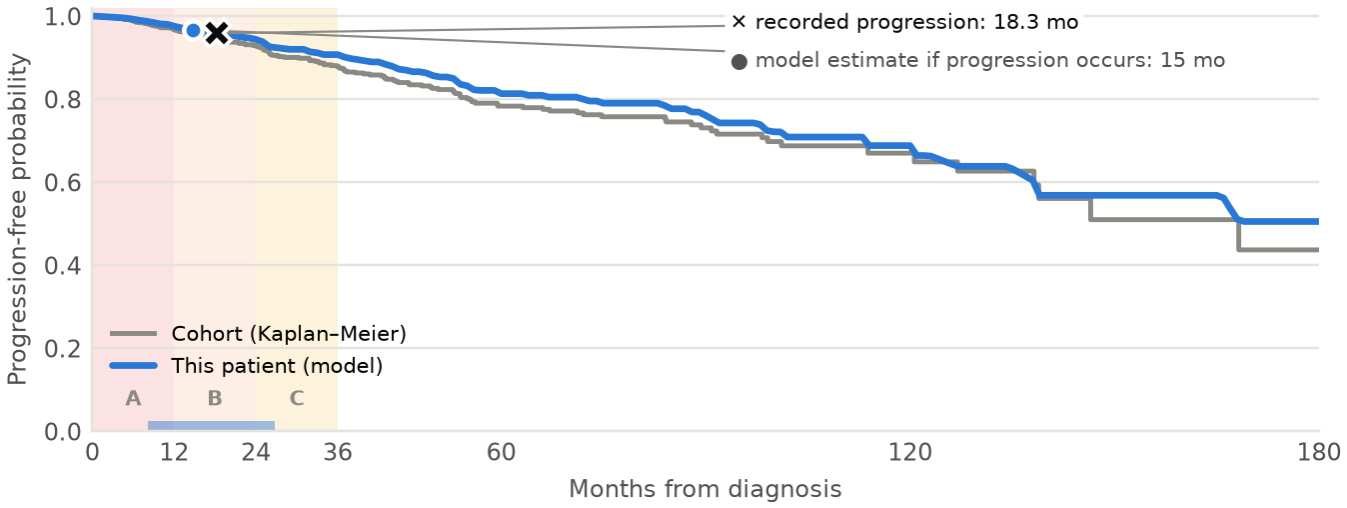
Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

18.3 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 3 mo



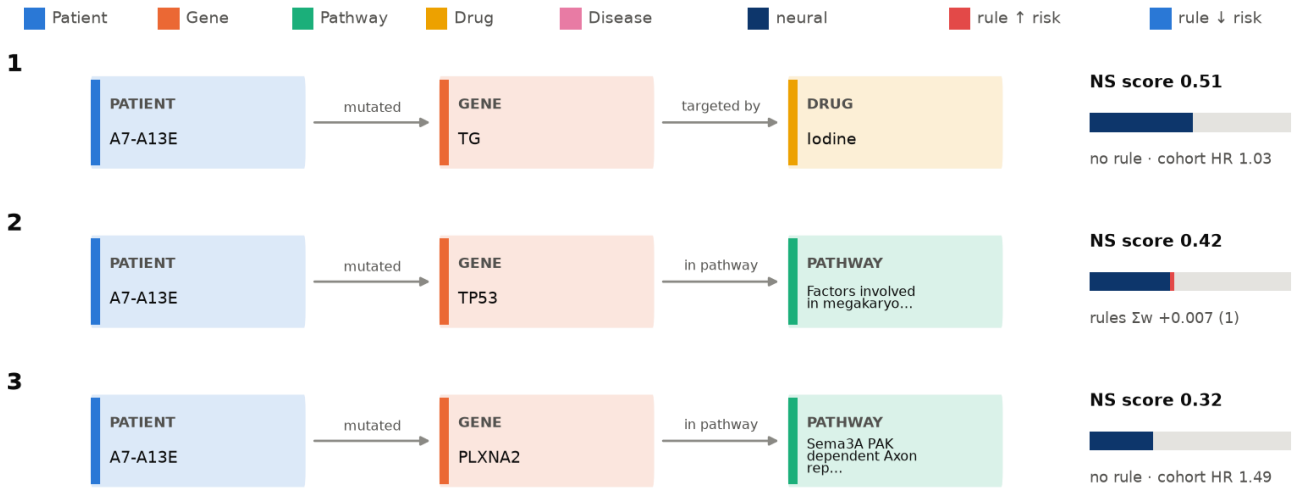
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a Basal tumour, AJCC stage II (T2, N1, MX). The fused neuro-symbolic model places them in the **50th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'Neurofascin interactions')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Estrogen-dependent gene expression')* ∧ *Altered(p, g)* ∧ *TargetedBy(g, 'Zinc')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.007 per SD). The highest-scoring knowledge-graph path runs from **TG** to **Iodine**. Progression was recorded at 18.3 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

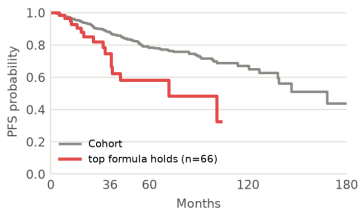


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A7-A13E** mutated → **TG** targeted by → **Iodine**	85	No learned formula fires on this path; its score is neural only. cohort: HR 1.03, n = 120 carriers	**0.51**
2	**TCGA-A7-A13E** mutated → **TP53** in pathway → **Factors involved in megakaryocyte development and platelet production**	67	↑ **faster progression** · Σw +0.007 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ∧ Altered(p, g) ∧ TargetedBy(g, 'Zinc') ⇒ Progress(p) cohort: HR 0.95, n = 428 carriers	**0.42**
3	**TCGA-A7-A13E** mutated → **PLXNA2** in pathway → **Sema3A PAK dependent Axon repulsion**	53	No learned formula fires on this path; its score is neural only. cohort: HR 1.49, n = 87 carriers	**0.32**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

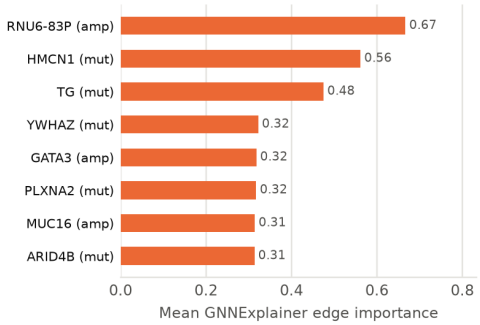
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.007 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ∧ Altered(p, g) ∧ TargetedBy(g, 'Zinc') ⇒ Progress(p) **[KG]**	RL-proposed	65 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	High-level amplification	GISTIC ±2	13	0.32
TP53 · driver	Mutation	p.P77Gfs*65, p.R175H	57	0.31
RNU6-83P	High-level amplification	GISTIC ±2	0	0.67
HMCN1	Mutation	p.I4988=	0	0.56
TG	Mutation	p.R1549W	1	0.48
YWHAZ	Mutation	3'UTR	15	0.32
PLXNA2	Mutation	p.S1617F	4	0.32
MUC16	High-level amplification	GISTIC ±2	8	0.31
ARID4B	Mutation	p.L83Qfs*8	3	0.31
TP53BP2	Mutation	p.P544=	12	0.31
NFASC	Mutation	3'UTR	3	0.31



11 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); TG → Iodine (DB05382); YWHAZ → Phenethyl Isothiocyanate (DB12695) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1AZ	63	II	Basal	0.74	no progression, followed 76.6 mo
TCGA-BH-A5IZ	51	II	Basal	0.73	no progression, followed 18.6 mo
TCGA-LL-A6FR	50	II	Basal	0.72	no progression, followed 16.1 mo
TCGA-LL-A7SZ	49	II	LumA	0.72	no progression, followed 19.5 mo
TCGA-A2-A3XT	45	II	Basal	0.71	no progression, followed 91.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A0YJ-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **2.5** · MSIsensor score: **0.47** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

90th

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 5% · 24 mo 10% · 60 mo 31%

Model-estimated progression time

≈ 16 months

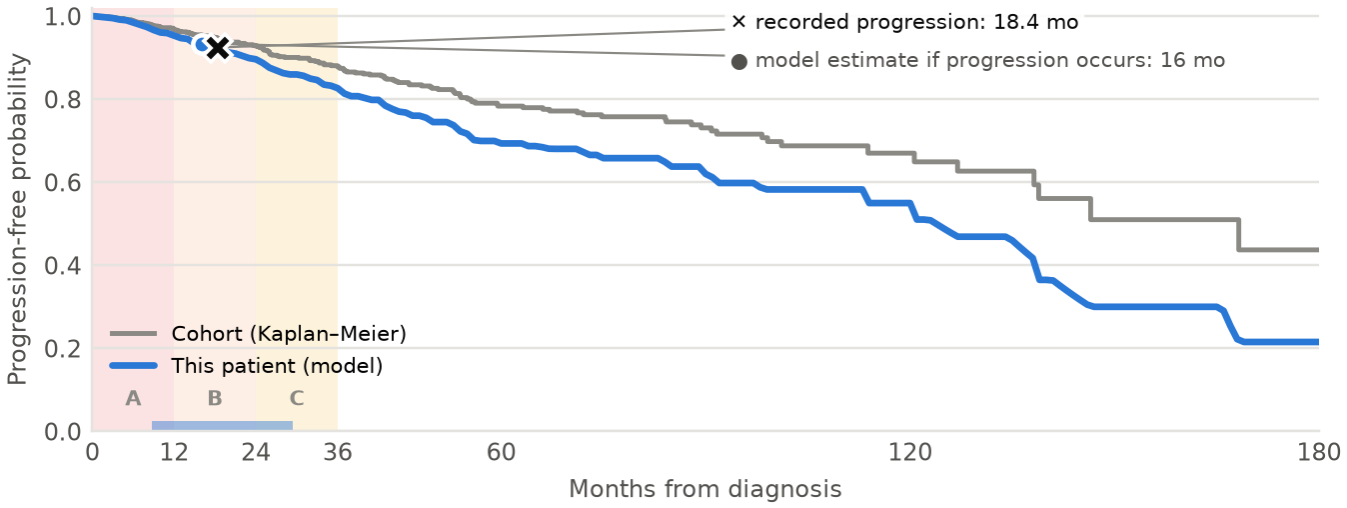
Window 12–24 months · IQR 9–29 mo if progression occurs

Recorded progression

18.4 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 2 mo



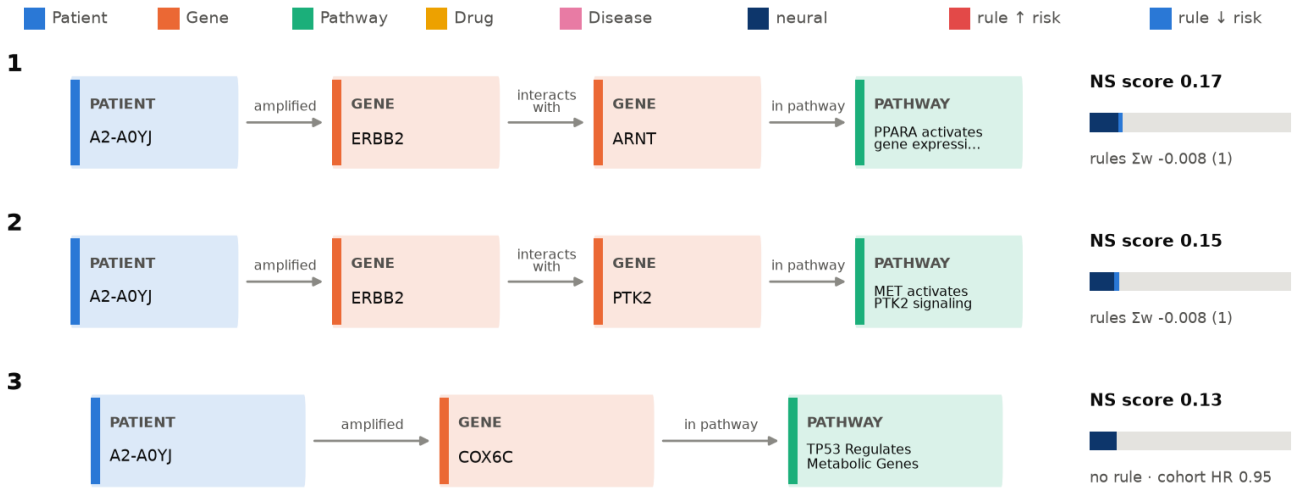
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a Basal tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **90th risk percentile** (high tier). The strongest learned formula that holds for them (*Age(p) < 40 ⇒ Progress(p)*) raises the progression hazard (weight +0.035 per SD). The highest-scoring knowledge-graph path runs from **ERBB2** to **PPARA activates gene expression**; the learned formulas it instantiates point to slower progression (Σw -0.008). Progression was recorded at 18.4 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

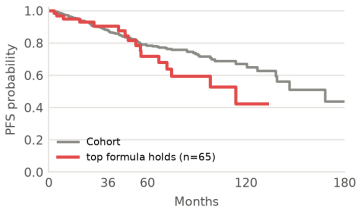


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A0YJ** amplified → **ERBB2** interacts with → **ARNT** in pathway → **PPARA activates gene expression**	24	↓ **slower progression** · Σw -0.008 over 1 formula(s) Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 1.17, n = 184 carriers	**0.17**
2	**TCGA-A2-A0YJ** amplified → **ERBB2** interacts with → **PTK2** in pathway → **MET activates PTK2 signaling**	21	↓ **slower progression** · Σw -0.008 over 1 formula(s) Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.85, n = 105 carriers	**0.15**
3	**TCGA-A2-A0YJ** amplified → **COX6C** in pathway → **TP53 Regulates Metabolic Genes**	22	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 458 carriers	**0.13**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

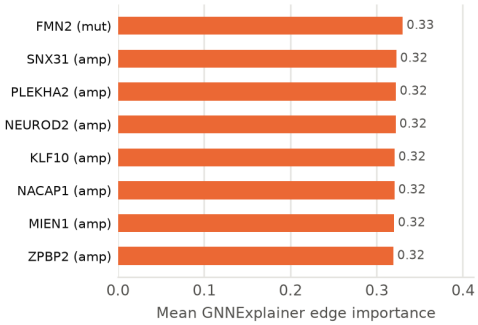
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.035 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	65 pts
-0.008 ↓ risk	Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	134 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R342*	57	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
FMN2	Mutation	p.V1487L	0	0.33
SNX31	High-level amplification	GISTIC ±2	0	0.32
PLEKHA2	High-level amplification	GISTIC ±2	1	0.32
NEUROD2	High-level amplification	GISTIC ±2	0	0.32
KLF10	High-level amplification	GISTIC ±2	0	0.32
NACAP1	High-level amplification	GISTIC ±2	0	0.32
MIEN1	High-level amplification	GISTIC ±2	0	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
AZIN1	High-level amplification	GISTIC ±2	1	0.32



66 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1B0	50	III	Her2	0.73	no progression, followed 53.6 mo
TCGA-B6-A1KN	57	III	LumB	0.70	no progression, followed 139.2 mo
TCGA-C8-A27A	48	II	LumB	0.68	no progression, followed 24.6 mo
TCGA-E2-A14Y	35	II	Basal	0.64	no progression, followed 69.3 mo
TCGA-C8-A130	52	III	LumB	0.64	no progression, followed 12.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-B6-A0WX-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **0.9** · MSIsensor score: **0.18** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

94th

High risk tier · fused neuro-symbolic model



Probability of progression

22% by 36 mo

12 mo 6% · 24 mo 12% · 60 mo 35%

Model-estimated progression time

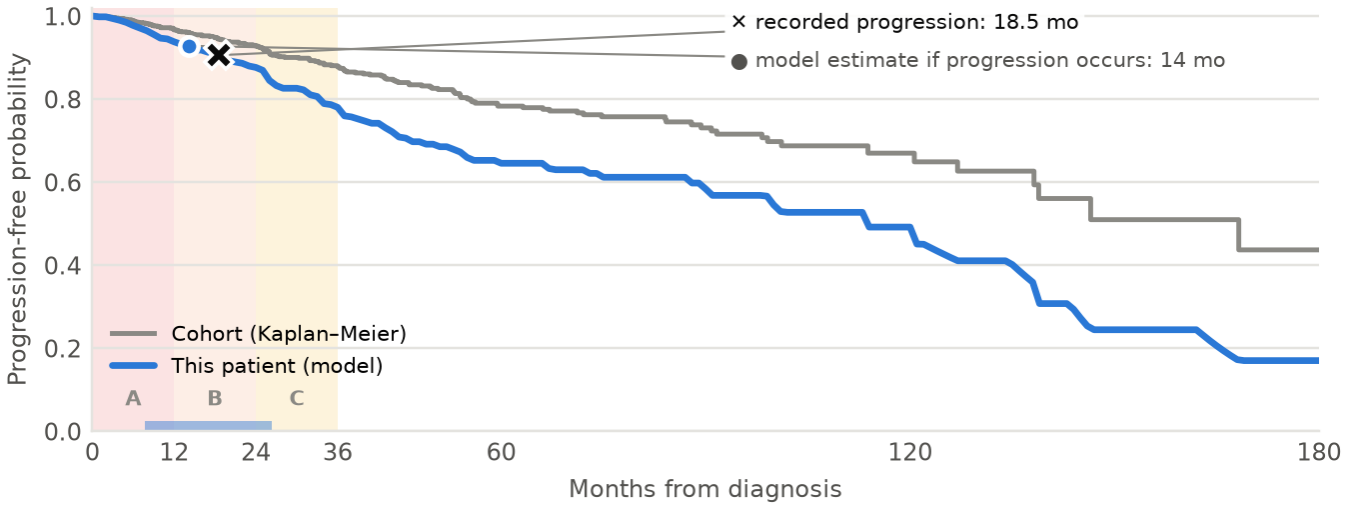
≈ 14 months

Window 12–24 months · IQR 8–26 mo if progression occurs

Recorded progression

18.5 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 4 mo



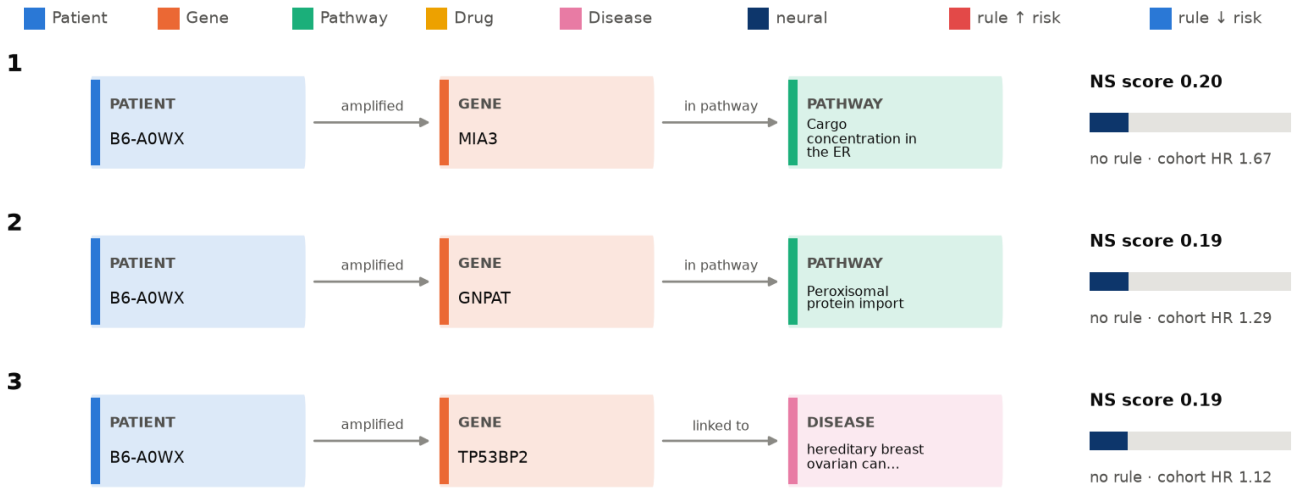
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a Basal tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **94th risk percentile** (high tier). The strongest learned formula that holds for them (*Stage(p, III)* ∧ *Amplified(p, FAM72C)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.098 per SD). The highest-scoring knowledge-graph path runs from **MIA3** to **Cargo concentration in the ER**. Progression was recorded at 18.5 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

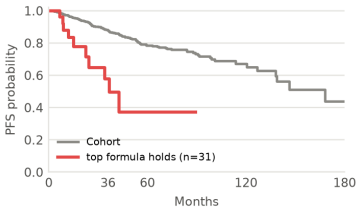


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0WX** amplified → **MIA3** in pathway → **Cargo concentration in the ER**	33	No learned formula fires on this path; its score is neural only. cohort: HR 1.67, n = 87 carriers	**0.20**
2	**TCGA-B6-A0WX** amplified → **GNPAT** in pathway → **Peroxisomal protein import**	32	No learned formula fires on this path; its score is neural only. cohort: HR 1.29, n = 86 carriers	**0.19**
3	**TCGA-B6-A0WX** amplified → **TP53BP2** linked to → **hereditary breast ovarian cancer syndrome**	31	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 800 carriers	**0.19**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

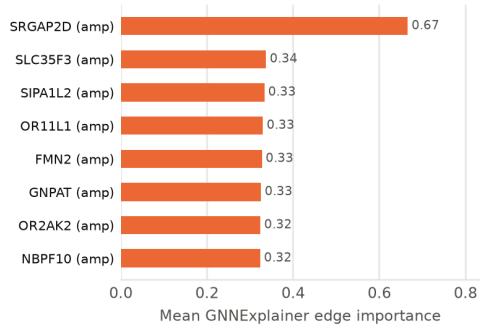
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.098 ↑ risk	Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	mined pair	31 pts
+0.026 ↑ risk	TStage(p, T3) ⇒ Progress(p)	single atom	102 pts
+0.012 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	mined pair	45 pts
+0.008 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts
+0.006 ↑ risk	Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ∧ Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	RL-proposed	37 pts
-0.001 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	147 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.L114*, p.R248W	57	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.67
SLC35F3	High-level amplification	GISTIC ±2	0	0.34
SIPA1L2	High-level amplification	GISTIC ±2	0	0.33
OR11L1	High-level amplification	GISTIC ±2	1	0.33
FMN2	High-level amplification	GISTIC ±2	0	0.33
GNPAT	High-level amplification	GISTIC ±2	3	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.32
NBPF10	High-level amplification	GISTIC ±2	0	0.32
DISP1	High-level amplification	GISTIC ±2	0	0.32
OR2L13	High-level amplification	GISTIC ±2	1	0.32
ZNF496	High-level amplification	GISTIC ±2	1	0.32



110 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); NID1 → Urokinase (DB00013); SV2A → Levetiracetam (DB01202) … and 62 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0IQ	40	III	Basal	0.86	no progression, followed 140.9 mo
TCGA-AR-A1AH	51	II	Basal	0.84	progressed at 37.3 mo (group D)
TCGA-BH-A0BA	51	III	LumA	0.83	no progression, followed 37.2 mo
TCGA-A7-A0CE	57	II	Basal	0.82	no progression, followed 35.3 mo
TCGA-AR-A24T	46	III	LumA	0.82	no progression, followed 105.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**35**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1MI**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A3XU-01** · Aneuploidy score: **10** · TMB (nonsyn., /Mb): **0.7** · MSIsensor score: **0.6** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

53rd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

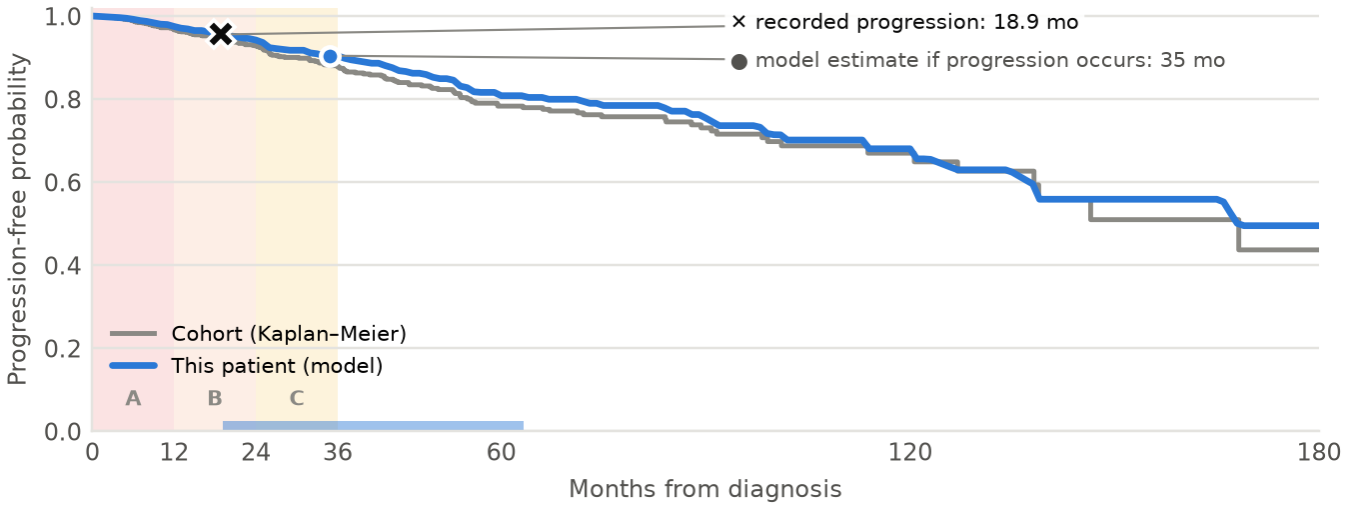
≈ 35 months

Window 24–36 months · IQR 19–63 mo
if progression occurs

Recorded progression

18.9 months

Group B · Progression in year 2
✗ recorded window differs · |error| 16 mo



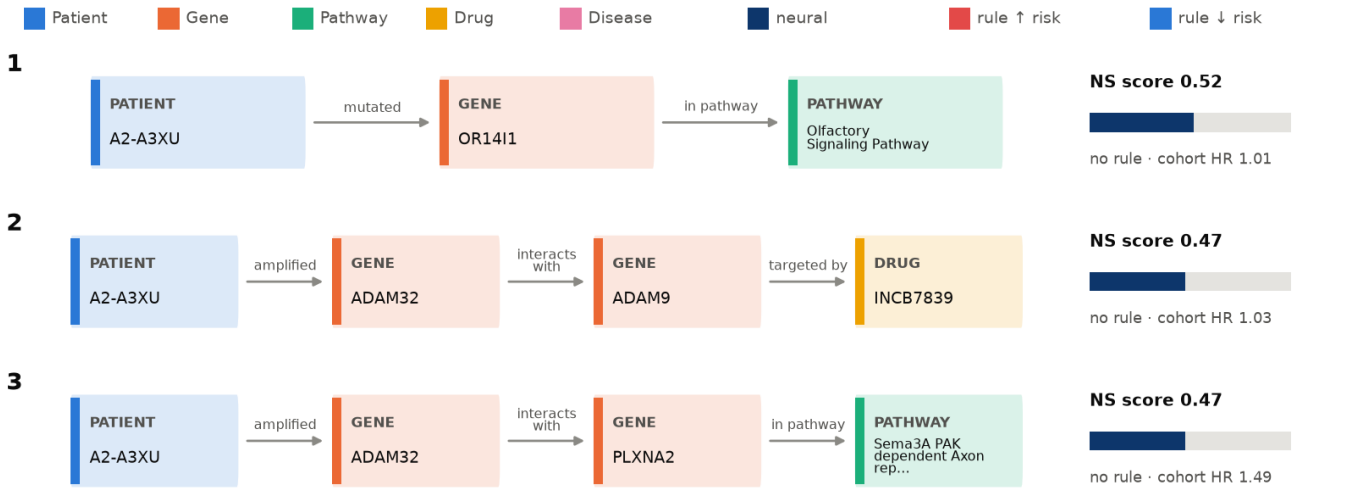
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 35-year-old patient has a Basal tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **53rd risk percentile** (intermediate tier). The highest-scoring knowledge-graph path runs from **OR14I1** to **Olfactory Signaling Pathway**. Progression was recorded at 18.9 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

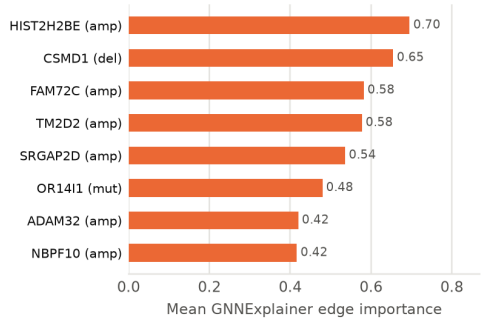
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A3XU** mutated → **OR14I1** in pathway → **Olfactory Signaling Pathway**	86	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 158 carriers	**0.52**
2	**TCGA-A2-A3XU** amplified → **ADAM32** interacts with → **ADAM9** targeted by → **INCB7839**	79	No learned formula fires on this path; its score is neural only. cohort: HR 1.03, n = 93 carriers	**0.47**
3	**TCGA-A2-A3XU** amplified → **ADAM32** interacts with → **PLXNA2** in pathway → **Sema3A PAK dependent Axon repulsion**	79	No learned formula fires on this path; its score is neural only. cohort: HR 1.49, n = 87 carriers	**0.47**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
HIST2H2BE	High-level amplification	GISTIC ±2	0	0.70
CSMD1	Homozygous deletion	GISTIC ±2	0	0.65
FAM72C	High-level amplification	GISTIC ±2	0	0.58
TM2D2	High-level amplification	GISTIC ±2	0	0.58
SRGAP2D	High-level amplification	GISTIC ±2	0	0.54
OR14I1	Mutation	p.A26T	1	0.48
ADAM32	High-level amplification	GISTIC ±2	0	0.42
NBPF10	High-level amplification	GISTIC ±2	0	0.42
HAS2	Mutation	p.L407I	1	0.37
HTRA4	High-level amplification	GISTIC ±2	0	0.34
ITGA10	High-level amplification	GISTIC ±2	3	0.34
PDE4DIP	High-level amplification	GISTIC ±2	0	0.32



21 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

SV2A → Levetiracetam (DB01202); SV2A → Brivaracetam (DB05541); SV2A → Seletracetam (DB05885); ADAM9 → INCB7839 (DB05033); PARP1 → Theophylline (DB00277); PARP1 → Zinc (DB01593); PARP1 → Carba-nicotinamide-adenine-dinucleotide (DB02498); PARP1 → NU1025 (DB02690); PARP1 → Nicotinamide (DB02701); PARP1 → 2-{3-[4-(4-Fluorophenyl)-3,6-Dihydro-1(2h)-Pyridinyl]Propyl}-8-Methyl-4(3h)-Quinazolinone (DB03072) … and 16 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1JM	59	II	Basal	0.66	no progression, followed 19.4 mo
TCGA-S3-AA15	51	II	Basal	0.66	no progression, followed 17.3 mo
TCGA-E2-A1LI	57	II	Basal	0.63	no progression, followed 102.6 mo
TCGA-A1-A0SO	67	II	Basal	0.63	no progression, followed 28.0 mo
TCGA-BH-A1EW	38	II	LumA	0.63	progressed at 55.7 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**Yes**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

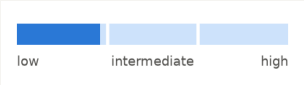
Sample: **TCGA-A7-A13G-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **0.18** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

31st

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

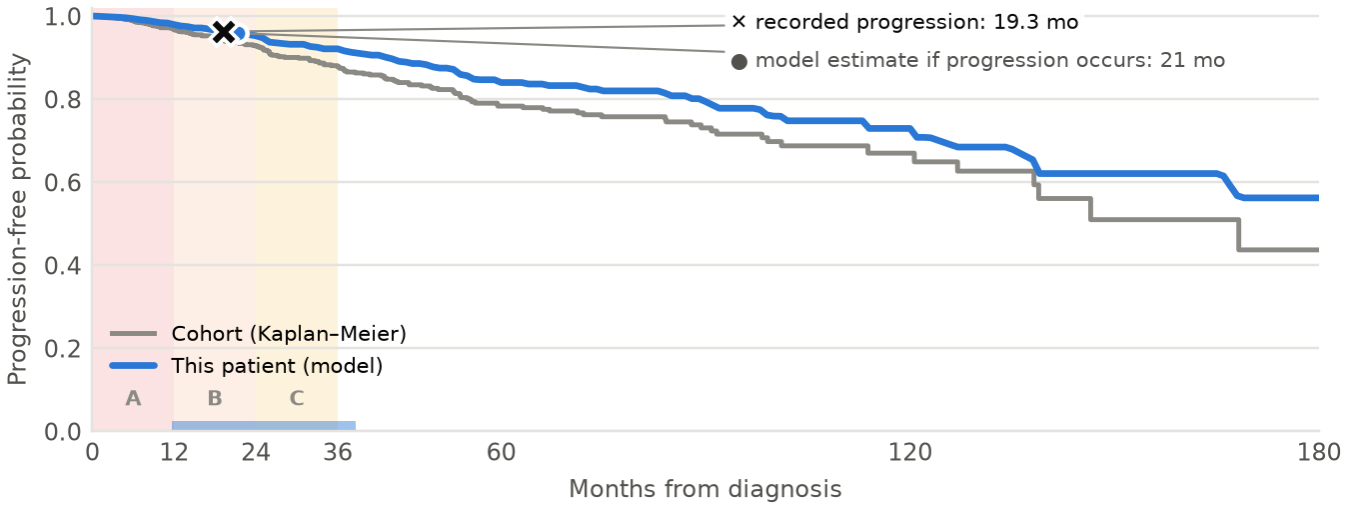
≈ 21 months

Window 12–24 months · IQR 12–39 mo
if progression occurs

Recorded progression

19.3 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 2 mo



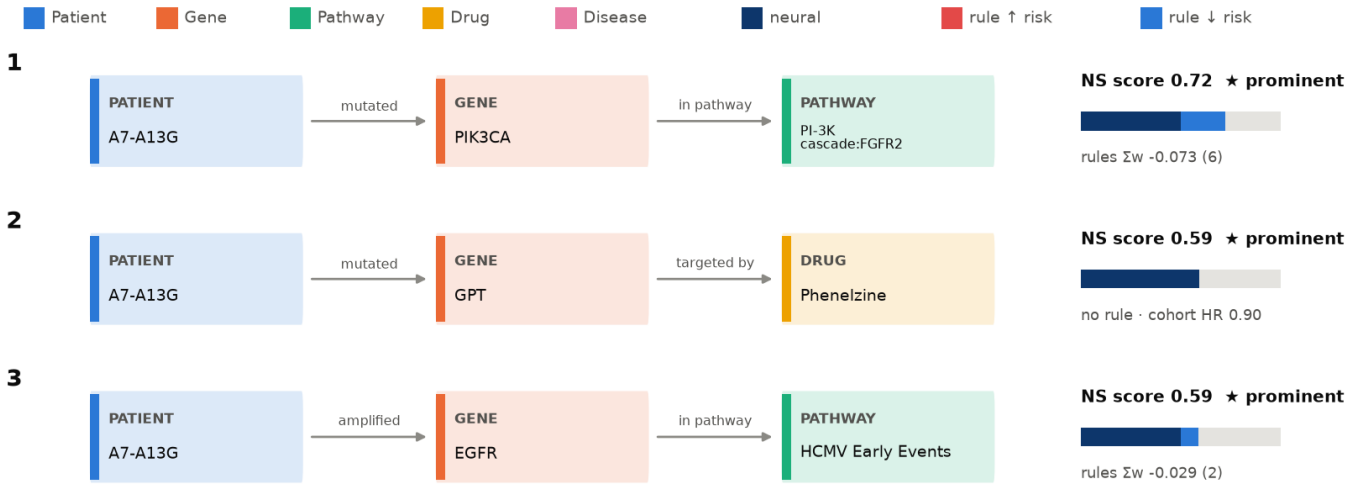
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage II (T2, N0, MX). The fused neuro-symbolic model places them in the **31st risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.099 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **PI-3K cascade:FGFR2**; the learned formulas it instantiates point to slower progression (Σw -0.073). Progression was recorded at 19.3 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

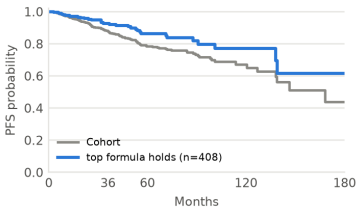


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A7-A13G** mutated → **PIK3CA** in pathway → **PI-3K cascade:FGFR2**	84	↓ **slower progression** · Σw -0.073 over 6 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.70, n = 404 carriers	**0.72**
2 ★	**TCGA-A7-A13G** mutated → **GPT** targeted by → **Phenelzine**	99	No learned formula fires on this path; its score is neural only. cohort: HR 0.90, n = 98 carriers	**0.59**
3 ★	**TCGA-A7-A13G** amplified → **EGFR** in pathway → **HCMV Early Events**	84	↓ **slower progression** · Σw -0.029 over 2 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.68, n = 32 carriers	**0.59**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

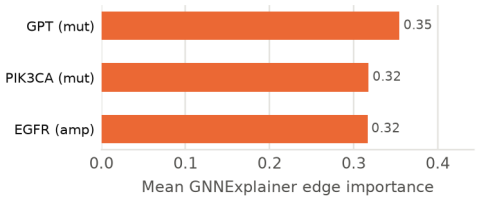
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.099 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	398 pts
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ TargetedBy(g, 'ATP') ⇒ Progress(p) [KG]	mined pair	162 pts
-0.011 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.009 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	mined pair	212 pts
-0.006 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR4') ⇒ Progress(p) [KG]	mined pair	233 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.A1046=, p.H1047R	70	0.32
EGFR · driver	High-level amplification	GISTIC ±2	73	0.32
GPT	Mutation	p.A114=	5	0.35



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); EGFR → Cetuximab (DB00002); EGFR → Lidocaine (DB00281); EGFR → Gefitinib (DB00317) … and 32 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1RD	67	II	LumA	0.77	no progression, followed 1.1 mo
TCGA-D8-A1XY	74	II	LumA	0.77	no progression, followed 16.5 mo
TCGA-C8-A1HE	59	II	LumA	0.76	no progression, followed 12.3 mo
TCGA-AN-A0XL	61	II	LumA	0.76	no progression, followed 5.4 mo
TCGA-OL-A66P	75	II	Her2	0.76	no progression, followed 14.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8510/3**

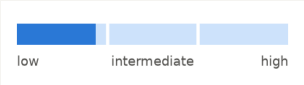
Sample: **TCGA-AR-A1AI-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **2.33** · MSIsensor score: **0.79** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

29th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 16%

Model-estimated progression time

≈ 16 months

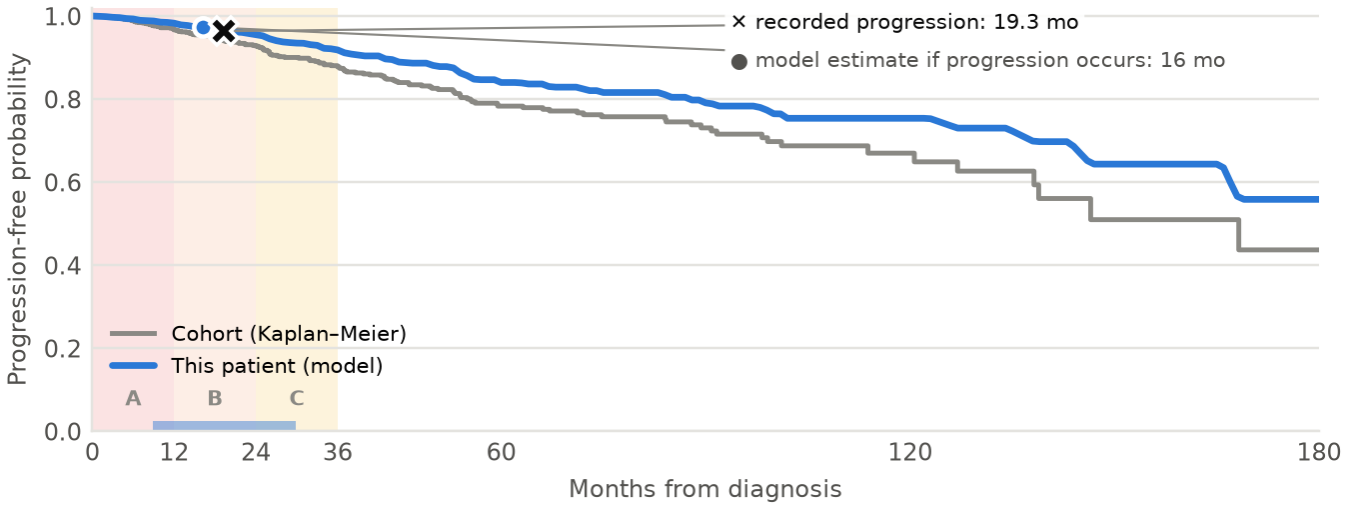
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

19.3 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 3 mo



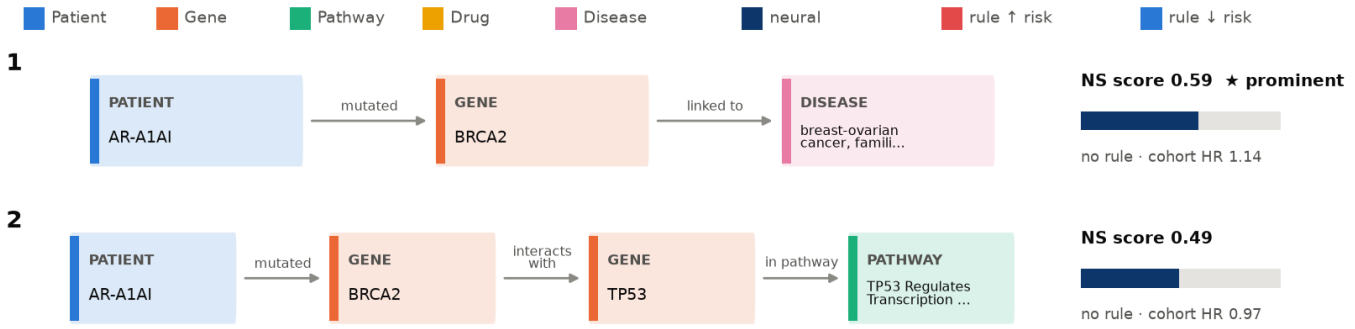
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **29th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **BRCA2** to **breast-ovarian cancer, familial, susceptibility to**. Progression was recorded at 19.3 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

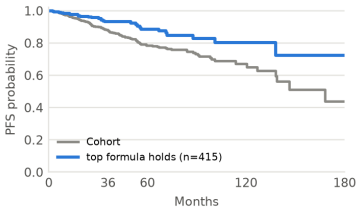


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AR-A1AI** mutated → **BRCA2** linked to → **breast-ovarian cancer, familial, susceptibility to**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.14, n = 72 carriers	**0.59**
2	**TCGA-AR-A1AI** mutated → **BRCA2** interacts with → **TP53** in pathway → **TP53 Regulates Transcription of DNA Repair Genes**	82	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 396 carriers	**0.49**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

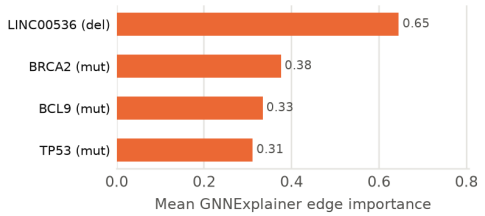
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
+0.023 ↑ risk	Radiation(p, no) ⇒ Progress(p)	single atom	348 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BRCA2 · driver	Mutation	p.R2336C	14	0.38
TP53 · driver	Mutation	p.F270I	57	0.31
LINC00536	Homozygous deletion	GISTIC ±2	0	0.65
BCL9	Mutation	p.F1204L	3	0.33



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XX	49	II	Basal	0.94	no progression, followed 47.3 mo
TCGA-E2-A1B6	44	II	Basal	0.92	no progression, followed 28.5 mo
TCGA-C8-A12V	55	II	Basal	0.91	no progression, followed 12.7 mo
TCGA-BH-A0RX	59	II	Basal	0.90	no progression, followed 5.6 mo
TCGA-E9-A1N8	48	II	Basal	0.90	no progression, followed 34.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**78**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**ADMIX**	Pathologic T	**T2**
Prior diagnosis	**Yes**	Pathologic N	**N3**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

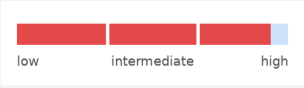
Sample: **TCGA-E2-A2P5-01** · Aneuploidy score: **26** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.01** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

93rd

High risk tier · fused neuro-symbolic model



Probability of progression

21% by 36 mo

12 mo 6% · 24 mo 13% · 60 mo 36%

Model-estimated progression time

≈ 17 months

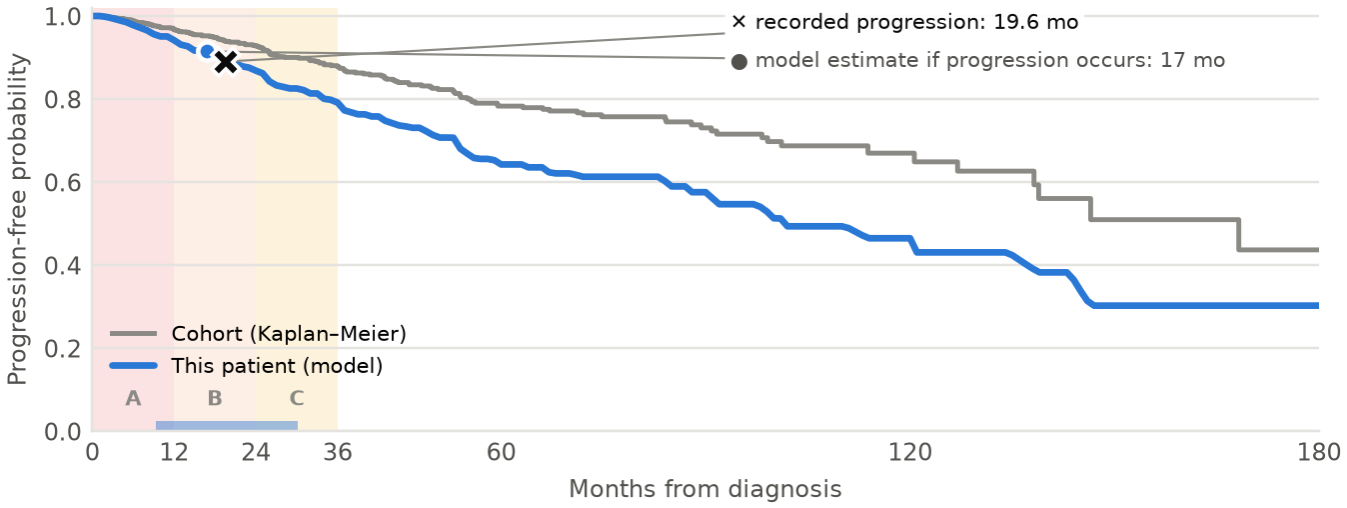
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

19.6 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 3 mo



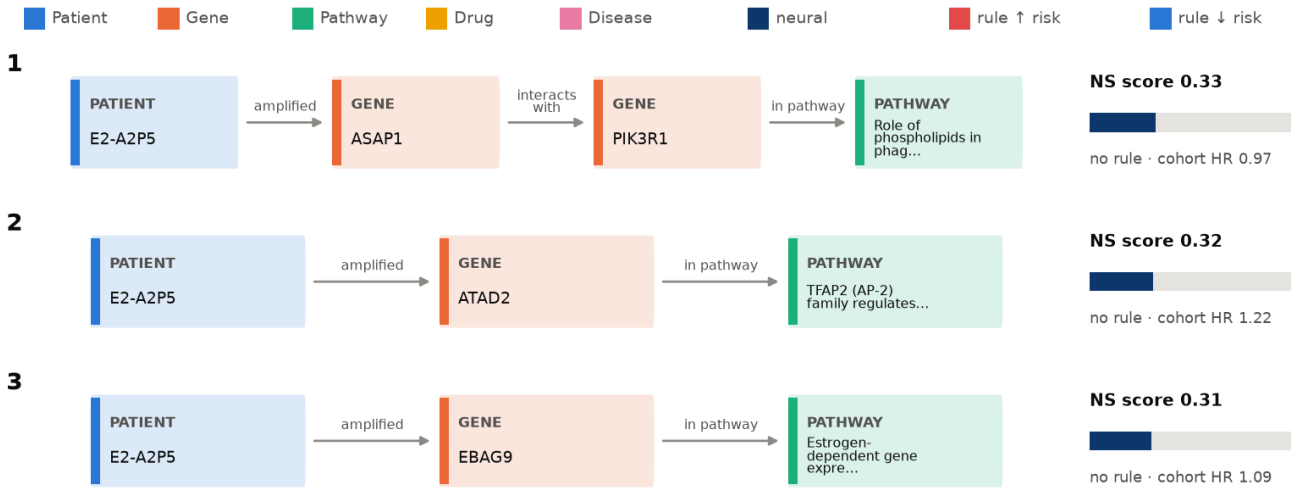
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 78-year-old patient has a subtype-unassigned tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **93rd risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N3) ⇒ Progress(p)*) raises the progression hazard (weight +0.098 per SD). The highest-scoring knowledge-graph path runs from **ASAP1** to **Role of phospholipids in phagocytosis**. Progression was recorded at 19.6 months; the model's estimate given progression was 17 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

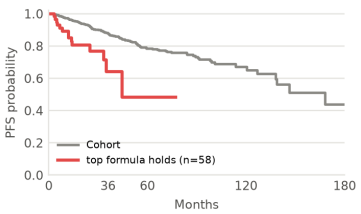


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-E2-A2P5** amplified → **ASAP1** interacts with → **PIK3R1** in pathway → **Role of phospholipids in phagocytosis**	54	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 324 carriers	**0.33**
2	**TCGA-E2-A2P5** amplified → **ATAD2** in pathway → **TFAP2 (AP-2) family regulates transcription of growth factors and their receptors**	53	No learned formula fires on this path; its score is neural only. cohort: HR 1.22, n = 262 carriers	**0.32**
3	**TCGA-E2-A2P5** amplified → **EBAG9** in pathway → **Estrogen-dependent gene expression**	51	No learned formula fires on this path; its score is neural only. cohort: HR 1.09, n = 459 carriers	**0.31**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

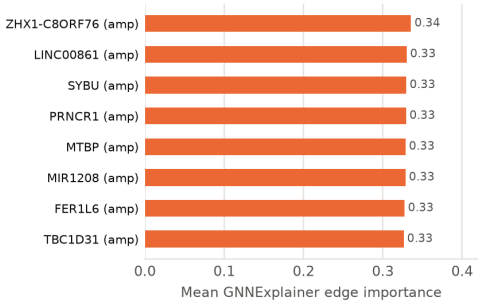
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.098 ↑ risk	NStage(p, N3) ⇒ Progress(p)	single atom	57 pts
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
ZHX1-C8ORF76	High-level amplification	GISTIC ±2	0	0.34
LINC00861	High-level amplification	GISTIC ±2	0	0.33
SYBU	High-level amplification	GISTIC ±2	0	0.33
PRNCR1	High-level amplification	GISTIC ±2	0	0.33
MTBP	High-level amplification	GISTIC ±2	0	0.33
MIR1208	High-level amplification	GISTIC ±2	0	0.33
FER1L6	High-level amplification	GISTIC ±2	0	0.33
TBC1D31	High-level amplification	GISTIC ±2	0	0.33
EFR3A	High-level amplification	GISTIC ±2	0	0.33
ZNF572	High-level amplification	GISTIC ±2	0	0.32
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.32



89 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); NDUFB9 → NADH (DB00157); TG → Iodine (DB05382); SQLE → Naftifine (DB00735); SQLE → Terbinafine (DB00857); SQLE → Butenafine (DB01091); SQLE → Ellagic acid (DB08846); KCNQ3 → Amitriptyline (DB00321); KCNQ3 → Diclofenac (DB00586) … and 4 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A3Z5	54	III	missing	0.65	no progression, followed 33.4 mo
TCGA-E2-A3DX	43	III	missing	0.60	no progression, followed 43.6 mo
TCGA-A2-A4RW	49	III	missing	0.59	no progression, followed 7.3 mo
TCGA-E9-A3Q9	78	III	missing	0.59	no progression, followed 32.9 mo
TCGA-A8-A09N	57	III	LumB	0.58	no progression, followed 1.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A18J · Group B

Report date 31 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**56**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**—**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A18J-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **2.5** · MSIsensor score: **0.08** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

64% by 36 mo

12 mo 19% · 24 mo 42% · 60 mo 88%

Model-estimated progression time

≈ 43 months

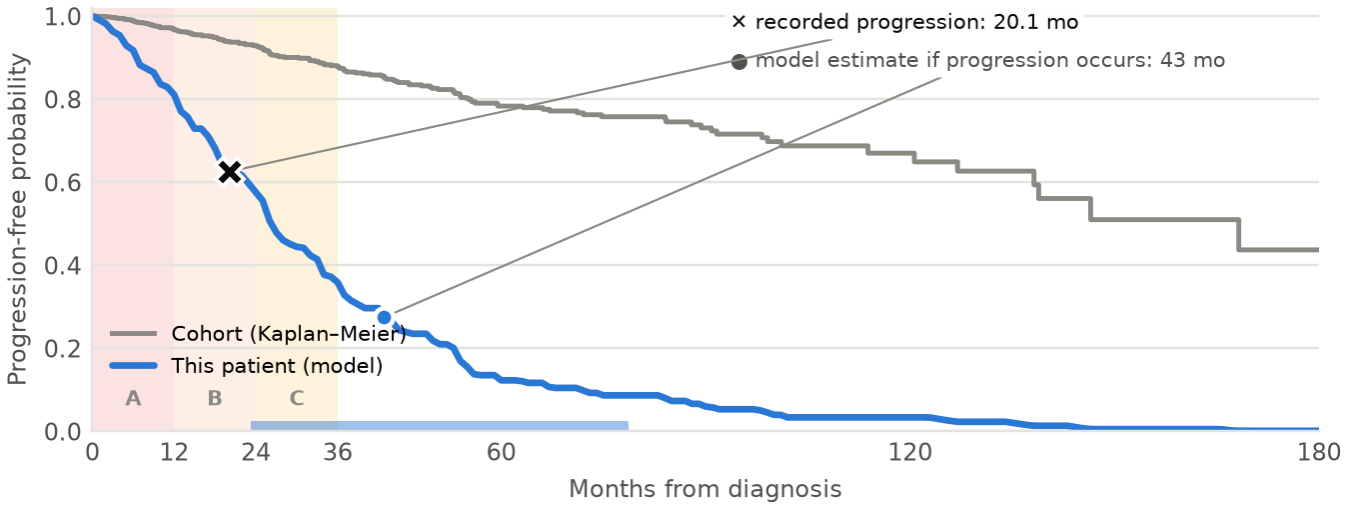
Window > 36 months · IQR 23–79 mo if progression occurs

Recorded progression

20.1 months

Group B · Progression in year 2

✗ recorded window differs · |error| 23 mo



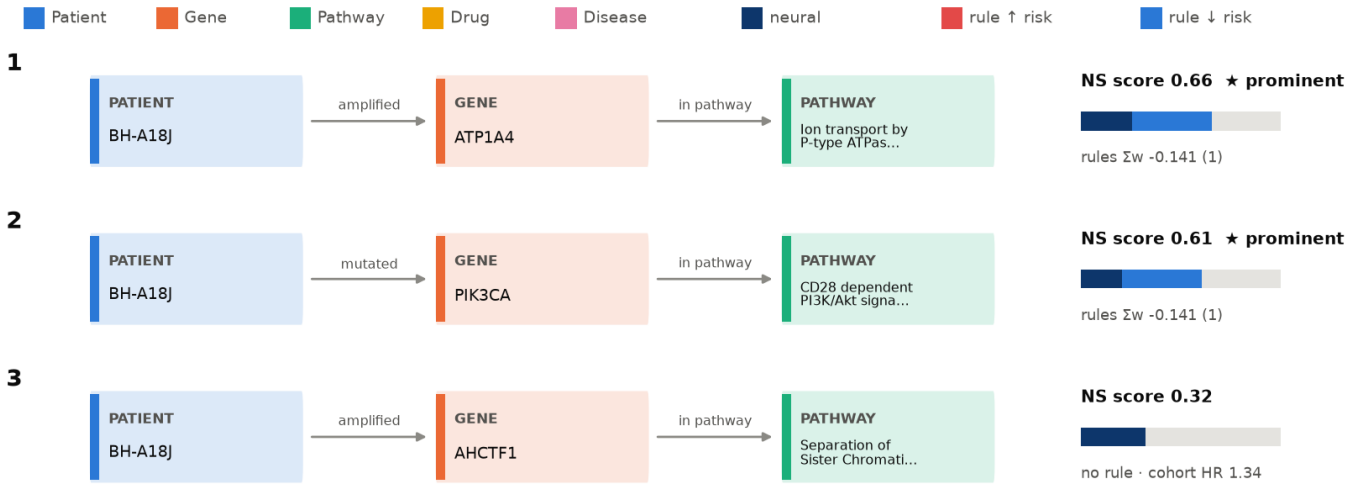
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 56-year-old patient has a LumA tumour, AJCC stage IV (T4, N2, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'VEGFA-VEGFR2 Pathway')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Ion homeostasis')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.141 per SD). The highest-scoring knowledge-graph path runs from **ATP1A4** to **Ion transport by P-type ATPases**; the learned formulas it instantiates point to slower progression (Σw -0.141). Progression was recorded at 20.1 months; the model's estimate given progression was 43 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

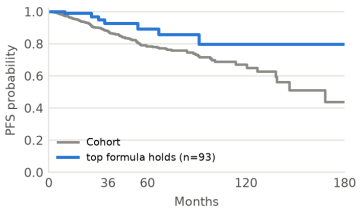


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A18J** amplified → **ATP1A4** in pathway → **Ion transport by P-type ATPases**	43	↓ **slower progression** · Σw -0.141 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 1.14, n = 94 carriers	**0.66**
2 ★	**TCGA-BH-A18J** mutated → **PIK3CA** in pathway → **CD28 dependent PI3K/Akt signaling**	34	↓ **slower progression** · Σw -0.141 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 0.98, n = 401 carriers	**0.61**
3	**TCGA-BH-A18J** amplified → **AHCTF1** in pathway → **Separation of Sister Chromatids**	54	No learned formula fires on this path; its score is neural only. cohort: HR 1.34, n = 211 carriers	**0.32**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

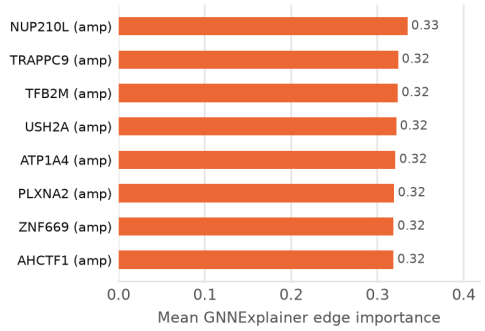
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.141 ↓ risk	Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) [KG]	RL-proposed	93 pts
+0.134 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	27 pts
+0.018 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	mined pair	42 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.Q318E	4	0.32
PIK3CA · driver	Mutation	p.E545K	70	0.31
NUP210L	High-level amplification	GISTIC ±2	0	0.33
TRAPPC9	High-level amplification	GISTIC ±2	2	0.32
TFB2M	High-level amplification	GISTIC ±2	2	0.32
USH2A	High-level amplification	GISTIC ±2	0	0.32
ATP1A4	High-level amplification	GISTIC ±2	4	0.32
PLXNA2	High-level amplification	GISTIC ±2	4	0.32
ZNF669	High-level amplification	GISTIC ±2	1	0.32
AHCTF1	High-level amplification	GISTIC ±2	7	0.32
ZNF124	High-level amplification	GISTIC ±2	1	0.32
FAM72C	High-level amplification	GISTIC ±2	0	0.32



40 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); ATP1A4 → Rubidium Rb-82 (DB09479); PARP1 → Theophylline (DB00277); PARP1 → Zinc (DB01593) … and 22 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CS	73	IV	LumA	0.61	no progression, followed 77.2 mo
TCGA-AN-A0XP	69	III	LumA	0.60	no progression, followed 0.3 mo
TCGA-A8-A09B	58	III	LumA	0.56	no progression, followed 12.0 mo
TCGA-A8-A08T	64	IV	LumA	0.54	progressed at 84.1 mo (group D)
TCGA-AO-A12A	47	II	LumA	0.52	no progression, followed 102.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A2-A0SW · Group B

Report date 31 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**82**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A0SW-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **1.73** · MSIsensor score: **1.59** · AJCC edition: —

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

68% by 36 mo

12 mo 25% · 24 mo 45% · 60 mo 86%

Model-estimated progression time

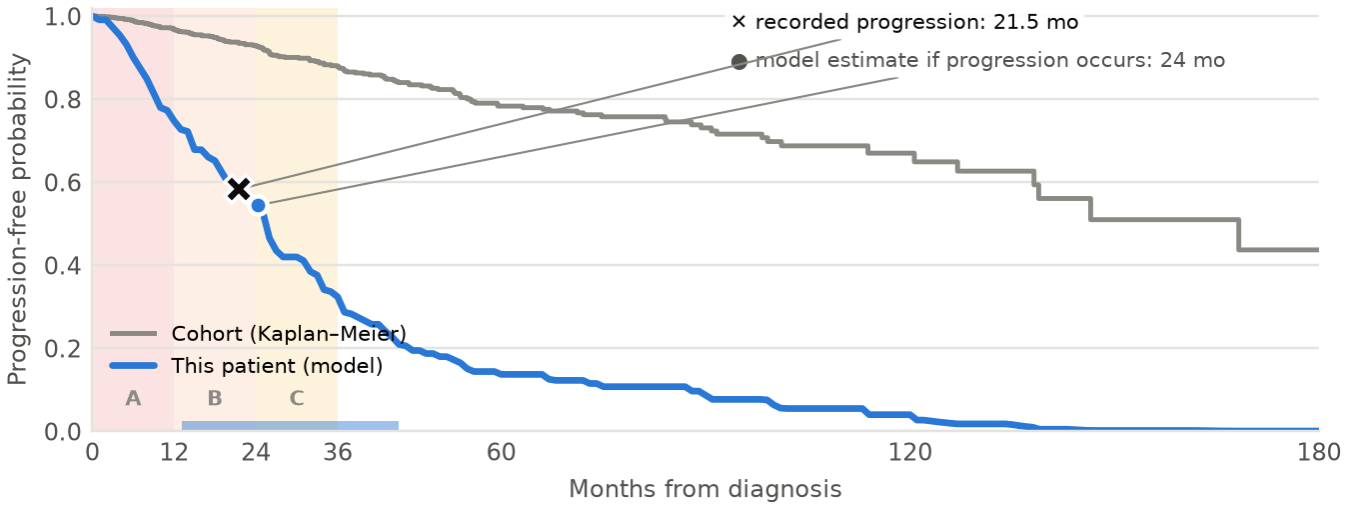
≈ 24 months

Window 24–36 months · IQR 13–45 mo
if progression occurs

Recorded progression

21.5 months

Group B · Progression in year 2
✗ recorded window differs · |error| 3 mo



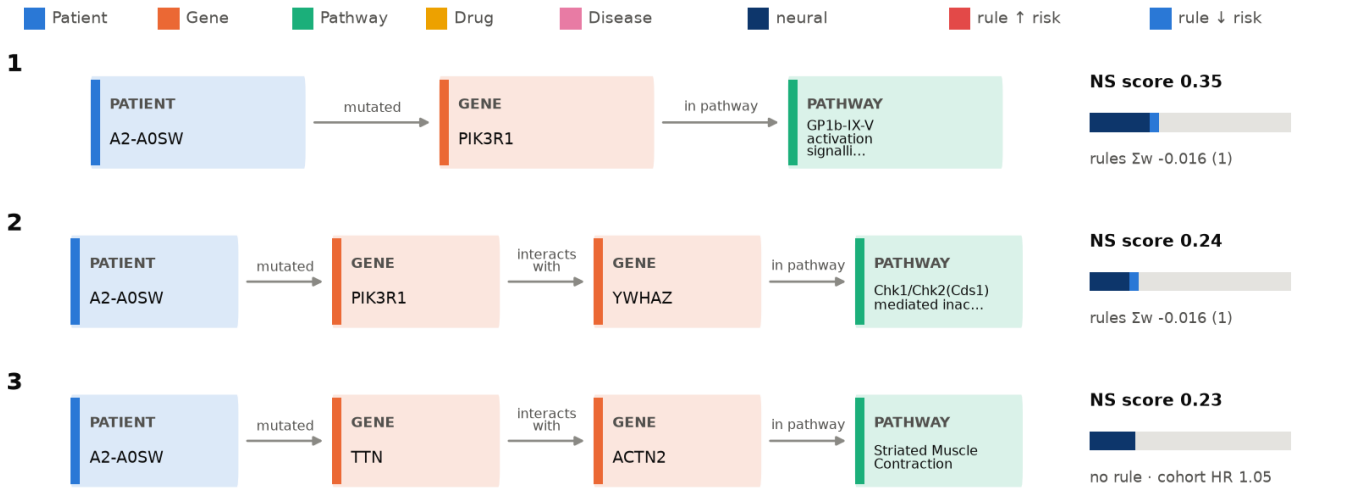
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 82-year-old patient has a LumB tumour, AJCC stage IV (T2, N2, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T2)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'PI3K events in ERBB2 signaling')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Regulation of PTEN gene transcription')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.016 per SD). The highest-scoring knowledge-graph path runs from **PIK3R1** to **GP1b-IX-V activation signalling**; the learned formulas it instantiates point to slower progression (Σw -0.016). Progression was recorded at 21.5 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

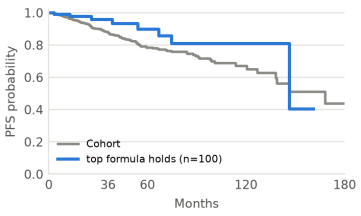


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A0SW** mutated → **PIK3R1** in pathway → **GP1b-IX-V activation signalling**	50	↓ **slower progression** · Σw -0.016 over 1 formula(s) TStage(p, T2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K events in ERBB2 signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of PTEN gene transcription') ⇒ Progress(p) cohort: HR 1.28, n = 126 carriers	**0.35**
2	**TCGA-A2-A0SW** mutated → **PIK3R1** interacts with → **YWHAZ** in pathway → **Chk1/Chk2(Cds1) mediated inactivation of Cyclin B:Cdk1 complex**	33	↓ **slower progression** · Σw -0.016 over 1 formula(s) TStage(p, T2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K events in ERBB2 signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of PTEN gene transcription') ⇒ Progress(p) cohort: HR 1.36, n = 96 carriers	**0.24**
3	**TCGA-A2-A0SW** mutated → **TTN** interacts with → **ACTN2** in pathway → **Striated Muscle Contraction**	38	No learned formula fires on this path; its score is neural only. cohort: HR 1.05, n = 321 carriers	**0.23**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

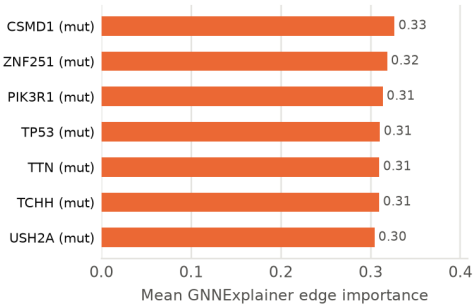
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.016 ↓ risk	TStage(p, T2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K events in ERBB2 signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of PTEN gene transcription') ⇒ Progress(p) **[KG]**	RL-proposed	98 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3R1 · driver	Mutation	p.E468_E469insGLYE	79	0.31
TP53 · driver	Mutation	p.R273C	57	0.31
CSMD1	Mutation	p.V2880=	0	0.33
ZNF251	Mutation	p.F265L	0	0.32
TTN	Mutation	p.*3454*	2	0.31
TCHH	Mutation	p.R68C	1	0.31
USH2A	Mutation	p.Q4382K	0	0.30



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486); PIK3R1 → Wortmannin (DB08059); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 2 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A07W	76	IV	LumB	0.73	no progression, followed 10.0 mo
TCGA-A7-A56D	84	II	LumA	0.70	no progression, followed 14.7 mo
TCGA-BH-A0BF	56	II	LumB	0.64	no progression, followed 43.5 mo
TCGA-E9-A226	45	III	LumB	0.64	progressed at 26.0 mo (group C)
TCGA-AR-A0U0	73	II	Basal	0.64	no progression, followed 65.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**84**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

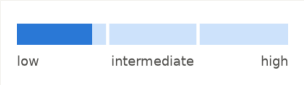
Sample: **TCGA-AC-A2B8-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **2.77** · MSIsensor score: **0.54** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

28th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 16%

Model-estimated progression time

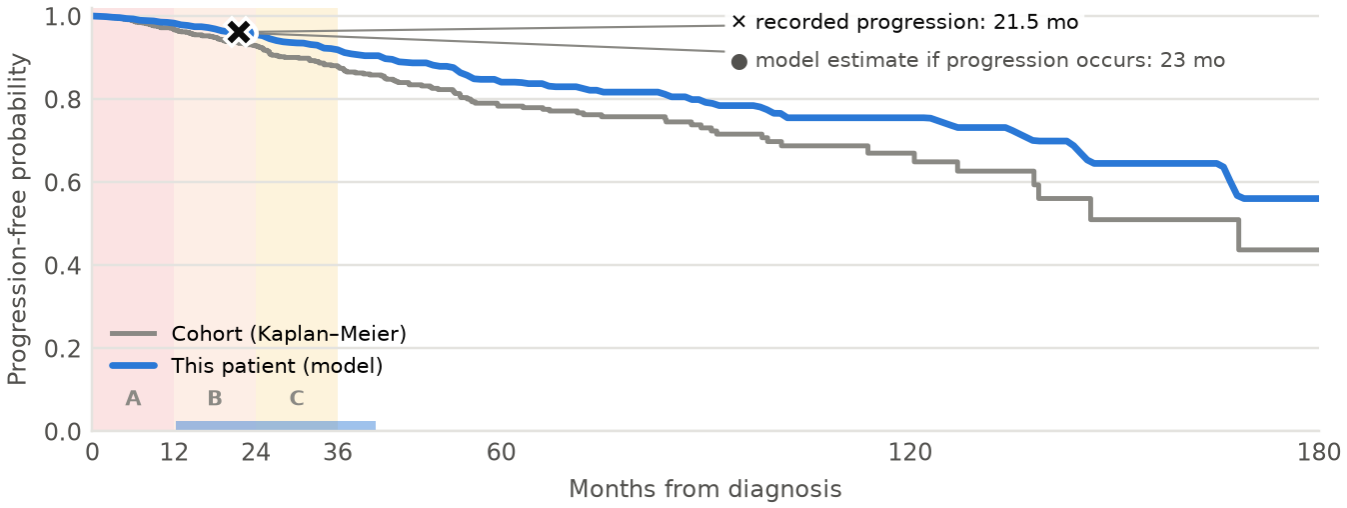
≈ 23 months

Window 12–24 months · IQR 12–42 mo
if progression occurs

Recorded progression

21.5 months

■ **Group B** · Progression in year 2
✓ **estimate in recorded window** · |error| 1 mo



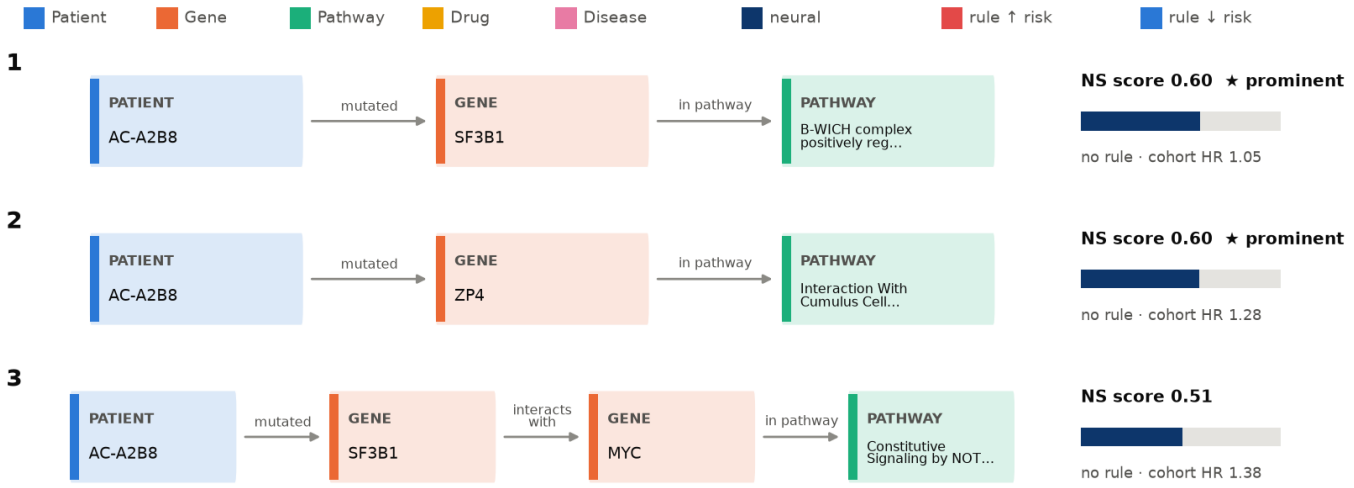
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 84-year-old patient has a LumA tumour, AJCC stage II (T3, N0, M0). The fused neuro-symbolic model places them in the **28th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **SF3B1** to **B-WICH complex positively regulates rRNA expression**. Progression was recorded at 21.5 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

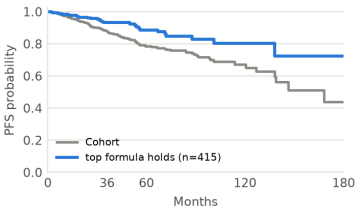


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AC-A2B8** mutated → **SF3B1** in pathway → **B-WICH complex positively regulates rRNA expression**	100	No learned formula fires on this path; its score is neural only. cohort: HR 1.05, n = 113 carriers	**0.60**
2 ★	**TCGA-AC-A2B8** mutated → **ZP4** in pathway → **Interaction With Cumulus Cells And The Zona Pellucida**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.28, n = 183 carriers	**0.60**
3	**TCGA-AC-A2B8** mutated → **SF3B1** interacts with → **MYC** in pathway → **Constitutive Signaling by NOTCH1 PEST Domain Mutants**	85	No learned formula fires on this path; its score is neural only. cohort: HR 1.38, n = 129 carriers	**0.51**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

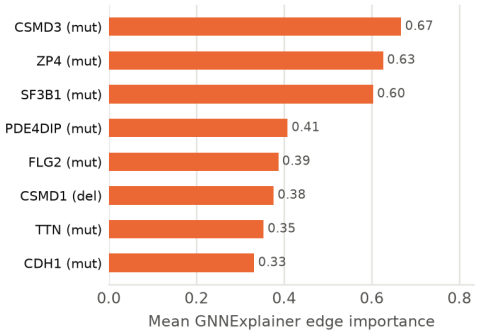
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
-0.032 ↓ risk	NStage(p, N0) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	194 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
SF3B1 · driver	Mutation	p.E622Q	4	0.60
CDH1 · driver	Mutation	p.C603del	17	0.33
PIK3CA · driver	Mutation	p.E545K	70	0.33
CSMD3	Mutation	p.T1007=	0	0.67
ZP4	Mutation	p.R149W	1	0.63
PDE4DIP	Mutation	p.H438=	0	0.41
FLG2	Mutation	p.G965A, p.Q944H	1	0.39
CSMD1	Homozygous deletion	GISTIC ±2	0	0.38
TTN	Mutation	p.R23443P	2	0.35



9 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

SF3B1 → H3B-8800 (DB14017); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-PE-A5DD	64	II	LumA	0.82	no progression, followed 64.2 mo
TCGA-AR-A5QM	62	II	LumA	0.81	progressed at 31.8 mo (group C)
TCGA-A2-A0CR	54	II	LumA	0.78	no progression, followed 107.9 mo
TCGA-E2-A10F	47	II	LumA	0.77	no progression, followed 28.9 mo
TCGA-A2-A0YD	63	II	LumA	0.77	no progression, followed 25.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-LQ-A4E4-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **—** · MSIsensor score: **0.06** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

90th

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 32%

Model-estimated progression time

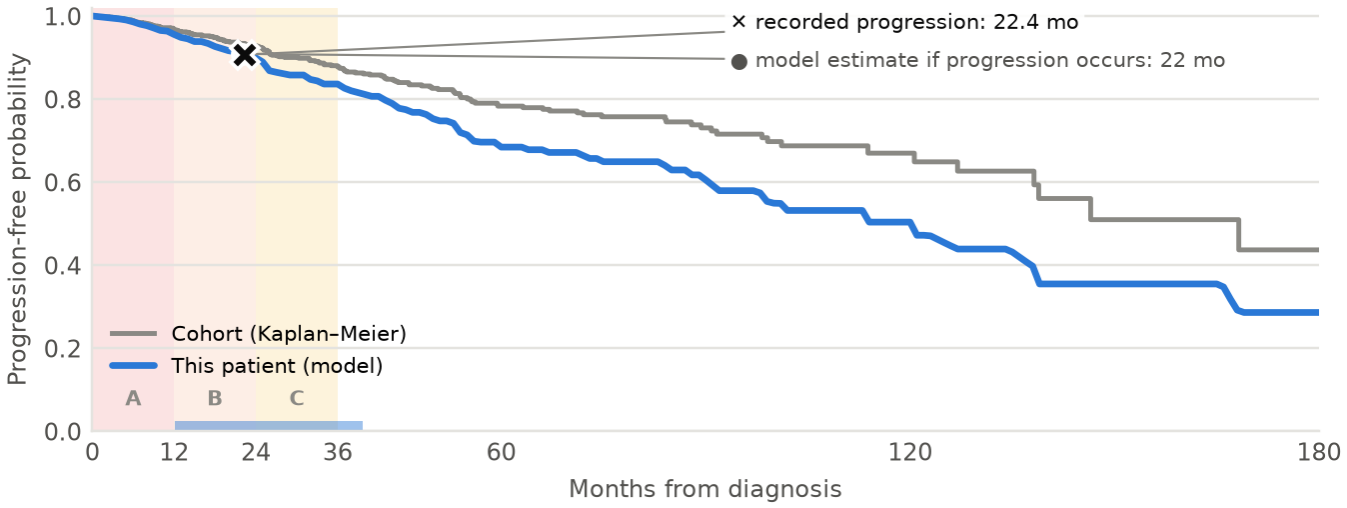
≈ 22 months

Window 12–24 months · IQR 12–40 mo
if progression occurs

Recorded progression

22.4 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 1 mo



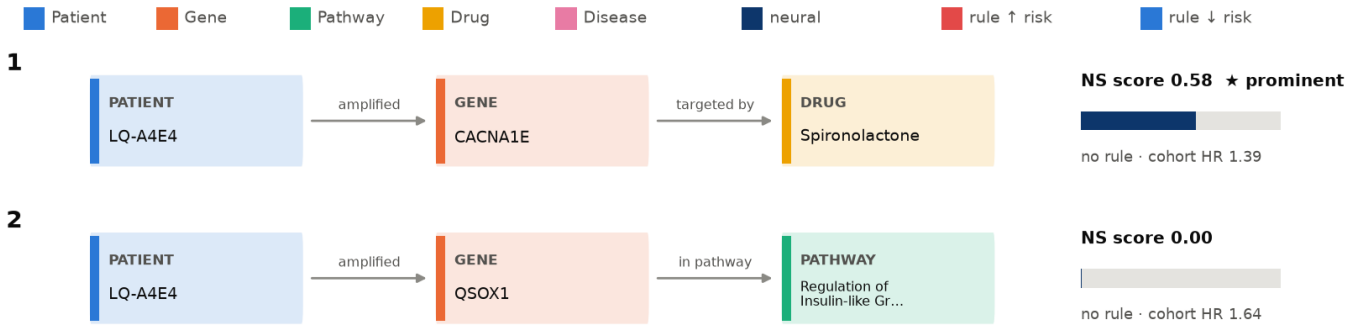
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a subtype-unassigned tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **90th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **CACNA1E** to **Spironolactone**. Progression was recorded at 22.4 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 1 of the 16 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

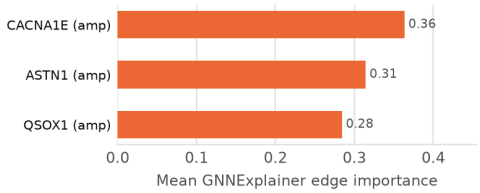
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-LQ-A4E4** amplified → **CACNA1E** targeted by → **Spironolactone**	96	No learned formula fires on this path; its score is neural only. cohort: HR 1.39, n = 182 carriers	**0.58**
2	**TCGA-LQ-A4E4** amplified → **QSOX1** in pathway → **Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)**	1	No learned formula fires on this path; its score is neural only. cohort: HR 1.64, n = 111 carriers	**0.00**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CACNA1E	High-level amplification	GISTIC ±2	12	0.36
ASTN1	High-level amplification	GISTIC ±2	0	0.31
QSOX1	High-level amplification	GISTIC ±2	4	0.28



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CACNA1E → Ergocalciferol (DB00153); CACNA1E → Enflurane (DB00228); CACNA1E → Topiramate (DB00273); CACNA1E → Spironolactone (DB00421); CACNA1E → Lamotrigine (DB00555); CACNA1E → Calcium citrate (DB11093); CACNA1E → Butamben (DB11148); CACNA1E → Calcium Phosphate (DB11348); CACNA1E → Bioallethrin (DB13746); CACNA1E → Calcium phosphate dihydrate (DB14481). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AC-A2QI	76	III	missing	0.93	no progression, followed 19.3 mo
TCGA-AC-A3W6	90	III	missing	0.90	no progression, followed 19.8 mo
TCGA-EW-A424	51	III	missing	0.88	no progression, followed 23.5 mo
TCGA-AC-A2FE	62	III	missing	0.74	progressed at 25.2 mo (group C)
TCGA-AC-A3TM	50	III	missing	0.72	no progression, followed 25.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A3XS-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.38** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

92nd

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 6% · 24 mo 13% · 60 mo 34%

Model-estimated progression time

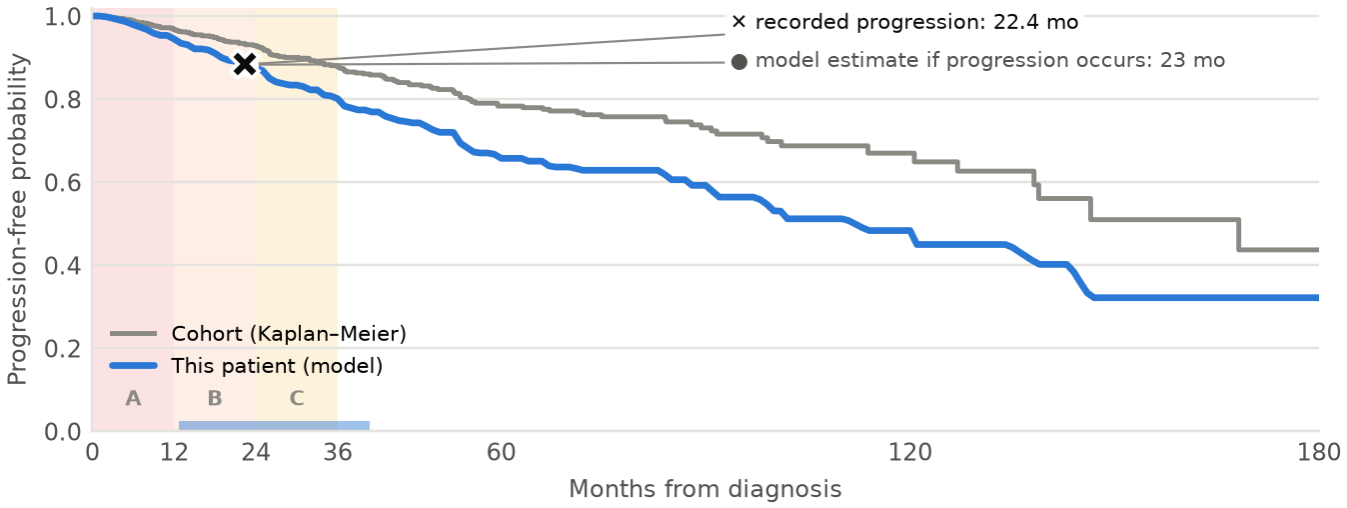
≈ 23 months

Window 12–24 months · IQR 13–41 mo
if progression occurs

Recorded progression

22.4 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 0 mo



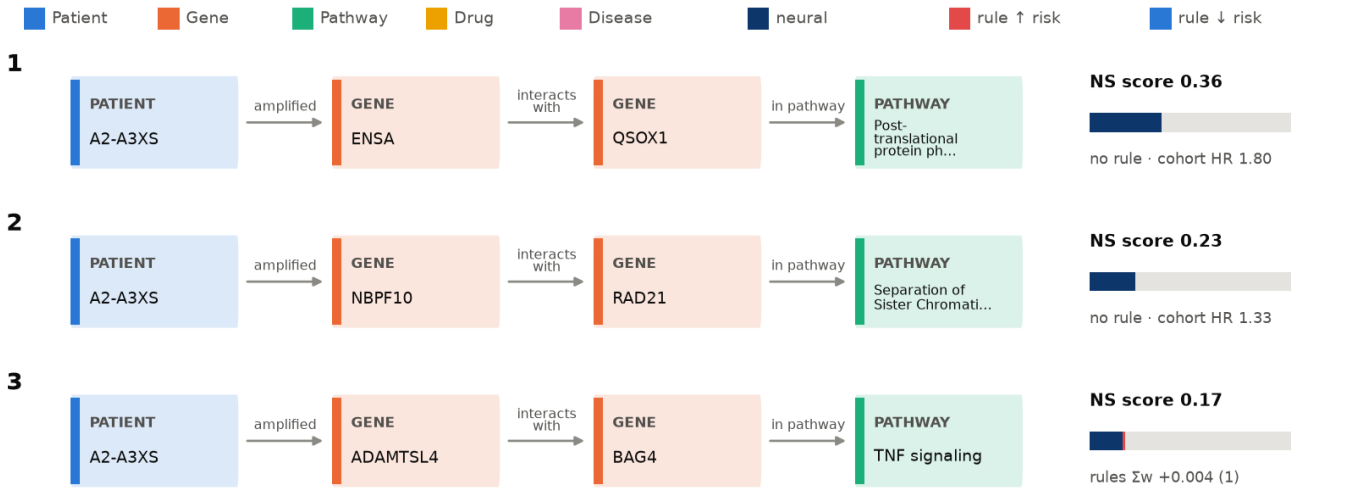
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a subtype-unassigned tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **92nd risk percentile** (high tier). The strongest learned formula that holds for them (*HighAneuploidy(p) ⇒ Progress(p)*) lowers the progression hazard (weight -0.005 per SD). The highest-scoring knowledge-graph path runs from **ENSA** to **Post-translational protein phosphorylation**. Progression was recorded at 22.4 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

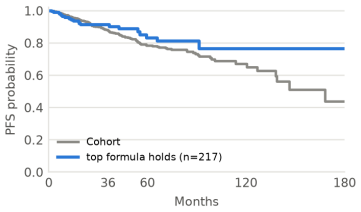


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A3XS** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Post-translational protein phosphorylation**	60	No learned formula fires on this path; its score is neural only. cohort: HR 1.80, n = 119 carriers	**0.36**
2	**TCGA-A2-A3XS** amplified → **NBPF10** interacts with → **RAD21** in pathway → **Separation of Sister Chromatids**	38	No learned formula fires on this path; its score is neural only. cohort: HR 1.33, n = 217 carriers	**0.23**
3	**TCGA-A2-A3XS** amplified → **ADAMTSL4** interacts with → **BAG4** in pathway → **TNF signaling**	27	↑ **faster progression** · Σw +0.004 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Defective B3GALTL causes Peters-plus syndrome (PpS)') ⇒ Progress(p) cohort: HR 0.97, n = 115 carriers	**0.17**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

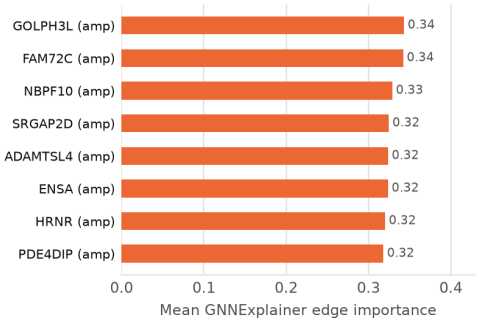
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts
+0.004 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Defective B3GALTL causes Peters-plus syndrome (PpS)') ⇒ Progress(p) [KG]	single atom	94 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GOLPH3L	High-level amplification	GISTIC ±2	0	0.34
FAM72C	High-level amplification	GISTIC ±2	0	0.34
NBPF10	High-level amplification	GISTIC ±2	0	0.33
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
ADAMTSL4	High-level amplification	GISTIC ±2	2	0.32
ENSA	High-level amplification	GISTIC ±2	1	0.32
HRNR	High-level amplification	GISTIC ±2	3	0.32
PDE4DIP	High-level amplification	GISTIC ±2	0	0.32
FLG	High-level amplification	GISTIC ±2	1	0.32
BCL9	High-level amplification	GISTIC ±2	3	0.32
ECM1	High-level amplification	GISTIC ±2	1	0.32
SV2A	High-level amplification	GISTIC ±2	7	0.32



24 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

HRNR → Zinc (DB01593); HRNR → Zinc acetate (DB14487); SV2A → Levetiracetam (DB01202); SV2A → Brivaracetam (DB05541); SV2A → Seletracetam (DB05885); TARS2 → Threonine (DB00156); CA14 → Benzthiazide (DB00562); CA14 → Cyclothiazide (DB00606); CA14 → Acetazolamide (DB00819); CA14 → Zonisamide (DB00909) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A400	43	III	missing	0.88	no progression, followed 7.1 mo
TCGA-E9-A1RG	62	III	LumB	0.62	no progression, followed 21.3 mo
TCGA-AO-A0JL	59	III	missing	0.62	no progression, followed 55.3 mo
TCGA-GM-A3XN	44	III	missing	0.61	no progression, followed 66.4 mo
TCGA-AR-A1AR	50	III	Basal	0.60	progressed at 8.8 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0X4-01** · Aneuploidy score: **0** · TMB (nonsyn., /Mb): **0.6** · MSIsensor score: **0** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

43rd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

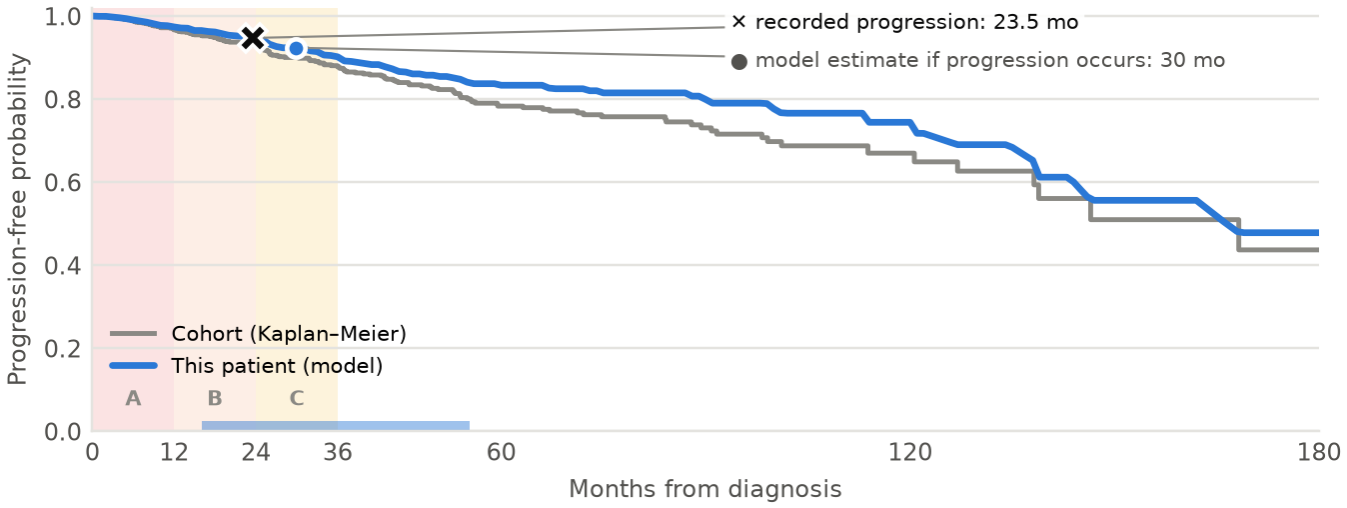
≈ 30 months

Window 24–36 months · IQR 16–55 mo
if progression occurs

Recorded progression

23.5 months

Group B · Progression in year 2
✗ recorded window differs · |error| 6 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **43rd risk percentile** (intermediate tier). The highest-scoring knowledge-graph path runs from **KMT2C** to **RUNX1 regulates genes involved in megakaryocyte differentiation and platelet function**. Progression was recorded at 23.5 months; the model's estimate given progression was 30 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

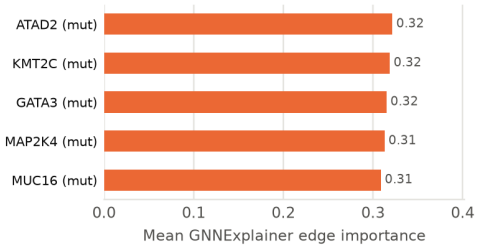
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A0X4** mutated → **KMT2C** in pathway → **RUNX1 regulates genes involved in megakaryocyte differentiation and platelet function**	96	No learned formula fires on this path; its score is neural only. cohort: HR 0.99, n = 261 carriers	**0.58**
2 ★	**TCGA-B6-A0X4** mutated → **ATAD2** in pathway → **TFAP2 (AP-2) family regulates transcription of growth factors and their receptors**	96	No learned formula fires on this path; its score is neural only. cohort: HR 1.20, n = 249 carriers	**0.57**
3 ★	**TCGA-B6-A0X4** mutated → **MAP2K4** in pathway → **Oxidative Stress Induced Senescence**	95	No learned formula fires on this path; its score is neural only. cohort: HR 1.08, n = 316 carriers	**0.57**
4 ★	**TCGA-B6-A0X4** mutated → **GATA3** linked to → **hereditary breast ovarian cancer syndrome**	94	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 800 carriers	**0.57**
5 ★	**TCGA-B6-A0X4** mutated → **MUC16** in pathway → **O-linked glycosylation of mucins**	92	No learned formula fires on this path; its score is neural only. cohort: HR 0.83, n = 114 carriers	**0.55**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.E640*	4	0.32
GATA3 · driver	Mutation	p.S404Qfs*109	13	0.32
MAP2K4 · driver	Mutation	p.R134W	12	0.31
ATAD2	Mutation	p.I827M	1	0.32
MUC16	Mutation	p.G13230R	8	0.31



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MUC16 → Oregovomab (DB04964). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A18N	88	II	LumA	0.79	no progression, followed 37.7 mo
TCGA-C8-A273	29	II	LumA	0.78	no progression, followed 16.9 mo
TCGA-BH-A1EX	67	II	LumA	0.78	progressed at 49.6 mo (group D)
TCGA-BH-A0E1	52	II	LumA	0.78	no progression, followed 15.7 mo
TCGA-D8-A27K	47	II	LumA	0.78	no progression, followed 48.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**59**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**—**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-EW-A1P7-01** · Aneuploidy score: **1** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **1.05** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

57th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

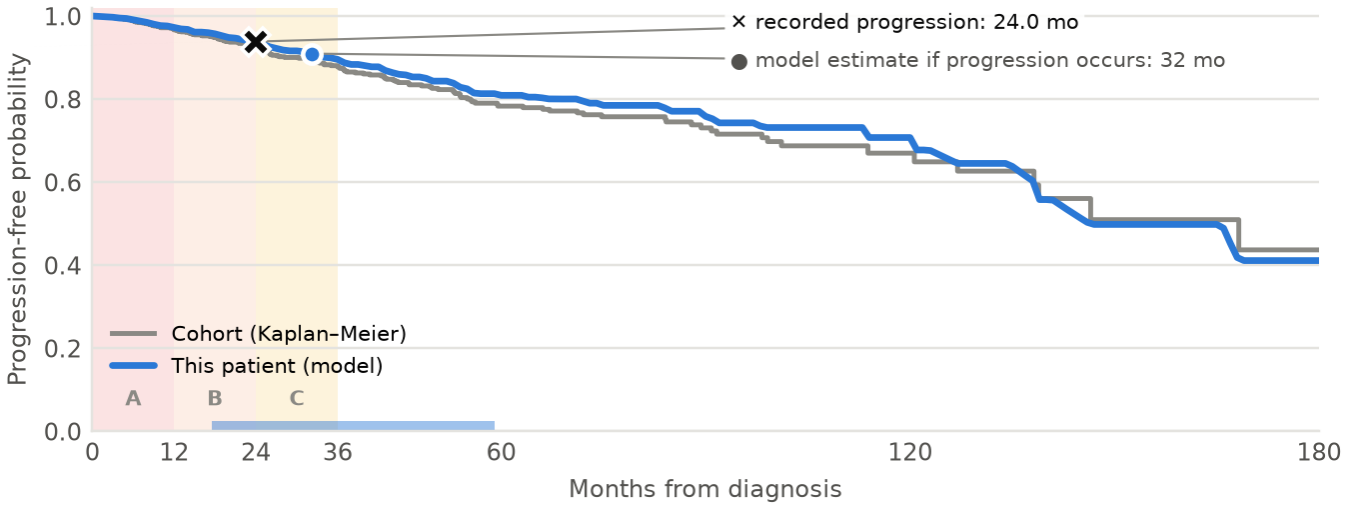
≈ 32 months

Window 24–36 months · IQR 18–59 mo
if progression occurs

Recorded progression

24.0 months

■ **Group B** · Progression in year 2
✗ **recorded window differs** · |error| 8 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 59-year-old patient has a Normal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **57th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **RB1** to **Phosphorylation of proteins involved in G1/S transition by active Cyclin E:Cdk2 complexes**; the learned formulas it instantiates point to slower progression (Σw -0.060). Progression was recorded at 24.0 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

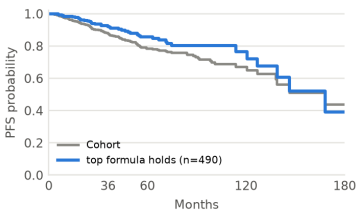


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-EW-A1P7** deleted → **RB1** in pathway → **Phosphorylation of proteins involved in G1/S transition by active Cyclin E:Cdk2 complexes**	97	↓ **slower progression** · Σw -0.060 over 1 formula(s) Deleted(p, RB1) ⇒ Progress(p) cohort: HR 0.97, n = 62 carriers	**0.77**
2 ★	**TCGA-EW-A1P7** deleted → **RB1** interacts with → **TP53** targeted by → **Triethyl phosphate**	87	↓ **slower progression** · Σw -0.060 over 1 formula(s) Deleted(p, RB1) ⇒ Progress(p) cohort: HR 0.95, n = 283 carriers	**0.70**
3 ★	**TCGA-EW-A1P7** mutated → **ZNF623** interacts with → **SETDB1** in pathway → **PKMTs methylate histone lysines**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 324 carriers	**0.59**
4 ★	**TCGA-EW-A1P7** mutated → **TTN** in pathway → **Platelet degranulation**	95	No learned formula fires on this path; its score is neural only. cohort: HR 1.03, n = 325 carriers	**0.57**
5 ★	**TCGA-EW-A1P7** deleted → **BRCA2** linked to → **breast carcinoma**	94	No learned formula fires on this path; its score is neural only. cohort: HR 1.19, n = 780 carriers	**0.57**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

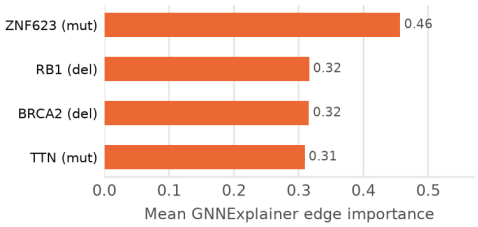
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
-0.060 ↓ risk	Deleted(p, RB1) ⇒ Progress(p)	single atom	35 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
RB1 · driver	Homozygous deletion	GISTIC ±2	25	0.32
BRCA2 · driver	Homozygous deletion	GISTIC ±2	14	0.32
ZNF623	Mutation	p.C377Y	0	0.46
TTN	Mutation	p.K2800M	2	0.31



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A6R9	61	II	Normal	0.77	no progression, followed 5.3 mo
TCGA-A8-A08C	65	II	missing	0.73	no progression, followed 29.0 mo
TCGA-E2-A14W	78	II	missing	0.73	no progression, followed 32.0 mo
TCGA-HN-A2NL	56	II	Basal	0.71	no progression, followed 2.6 mo
TCGA-D8-A1XW	53	II	Normal	0.71	no progression, followed 43.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.